

ECONOMICS & LAW

NIT, TUHH

Jan. 2026 - Feb. 2026

Prof. Dr. Roe Sarel



General

Equilibrium



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Source: <https://plainmoneytalk.files.wordpress.com/2011/09/stock-vector-a-cartoon-military-general-with-a-helmet-on-62644885.jpg>

ROUND OF INTRODUCTION

- Prof. Dr. Roee Sarel
 - The “ee” is the same as in “tree”
 - ..and it’s pronounced “Ro-EE”. Let’s see if you got it
- Prof. of Private Law and Law & Econ
- Past positions:
 - Post-Doc, ILE
 - Post-Doc, FS (lecturer & lab manager)
- Econ. Ph.D (FS), MBA & LL.B (HUJI)
- Certified Lawyer (Israel), experience as litigator in a large law firm
- Research: 1/3 theory, 1/3 experiments, 1/3 empirical. Topics: everything.



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Part I:
Principles of
Economics

Administrative

Part II
TBA



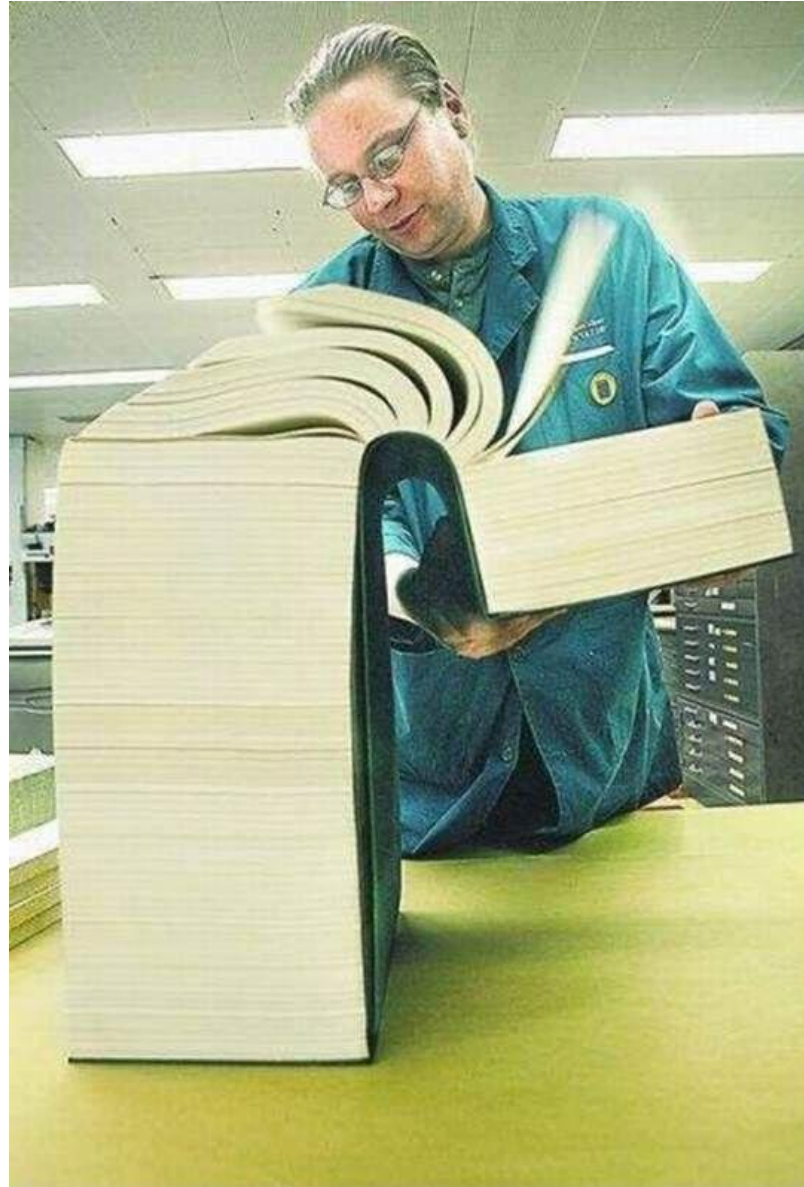
ADMINISTRATIVE

- ◉ Seven weekend sessions
- ◉ Evaluation: 100% exam.
- ◉ Problem sets - for practice only, do not affect the final grade.
- ◉ There is a possibility to write the master's thesis with me as a supervisor - if you are interested, let me know.
- ◉ If you need to contact me, please send me an email to: roee.sarel@uni-hamburg.de

How much I will assume you know about Economics



How much
you will
know when
the course is
over!



If you had 4 apples and 5 oranges in one hand and 6 apples and 7 oranges in the other, what would you have?

Very large hands.

**THIS IS OUR
FRIEND,
GENERAL
EQUILIBRIUM.**

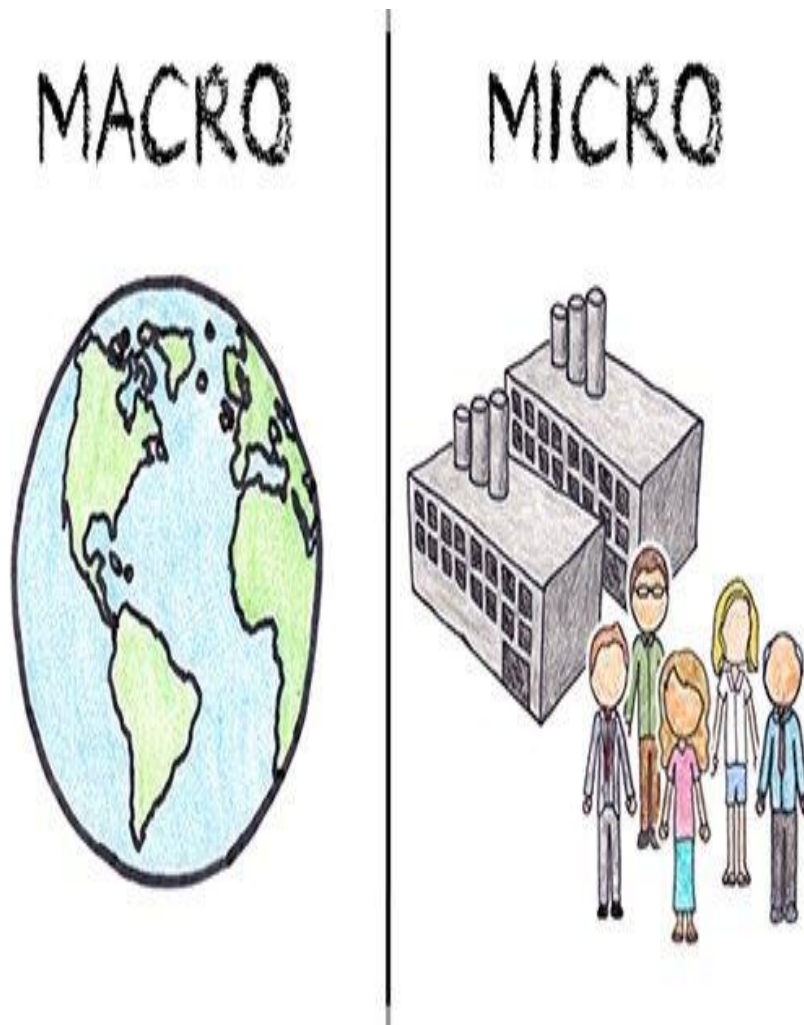
**HIS JOB IS TO
WARN US
WHENEVER
MATH IS
COMING.**



What is
Economics?

Introduction: Economics 2

- Main sub-fields of Economics:
 - Microeconomics: How economic units (individuals/firms) make decisions
 - Macroeconomics (e.g. GDP, International trade...)
- Other sub-divisions:
 - Theoretical vs. Behavioral/Experimental vs. Empirical/Observational
 - Particular fields (e.g. Financial, Labor, Environmental, Law&Econ, Health)
- → The course is “Economics & Law”, i.e. not only “Law & Economics”
 - But there is some overlap.



<https://www.quora.com/What-are-the-differences-between-Macro-and-micro-economy>

Introduction: What is Microeconomics

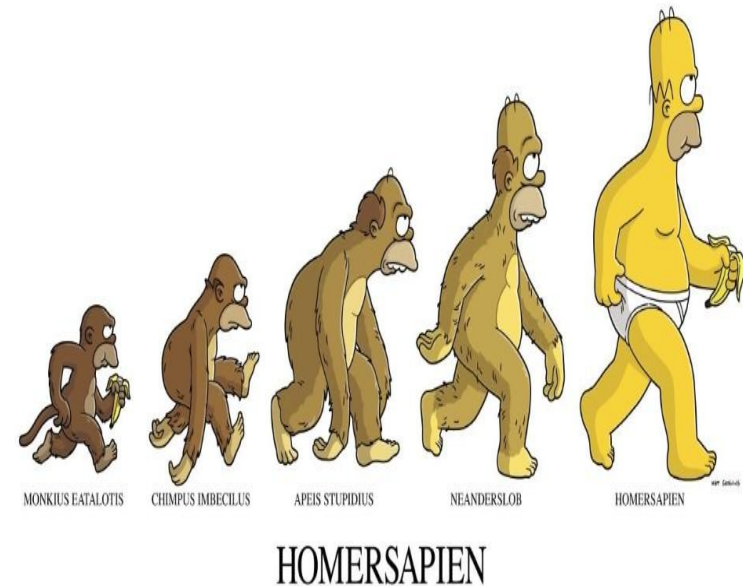
- Decision making of “economic units”
- Examples:
 - How does a monopoly decide how many units to produce?
 - How is the wage of workers determined?
 - How do firms decide on the amount of cancellation fees?
 - How do managers respond to incentive schemes?
 - how do potential criminals respond to changes in the probability of getting caught? And to the penalty?
 - How do politicians decide which policy is the best for fighting Covid-19?

Introduction: Assumptions of Microeconomics

- People want certain things and will change their behavior to get those things – i.e. people will respond to **incentives**.
- Economics adopts the assumptions of “rational choice theory”
 - Individuals are **‘selfish’** – they do what is best for themselves
 - → This means they are maximizing some function which describes their preferences - a **“utility function”**.
- Traditional (neoclassical) economics assumes that people behave as a ***“Homo(o)economicus”***

A Homo(o)economicus...

- Is selfish
- Make choices given *scarcity*
- By: **rational choice**
 - Interplay of preferences (“values”) and constraints
- Ranks Alternatives **can allot utility to every option*
- Strives for utility maximisation



Introduction: Example

- Dr S. is a lecturer of economics & law. He has little free time.
- It is Saturday morning and there is only time for 1 episode before he has to teach.
- Alas, there are 2 options: watch Squid Game, or an episode of “Tatort”
- **OH NO! what to do??**



Introduction: Example

- If he doesn't watch Tatort, he can watch it later.
- If he doesn't watch Squid Game , some student will spoil the ending.
- Dr. S. hates spoilers... so his **utility** from watching Squid Game is higher. Since he maximizes his utility, he will watch Squid Game.



Introduction: Example

- This logic can be applied to the ranking of any set of options, for example:
 - Which product to buy
 - Whether to wear a mask
 - How many hours to dedicate for studying
- Economics assumes that people are able to consider all the possible actions, rank them and choose the best option
 - Later we will explain how ranking is done when the choices involve a risk (e.g. when buying a lottery ticket).
 - Recall: the choices are done while facing *constraints*: e.g. limited budget, laws, and regulations.

How can laws impact a Homo(o)economicus?

- Economics assumes that *preferences* are constant, but constraints can be changed (also by law, e.g. a tax, subsidy, fine)
- ...which affects the costs and benefits related to choosing a certain option...
-which affects individual choices

Lawyers/Judges ...

- Apply the law
- Know (where to look for) the details of the law
- Interpret the law
- Seek justice and fairness ...



Economists...

- Make assumptions
- ...models, models, models...
- ...Costs and benefits...
- Efficiency....



What do Law & Economics scholars do?

**LAW &
ECONOMICS**

LAW

ECONOMICS

What the Hell is this?

LAW & ECONOMICS (FOR SHORT, “L&E”)

- A.k.a. “The economic analysis of law”
 - Legal questions, answered with economic tools / assumptions.
- L&E strives to explain how legal rules impact what happens in the world, i.e. the consequences of legal rules.
- Three main questions:
 - **Behavioral prediction:** How will people behave if the law says “X”?
 - **Efficiency:** which laws promote “welfare” in the world?
 - **Distribution:** how will wealth be distributed between individuals?

What does economics allow us to analyze?

- Economics allows us to analyze how the self-maximizing behavior of individuals leads to a market equilibrium.
- For example, if everyone watches GOT then the rating of Twin Peaks would be zero.. So Twin peaks will not be “produced”.
- But people have different preferences, so to reach a conclusion we must analyze three main things:
 - **Demand:** WTP, “law of demand” (Price $\uparrow$, quantity demanded $\downarrow$)
 - **Supply:** production costs (fixed, variable, average, marginal)
 - **Equilibrium:** interaction of supply and demand
- In an “equilibrium” we can figure out how many “products” are sold, and for which price.

Law vs Economics

CONNECTICUT



It's illegal for a barber to hum a tune while cutting your hair.
— IDIOTLAWS.COM

- The scholarly movement of Law & Economics applies economic thinking to analyzing how the law impacts behavior
 - Positive: what the law is
 - Normative: what the law should (be)
- Example:
 - Legal scholars: “How should we interpret “hum”? The plaintiff argues that *singing* is allowed, whereas the defendant disagrees....”
 - (Law&)Economics: “If humming isn’t allowed, barbers will instead play music in a high volume. This can hurt the clients. Thus, it is better to allow humming”.
- *Note: our course is Economics&Law (not Law&Economics)!*

Part 1: Principles of Economics

1. Basics

2. Necessary
Conditions for
Functioning Markets

3. Market Failure &
Market intervention

4. International Trade

5. Political Economy



Market Equilibrium*



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- 1 Demand**
- 2 Supply**
- 3 Equilibrium**
- 4 Misc.**

Demand

Demand: WTP

- Willingness to pay (WTP): largest amount of money the consumer is willing to pay for the product
- Determined by tastes, income and other products
- Substitutes and complements:
 - **Substitute goods:** fulfill similar need (e.g. Pepsi and cola)
 - **Complementary goods:** increase the value of another product (e.g. computer and keyboard).
- How is WTP affected by substitutes and complementary goods?
 - Substitutes: when price of X $\uparrow$ demand for Y $\uparrow$
 - E.g. if cola becomes expensive, I want Pepsi instead
 - Complementary: when price of X $\uparrow$ demand for Y $\downarrow$
 - E.g. if keyboards are expensive, I want less computers.

Demand: Buyer's demand curve

- How many cups of coffee is one willing to buy per day?

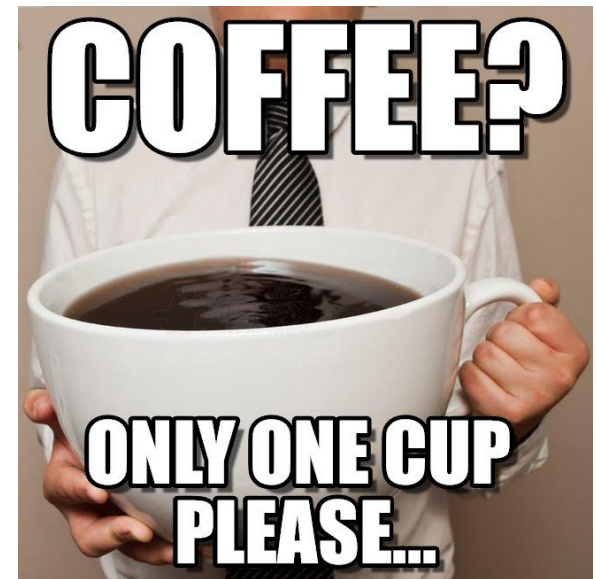
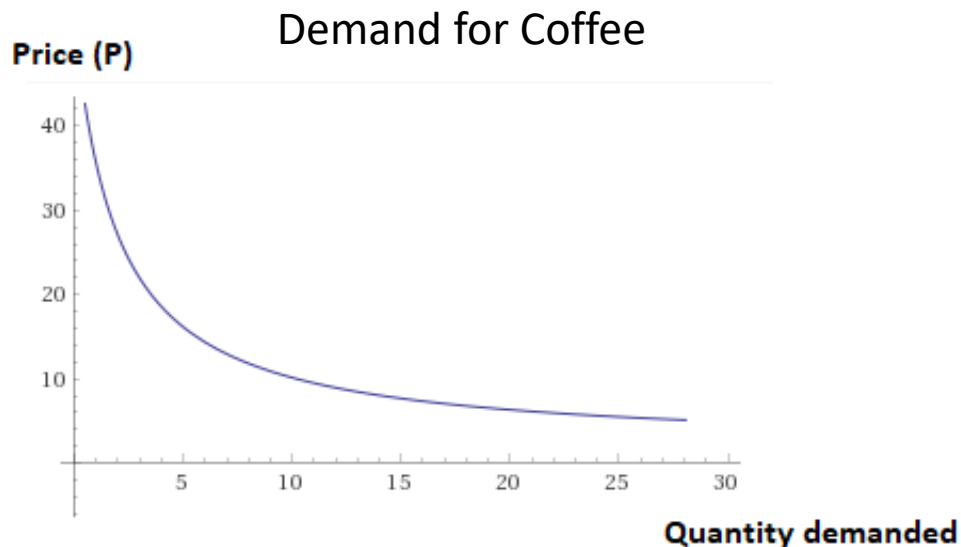
Demand: Buyer's demand curve

- How many cups of coffee is one willing to buy per day?

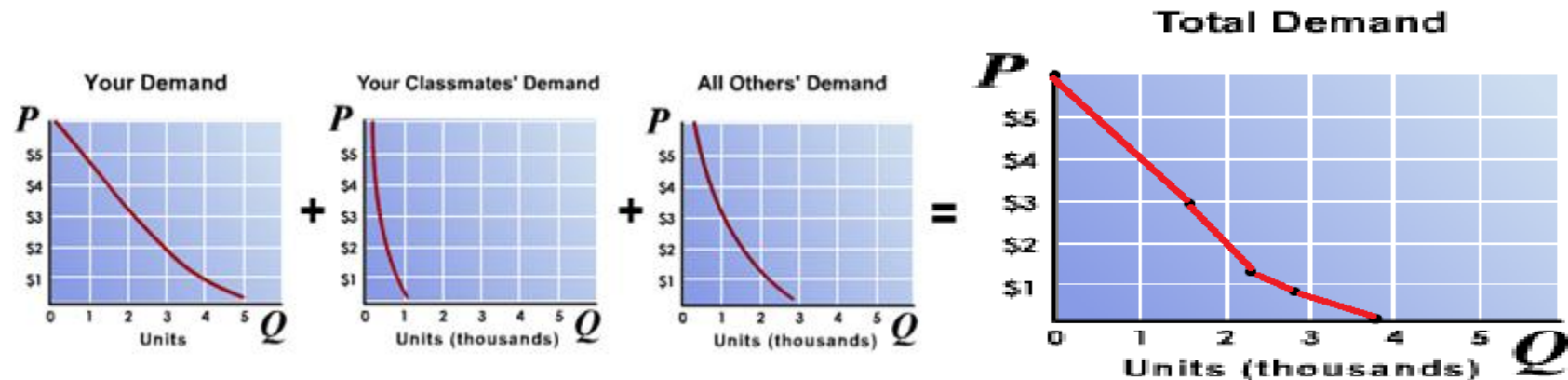


Demand: One buyer's demand

- How many cups of coffee is one willing to buy per day?
- Demand curve: WTP for a quantity at a given price
- Decreasing demand = “law of demand” (lower WTP for additional unit) → Homogenous products!

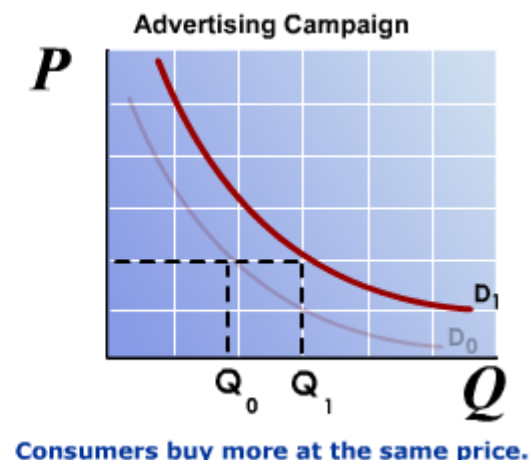
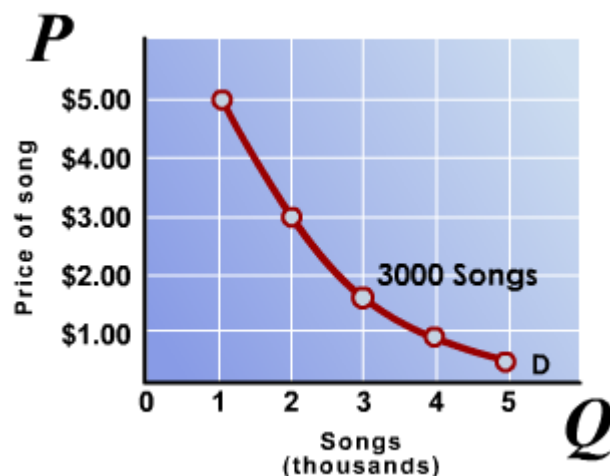


Demand: Total demand curve



- Total demand = sum of all individual demands in the market
- (Total) Demand Curve: quantity demanded by buyers at any given price.
- Note the inverted relationship between price and quantity (due to the law of demand).

Demand: Slides v. Shifts



- Example: Copies of a CD with the song Gangnam Style



- Slides: moving right along the curve indicates increase in quantity demanded as a result of price decrease
- Shifts: entire curve shifts due to shocks (technology, income, advertising, price of substitutes and complements)
 - Shift upwards: e.g. advertising increases WTP for each given price
 - Shift downwards: online shopping reduces demand for CD

Elasticity of Demand: Example - the Cola wars

- Coca-cola founded in 1886 in Atalanta, GA.
- “Brad’s drink” invented in 1893 in New Bern, NC and renamed “Pepsi-cola” in 1898.
- Early 1930’s: The Great Depression
 - 1931: Coke dominant. Pepsi near bankruptcy.
 - Pepsi president buys 12oz recycled beer bottles
 - Tries to sell at 10c, twice the price of 6oz Coke. Fails.
 - 1933. Brilliant idea: Why not sell 12oz Pepsi @ 5c = price of 6oz Coke
 - Jingle: [Twice as Much for a Nickel Too](#)

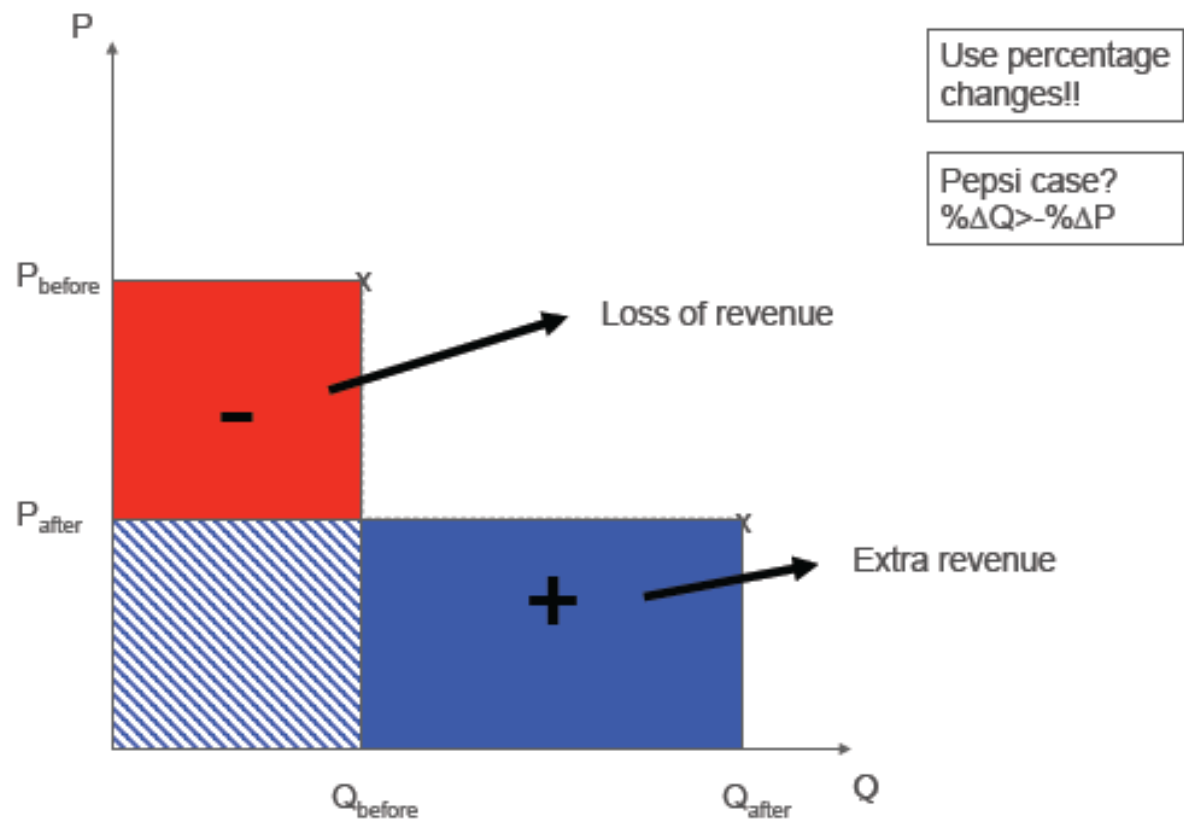


The Cola wars

Sales explode!
Pepsi comes out
of bankruptcy.

**The question:
why was it
profitable to
reduce the
price?**

The Cola Wars: Revenue Increased?? Comparing the Volume Effect and the Price Effect?



Warning: math ahead...



“Everything that can be counted
does not necessarily count;
everything that counts cannot
necessarily be counted”

(Albert Einstein)

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Demand: Elasticity

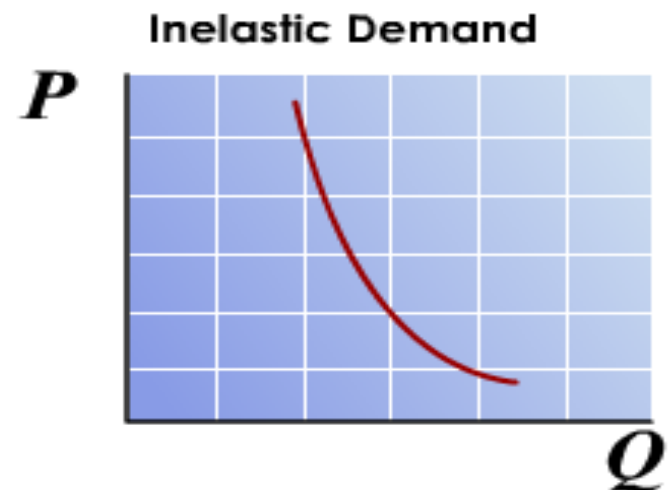
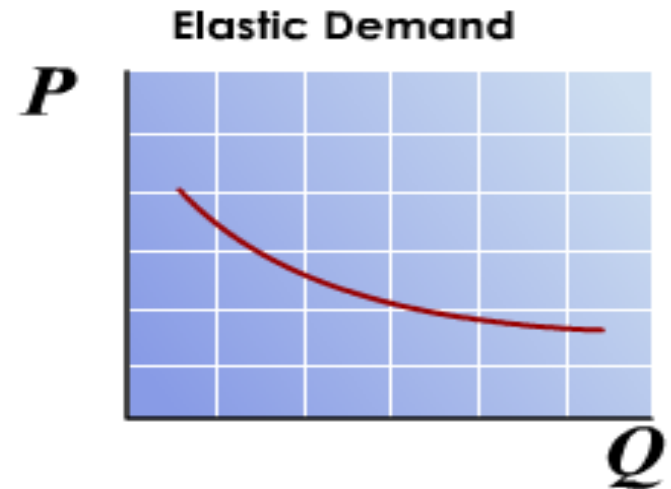
- What is (price) elasticity of demand?
 - A measurement of the **sensitivity** of demand to a change in price.
 - The price elasticity of demand (E_d) is the percent change in demand divided by the percent change in price

$$E_D = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

$$Ed = \frac{\frac{\Delta q}{q_{old}}}{\frac{\Delta p}{p_{old}}} = \frac{p_{old}}{q_{old}} * \frac{\Delta q}{\Delta p}$$

$$\Delta q = q_{new} - q_{old}$$

$$\Delta p = p_{new} - p_{old}$$



Demand: Elasticity - example

- Suppose that Cola-Zero costs 1.20€
- For this price, Roee is willing to buy 10 bottles per week
- Suppose that if the price increases to 1.50 €, he is only willing to buy 9 bottles per week. What is the elasticity at the old price?

	Price (P)	Quantity (Q)
Old	1.2	10
New	1.5	9
Change	1.5-1.2 = 0.3	9-10=-1

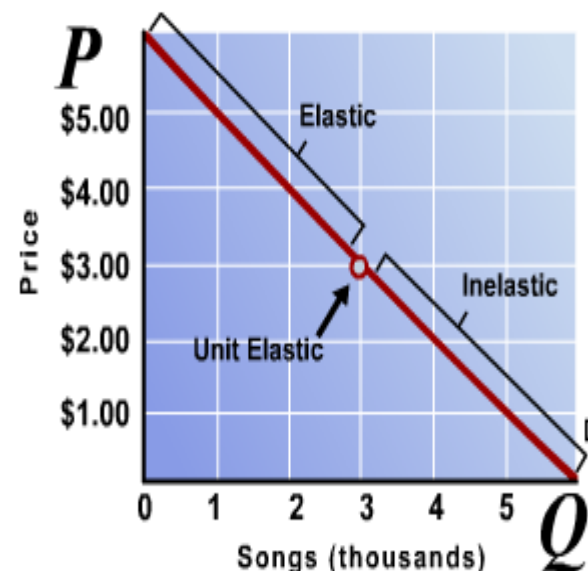
$$Ed = \frac{p_{old}}{q_{old}} * \frac{\Delta q}{\Delta p} = \frac{1.2}{10} * \frac{-1}{0.3} = -0.4$$

Demand: Elasticity

- Why is elasticity a **negative number**?
 - Q and P are inversely related. Thus, when you increase price then ΔQ is negative.
 - Sometimes people use a positive sign for simplicity.
 - You may come across different variations of the Elasticity formula:
 - Loosely: $\% \text{change quantity} / \% \text{change in price}$
 - Point elasticity: elasticity at a certain point.
 - Arc elasticity: relies on mid-point between two points

Demand: Elasticity as sensitivity

- Intuition:
 - If the change in quantity is **bigger** than the change in price, $E_d < -1$ (nominator larger than denominator)
 - If the change is equal, $E_d = -1$ (“unit elastic” or “isolastic”)
 - Elasticity changes as you move along the demand curve



Value	Description
$E_d = -\infty$	Perfectly elastic. Horizontal demand curve
$-\infty < E_d < -1$	Relatively elastic
$E_d = -1$	Unitary Elastic
$-1 < E_d < 0$	Relatively inelastic
$E_d = 0$	Perfectly inelastic. Vertical demand curve

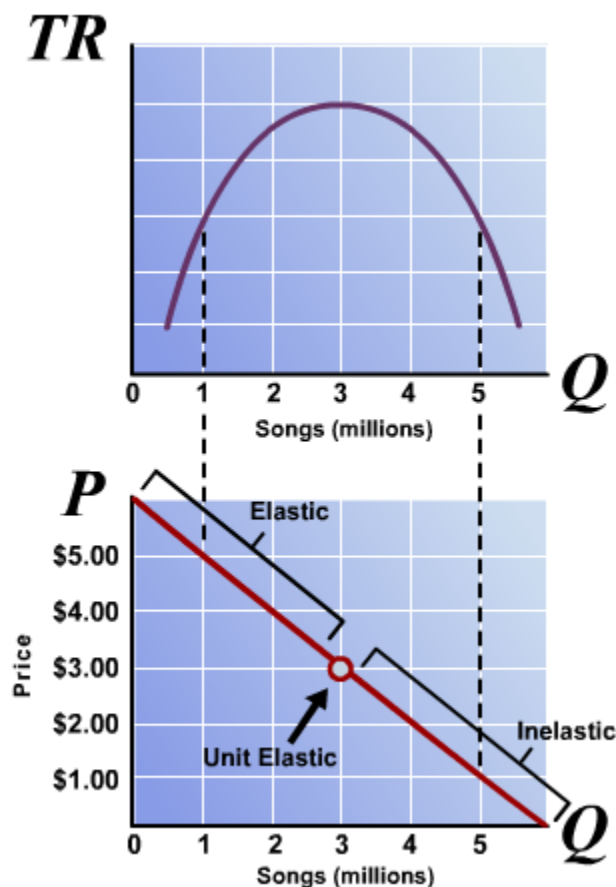
$$E_D = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

Demand: Elasticity

- What influences elasticity?
 - Availability of close substitutes:
 - if I don't mind switching from Pepsi to cola, an incremental increase in price might cause me to switch (=elastic demand)
 - Luxury or necessity:
 - Antibiotics – It can save my life, so I will buy it even if the price increases
 - A 5 star hotel– I might respond to a small increase by not booking the trip

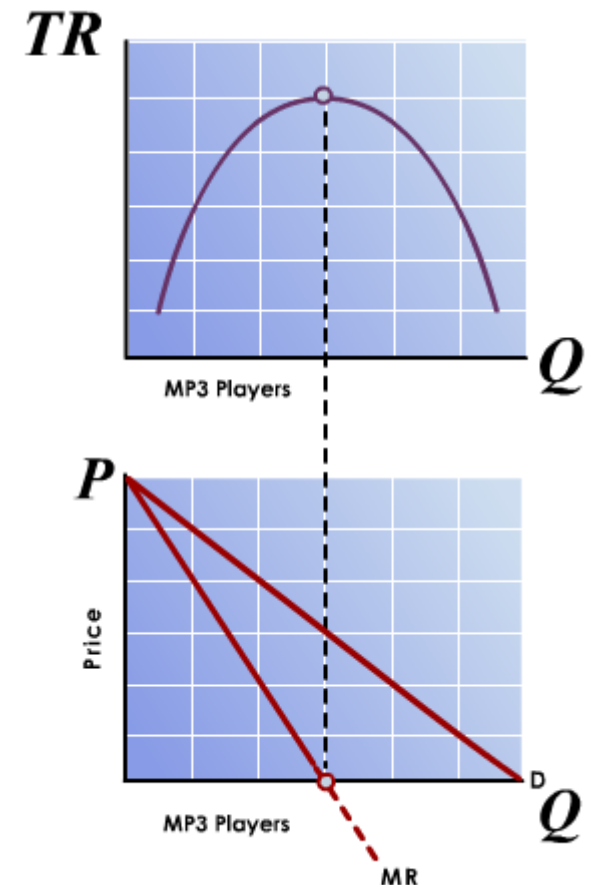
Demand: Total Revenue

- Total revenue = Price * Quantity sold
- Typically, TR curve is “inverse-U” shaped
- TR depends on the elasticity of demand
 - TR increases as long as the percentage change in the quantity demanded is greater than the percentage change in price.
 - In other words: TR increases when the demand is elastic.
 - But at some point, the increase in production (or the decrease in price) no longer “pays off” –
 - The firm produces more, but has less revenue overall
 - The firm lowers the price, but not enough customers buy



Demand: Marginal Revenue

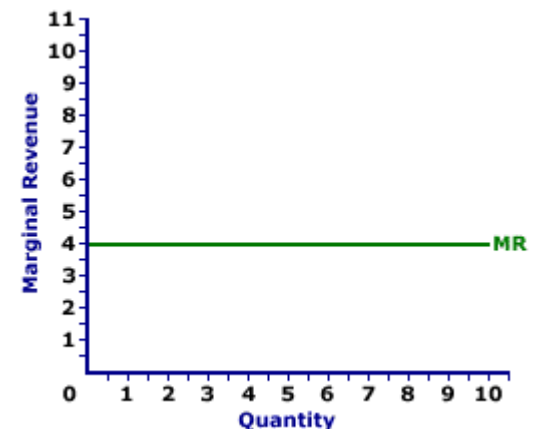
- TR curve: relation between price and sales. To sell ever-greater quantities, the firm needs to offer ever-lower prices.
- MR: term we use to denote the change in TR that results from selling an additional unit of a good.
- MR is positive when TR increases and negative when TR decreases.
- In the picture on the right, the MR curve is downward sloping. Is this always the case?



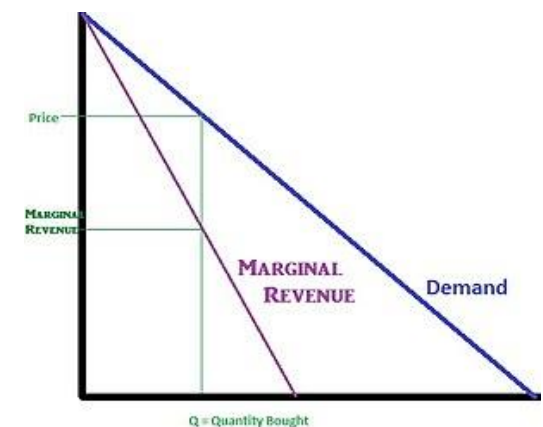
Demand: Marginal Revenue

- In Perfect competition: $MR=P$ (a horizontal line), because the firm is a “price taker”
- Conversely, in a monopoly, the firm controls the price. Then, due to decreasing demand, firm must decrease price for larger quantity
- This decrease is not only for the marginal unit – but for all units.
- Example:
 - Today monopoly sells 1 unit for 10 euro. $MR=10$
 - WTP is lower for 2 units, say 9 euro per unit
 - Monopoly can lower price to 9 euro, but then $MR=9 \cdot 2 - 10 \cdot 1 = 18 - 10 = 8$.
- Thus: $MR < \text{Demand}(P)$ for a monopoly
 - We will see the importance of this when discussing efficiency & market failure.

Perfect competition



Monopoly



Side note: Marginal State of Mind

- It is helpful to use a table to understand the difference between “total” and “marginal”

P	Q	TR	MR
10	1	$10 \cdot 1 = 10$	$10 - 0 = 10$
9	2	$9 \cdot 2 = 18$	$18 - 10 = 8$
8	3	$8 \cdot 3 = 24$	$24 - 18 = 6$

- Why is it important to look at the marginal revenue?
- Recall: A homo(o)economics *ranks* the options
 - The benchmark is what he can (already) get from the other alternative.
 - Only the additional (=marginal) benefit matters



Elasticities in the “real world”

They're talking price elasticity



About the KL-Singapore route:
"Once we get the price elasticity right, I think it will be a huge route"
- Tony Fernandes, Group CEO of Air Asia, January 2008.



"We feel quite confident with the response we've seen to our pricing actions," Rosenfeld said. "Price elasticity has been essentially where we expected it to be."
- Irene Rosenfeld, Kraft CEO (now Chairman), September 2011



"We've done price elasticity studies"
Jeff Bezos
- September 2012



"The price elasticity of heroin use is about 0.5"
Steve Levitt - 2003

Exercise 1: point elasticity and arc elasticity

<u>Point</u>	Price	Quantity
A	60	3000
B	70	2800
C	80	2600

- What is the Point Elasticity of Demand (PED) from point A to point B?
- What is the PED from point B to point C?
- What happens if we calculate using arc elasticity instead (where we calculate such that the “point” is the hypothetical average between the two points)

Exercise 1: point elasticity

Point	Price	Quantity
A	60	3000
B	70	2800
C	80	2600

- What is the Point Elasticity of Demand (PED) from point A to point B?

$$PED = \frac{p}{q} * \frac{\Delta q}{\Delta p} = \frac{60}{3000} * \frac{-200}{10} = -0.4$$

- What is the PED from point B to point C?

$$PED = \frac{p}{q} * \frac{\Delta q}{\Delta p} = \frac{70}{2800} * \frac{-200}{10} = -0.5$$

Arc ED from point B to point C:

$$AED_{A \rightarrow B} = \frac{(p_A + p_B)/2}{(q_A + q_B)/2} * \frac{\Delta q}{\Delta p} = \frac{65}{2900} * \frac{-200}{10} = -0.448$$

$$AED_{B \rightarrow C} = \frac{(p_B + p_C)/2}{(q_B + q_C)/2} * \frac{\Delta q}{\Delta p} = \frac{75}{2700} * \frac{-200}{10} = -0.556$$

Exercise 2: Ice cream on the beach

Ice cream vendors sell ice cream on the Elbstrand. It is known that consumers are willing to spend 100 euros per day on ice cream, irrespective of its price.

Which of the following is correct?

- Demand is perfectly elastic
- Demand is unit-elastic
- Demand is perfectly inelastic

Would your answer change if instead consumers want 100 ice creams, irrespective of price?

Exercise 2 : Ice cream - solution

- When consumers spend a fixed amount (i.e. total expenditure is fixed), demand is unit-elastic
 - Note: for simplicity, use arc-elasticity formula (average price p^* and average quantity q^*)
- Example: consumers want to spend always 100

Total expenditure	P	Q	PED
100	5	20	
100	4	25	?

$$AEd = \frac{p^*}{q^*} * \frac{\Delta q}{\Delta p}$$

$$\frac{4.5}{22.5} * (-5) = -1$$

- If instead consumers want a fixed quantity, irrespective of price, demand is perfectly inelastic

Demand summary questions

Question	Answer
1. Why is the demand curve downward sloping?	1. Law of demand
2. John is willing to pay 5 euros for a shirt, but would agree to buy it for 6. What is his WTP?	2. WTP = highest willingness (=6)
3. When is the marginal revenue function downward sloping?	3. Monopoly
4. What is the price elasticity of demand?	4. $\% \text{Change in } Q / \% \text{Change in } P$
5. Suppose Apple sells 1M iPhones for 100 euros each and at that point the demand is unit elastic. Should apple increase production?	5. No. Apple should only increase when the demand is elastic.
6. Suppose there is a monopoly, cost of production is 0, and the elasticity at some point is -0.2. Is this the profit maximizing price? What should the firm do?	6. Demand is inelastic – firm should decrease quantity until unit-elasticity point is reached

Supply

Supply: Fixed v. Variable costs 1

- Total costs = Fixed costs + Variable costs
- Fixed costs:
 - Independent of the a firm's output
 - Costs that a firm incurs even when it produces zero output
 - Include cash cost & opportunity cost
 - Cannot be changed in the **short run**.
 - In some contexts, a fixed cost is a “sunk cost”
- Variable costs – depends on the level of production.

Supply: Fixed v. Variable costs 2

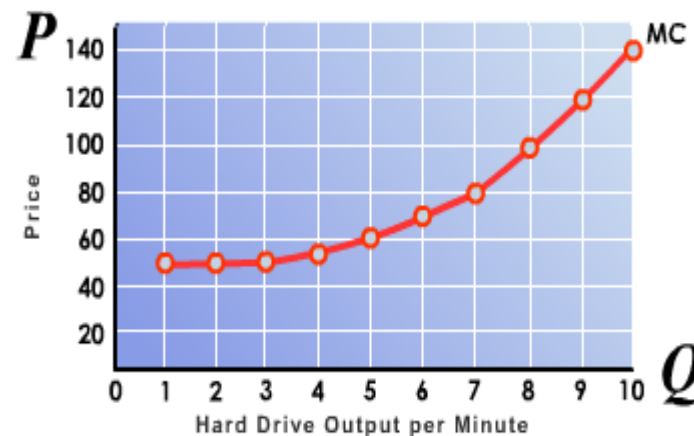
- Fixed or variable cost?
 - Tuition fees
 - Money spent on building the Elbphilharmonie
 - Example – Apple:
 - Firm rents a machine for 10 years to produce iPhones.
 - Hires 1000 workers to operate the machine.
 - Does it matter if the machine can be used to create also parts for Macbook?
 - Overtime must be paid if workers work too much
 - Roee is teaching at NIT.
 - Cost of Roee's train ticket
 - Roee could have used the time to watch Netflix.

Supply: Marginal cost

- Variable costs = sum of MC.
- MC = increase in variable costs if the firm expands production by 1 unit
- Marginal cost includes cash costs and opportunity costs
- Example:
 - Suppose Apple earned last year 1M euros, and must decide between:
 - Depositing the 1M in bank, for 8% per year.
 - Buy more metal to produce 1000 more units
 - What is the marginal cost of buying more metal?
 - Cash cost: money paid to purchase the metal
 - Opportunity cost: 8% interest which could have been earned

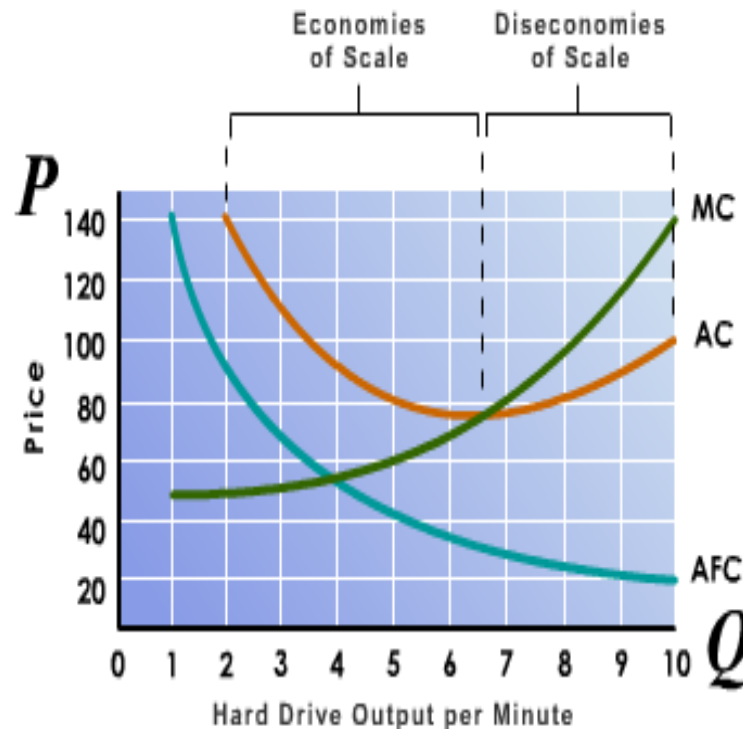
Supply: Marginal cost

- The marginal cost curve is usually increasing. Why?
 - E.g. paying for overtime becomes expensive
 - The “specialists” are used first (see the upcoming slides..)
- When is the marginal curve decreasing (at some parts of it)?
 - Economies of scale
- When the marginal cost is decreasing, companies should increase production, but these extra profits will usually not persist



Supply: Average cost

- Average cost = $\frac{\text{Fixed costs} + \text{Variable Costs}}{\text{Num Of Units}}$
- Why is the AC curve U-shaped?
 - Mixture of FC that do not change, and increasing MC
 - Example: suppose Apple uses its 1M\$ worth factory to produce:
 - 1 iPhone (high AFC, low MC)
 - 100 iPhones (medium AFC and MC)
 - 10M iPhones (low AFC, high MC)
 - Suppose Price = MC = 60 < AC
 - The firm is “losing money” because the revenue is below the AC.
 - Should the firm stop producing in the short run?
 - And in the long-run?



Exercise 3: Average v. Marginal costs

- Suppose the following production costs

Quantity	Average variable costs	Total variable costs	Marginal cost
1	10	?	?
2	15	?	?
3	20	?	?
4	25	?	?
5	30	?	?

- What are the total variable costs for each quantity?
- What are the marginal costs for each quantity?
- Suppose that the product is sold for 36€ in the short run. How many units should the producer produce?

Exercise 3: Average v. Marginal costs - solution

- Suppose the following production costs

Quantity	Average variable costs	Total variable costs	Marginal cost
1	10	10	10
2	15	30	20
3	20	60	30
4	25	100	40
5	30	150	50

- $P=36$. Thus Producer produces 3 units, where $MC < P$

A short introduction to production problems

- One of the questions that Economics can answer is how to design the production of goods.
- For example:
 - Which products should be produced and how many?
 - E.g. depending on how much demand there is.
 - Alternative: A central planner that sets quotas
 - How should the products be produced (which inputs)?
 - Each unit of production must know its *relative advantages*.
 - Example: some products are produced in developing countries because the wages are lower
 - To whom should the products be allocated?
 - More of a political question

The problem of scarcity

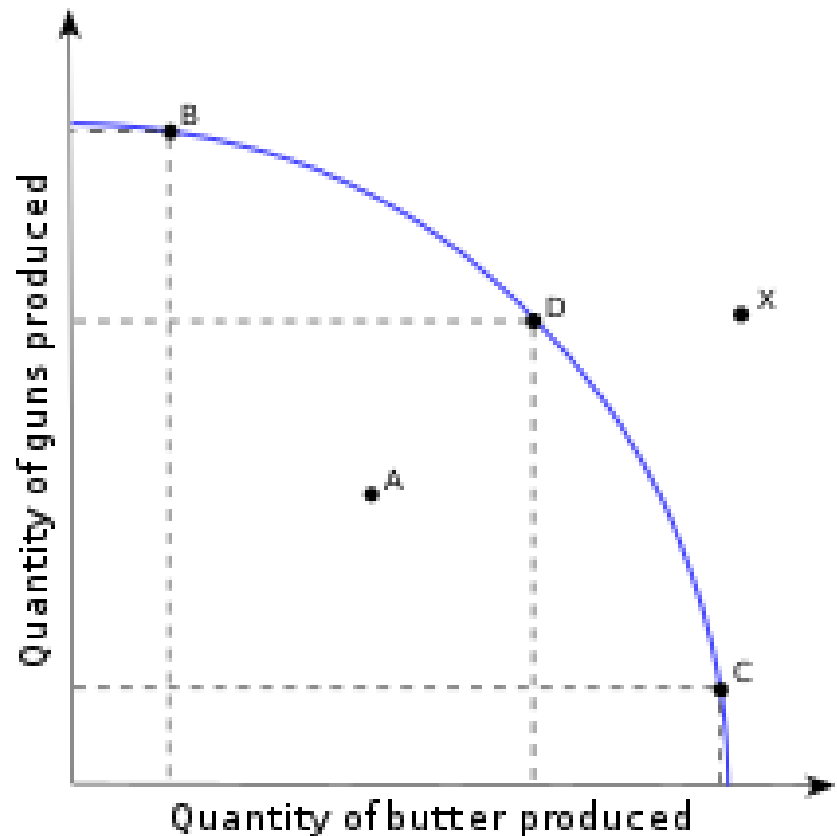
- **The Problem:** There is a difference between what individuals want and what individuals can have, given the available amount of resources.
 - Doesn't apply to non-scarce resources such as air.
- **The problem of scarcity implies that any decision to produce one good involves giving up the production of another good.**
- This basic rules provides the foundation to fundamental questions in economics

The problem of scarcity

- To address the problem, production should have:
 - Full employment: no wasted production factors
 - Implies no unemployment: because you could hire people and they produce w/o giving up something else
 - Structural unemployment: I have two factories requiring two workers each, but I have 3 workers. The third can't be exploited.
 - Efficient production:
 - Allocation of workers to their specialization
 - E.g. don't send mechanics to work as doctors.

The production-possibility frontier

- Suppose that a firm can only produce two products: guns and butter
- The frontier describes all the efficient production combinations given full employment
- The frontier is concave
 - At first, if you give up 1 gun you get a lot more butter
 - Then, if you give up 1 gun you get less butter.
 - Why is that? (we will get back to this in a few slides)
- Point X – cannot be achieved
- Point A – can be achieved, but inefficient



The production-possibility frontier

Output table	Butter	Guns
Worker A	2	4
Worker B	1	1

- Worker A has an absolute advantage – can produce more of both.
- But which worker should produce what?

The production-possibility frontier

Output table	Butter	Guns
Worker A	2	4
Worker B	1	1

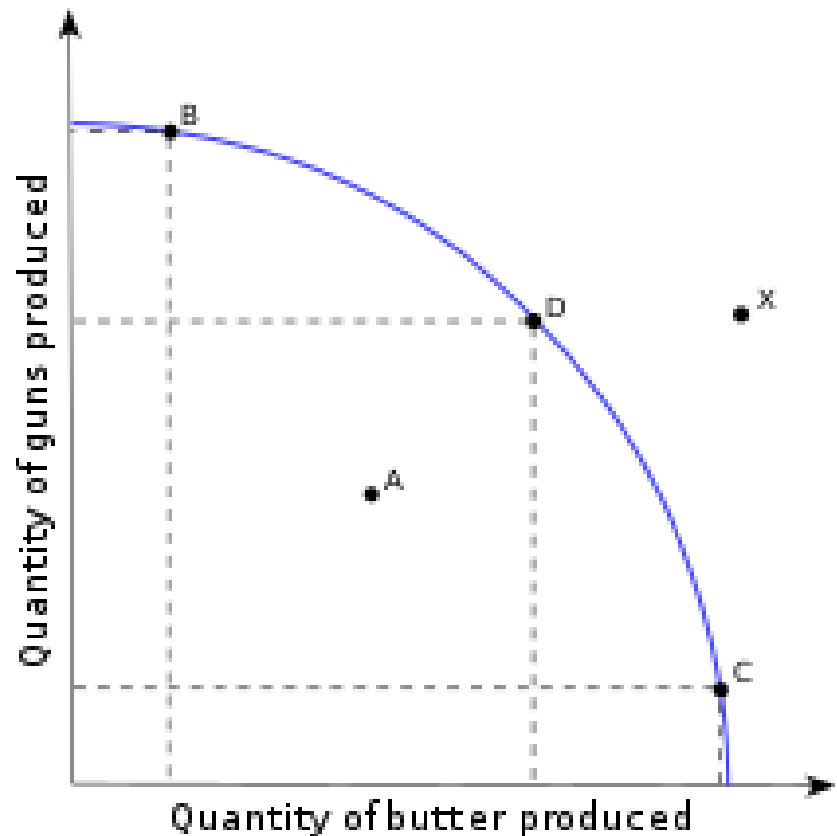
Opportunity cost	How many Guns to give up to produce 1 Butter	How many Butters to give up to produce 1 gun
Worker A	2	0.5
Worker B	1	1

- Worker A has a lower cost to produce guns ($0.5 < 1$) and Worker B has a lower cost to produce Butter ($1 < 2$).
- Thus -> Allocate Worker A to guns and Worker B to butter.

**We will get back to the importance of this insight when we talk about international trade*

The production-possibility frontier

- Suppose I have 100 Workers of type A and 100 workers of type B.
- First, I allocate all A to produce guns. This way, I don't give up much butter.
- Then, If I want more guns, I must start giving up more butter by using workers of type B.
- → Increasing opportunity cost

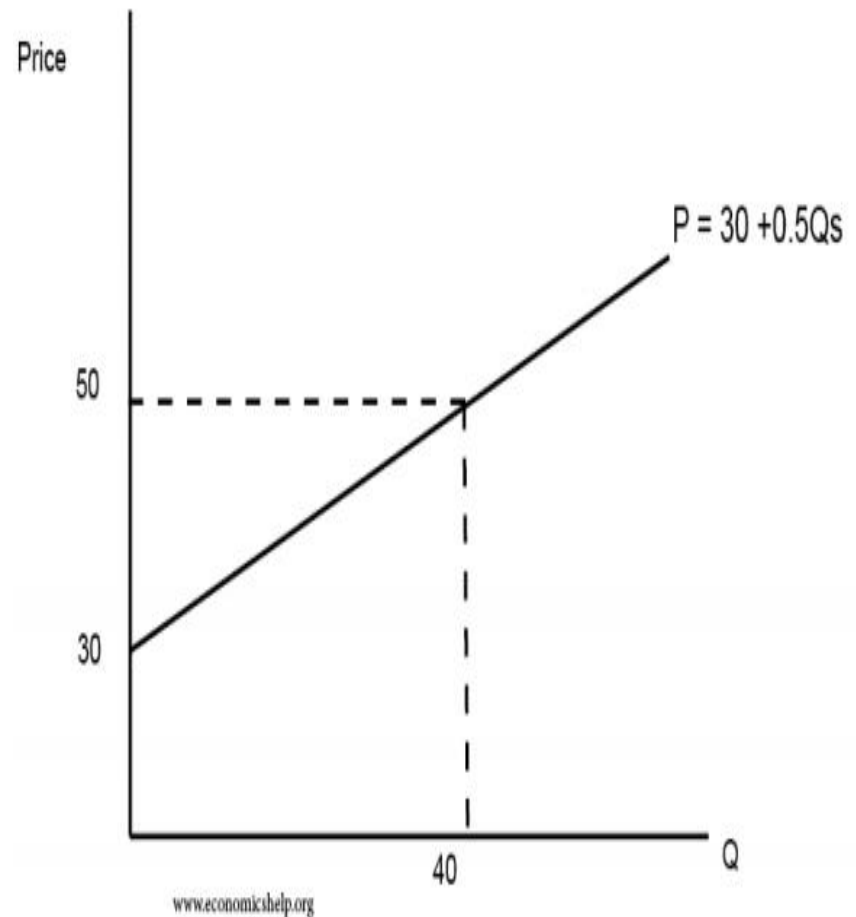


Connection to the marginal cost

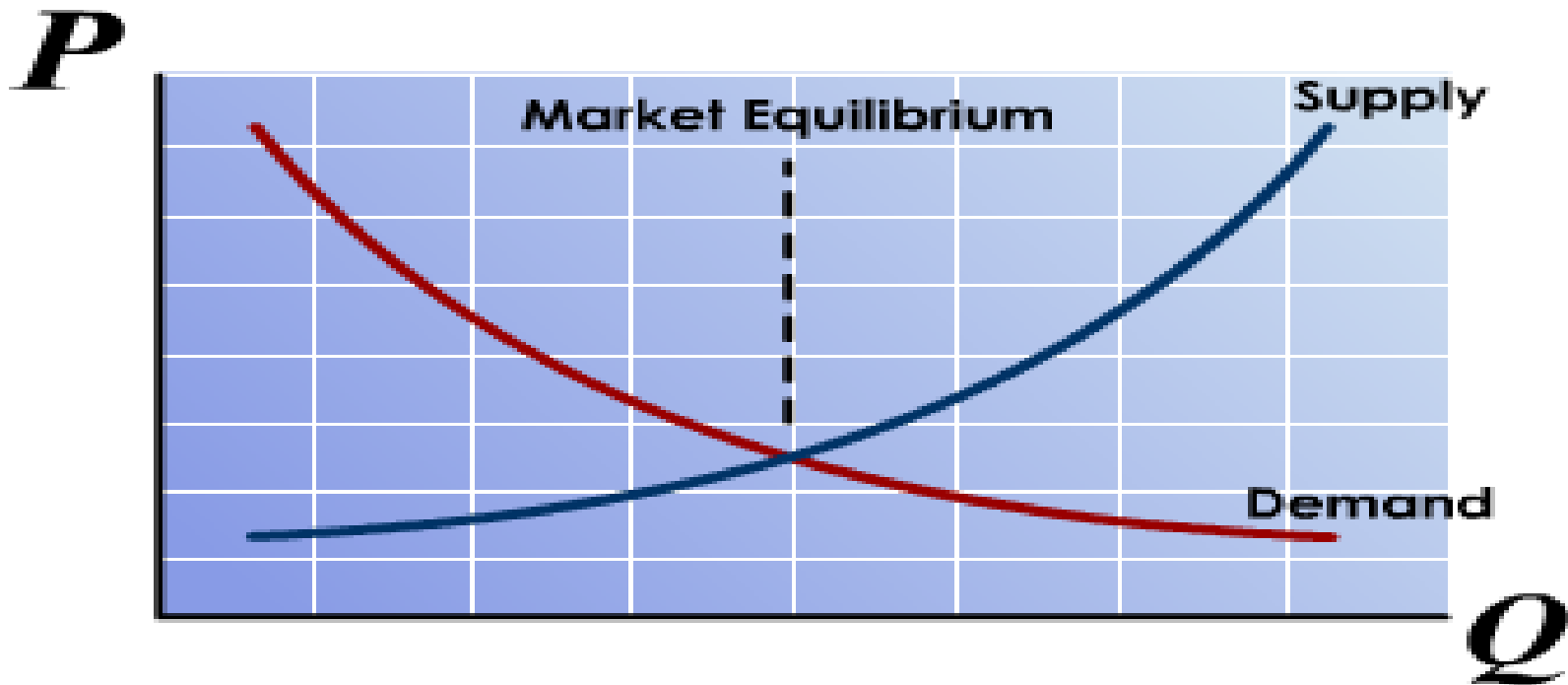
- One reason for the marginal cost of production to be increasing, is the need to allocate less specialized employees
 - Or instead: “too many chefs for one kitchen”
- Alternatively, think of paying “over time” - you first ask all the workers to fill-up their hours (‘type A’). Then, you must use those who already filled up their hours (‘type B’).
- On the market level: might be because more inefficient firms enter, even if there is no within-firm increasing marginal costs

The supply curve

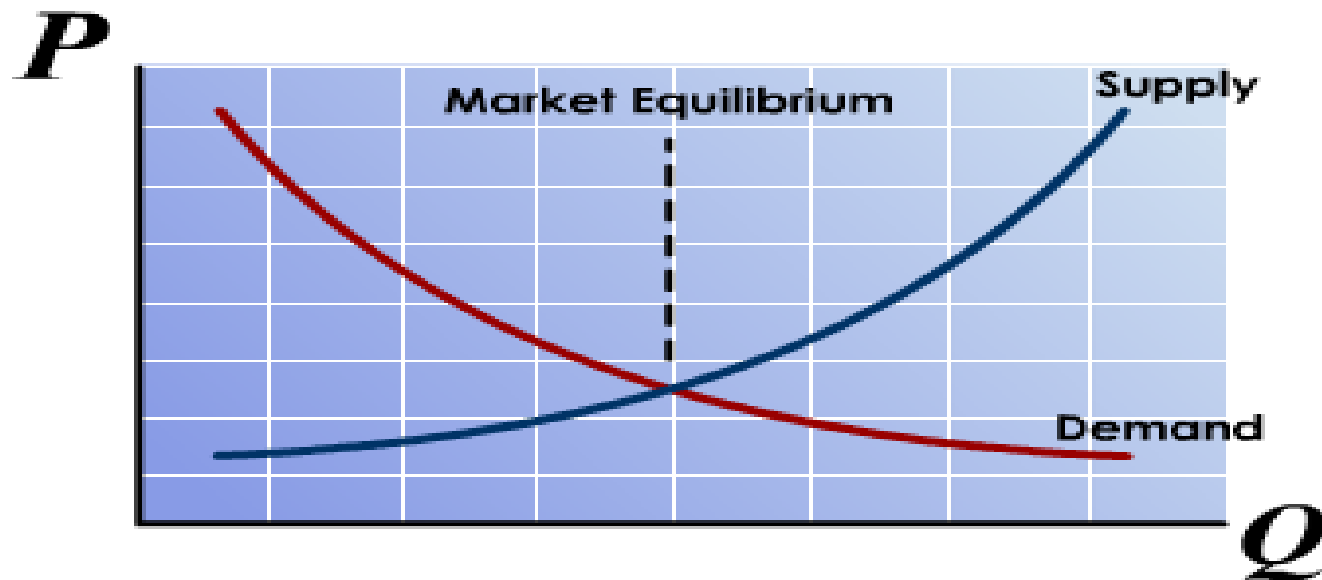
- The supply curve captures the relationship between the willingness to produce and the price
- The Willingness to produce depends on the **marginal cost**
- Why is the supply curve upward-sloping?



Equilibrium



Equilibrium: firm & industry supply curve



- Where does price come from?
- Industry supply interacts with demand
- In equilibrium, **Supply = Demand.**

Equilibrium: Example

- The demand and supply curves for coffee are given by:

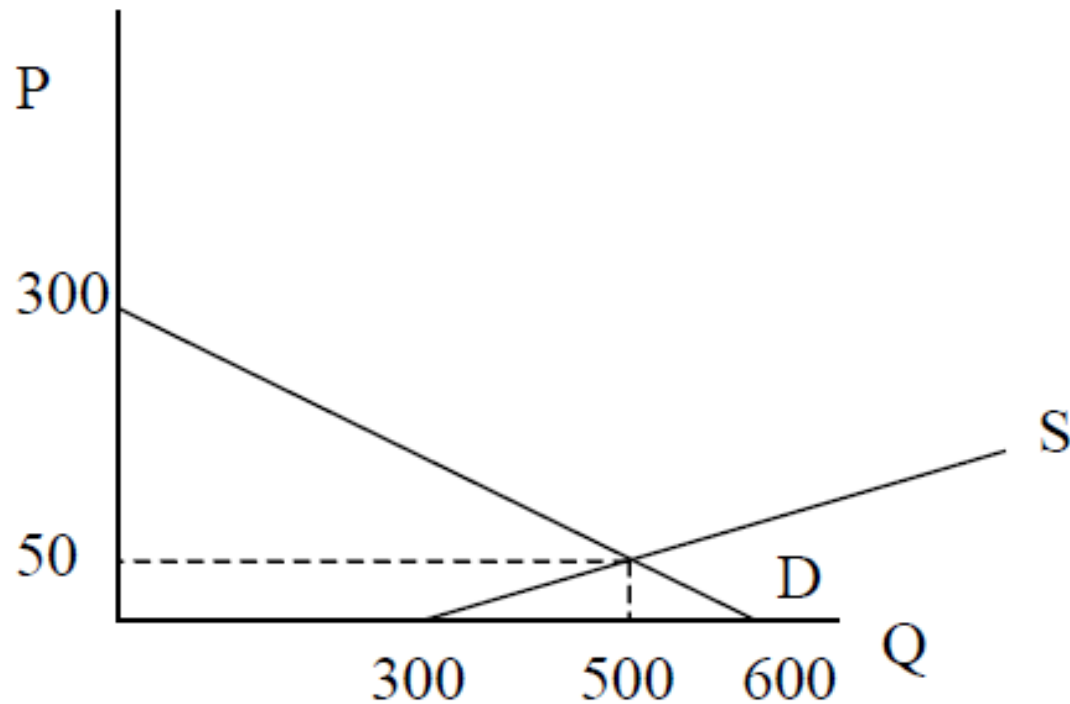
Demand: $Q^d = 600 - 2p$

Supply: $Q^s = 300 + 4p$

- What is the equilibrium price? What is the equilibrium quantity?
- We know that supply = demand, i.e. $Q^d = Q^s$.
- Thus, $600 - 2p = 300 + 4p$
- And $p = 50$ such that $Q = 300 + 4 * 50 = 500$

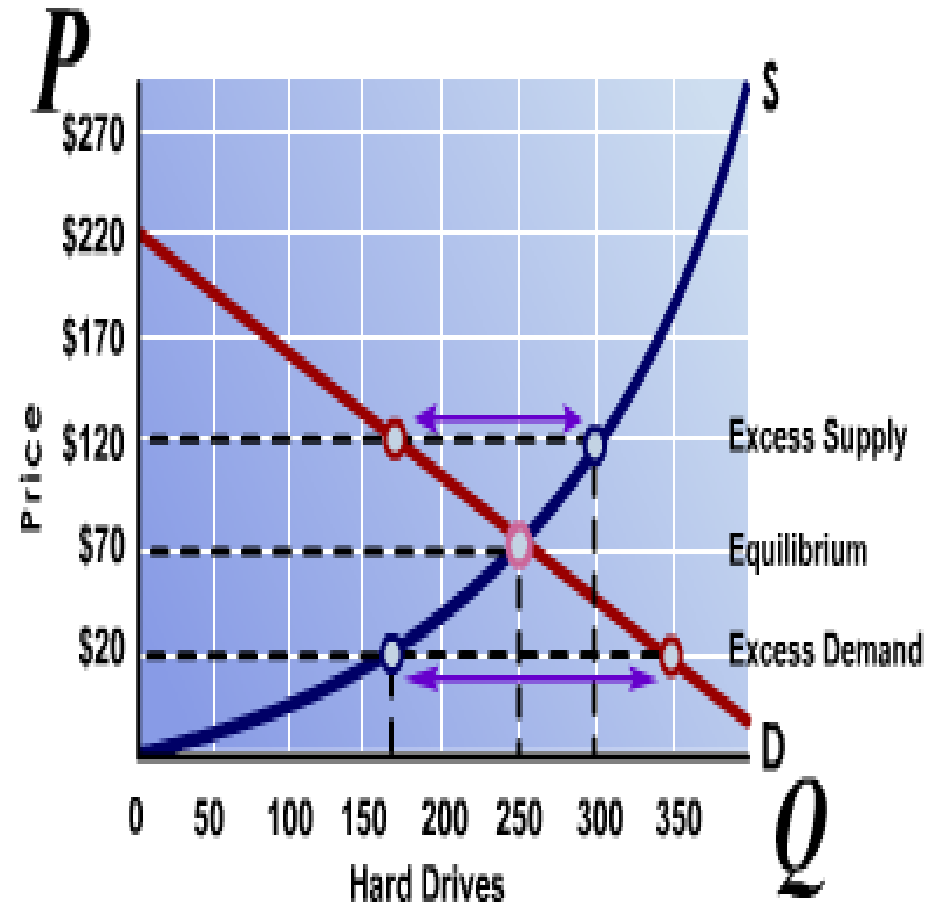
Equilibrium: Example

- We can also determine the equilibrium graphically

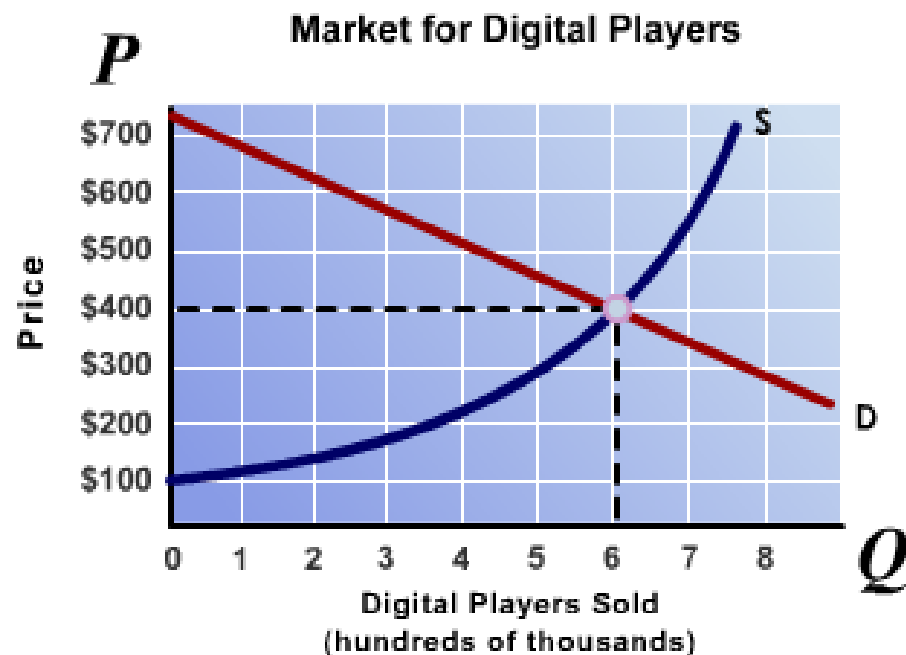
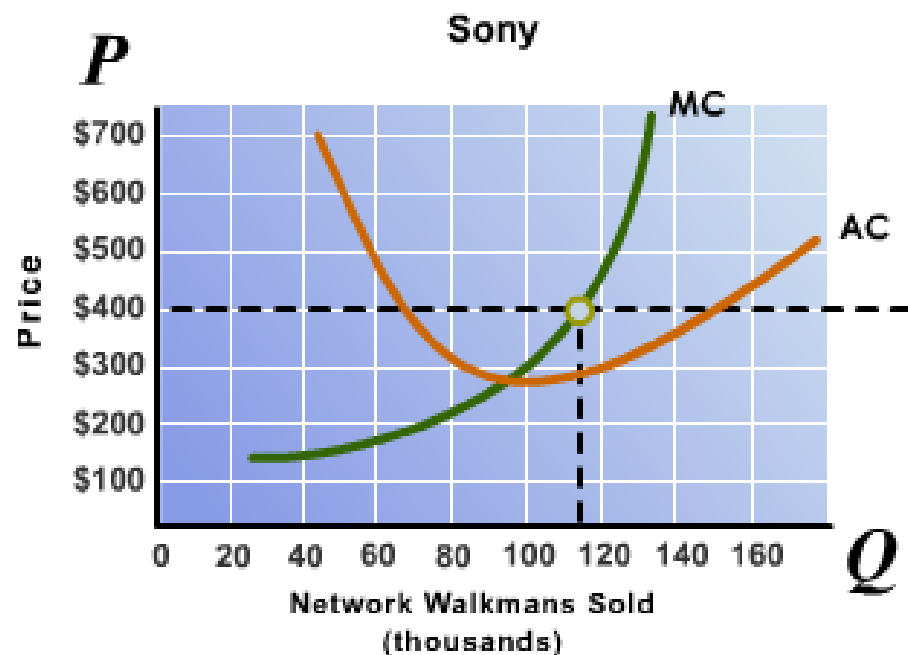


Equilibrium: excess supply and excess demand

- What happens if there is excess supply? Excess demand?
 - If $D > S$, production/entry is encouraged
 - If $S > D$, reducing production/exit is encouraged
 - Eventually, market reaches equilibrium

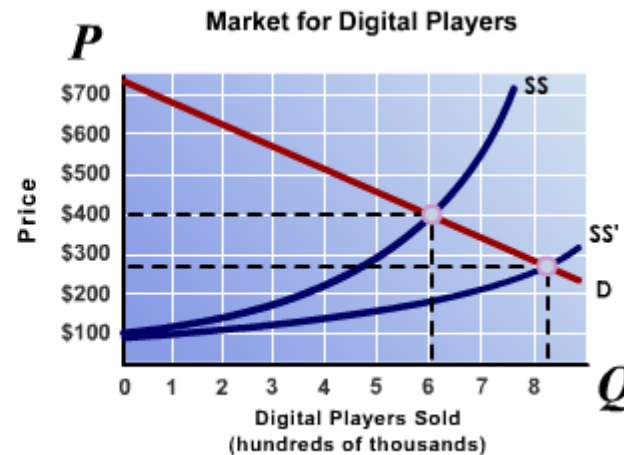
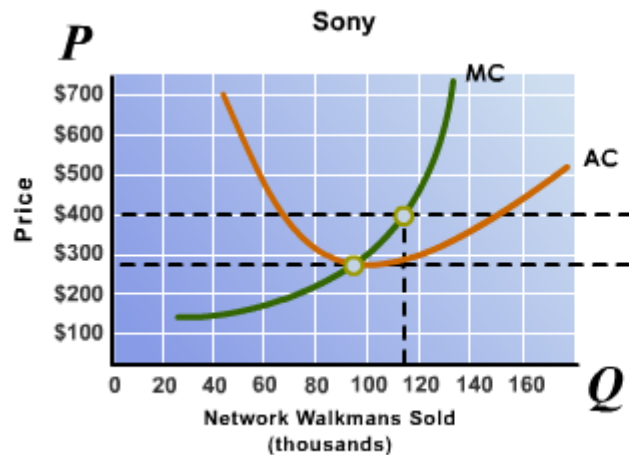


Equilibrium: long run equilibrium



- In the example, $P=400=MC$ but AC is lower.
 - Why not decrease production to 70K, where $P=AC$?
 - Why not Increase production to (approx.) 145K, where $P=AC$?
- Is 400\$ a long term equilibrium?

Equilibrium: long run adjustments



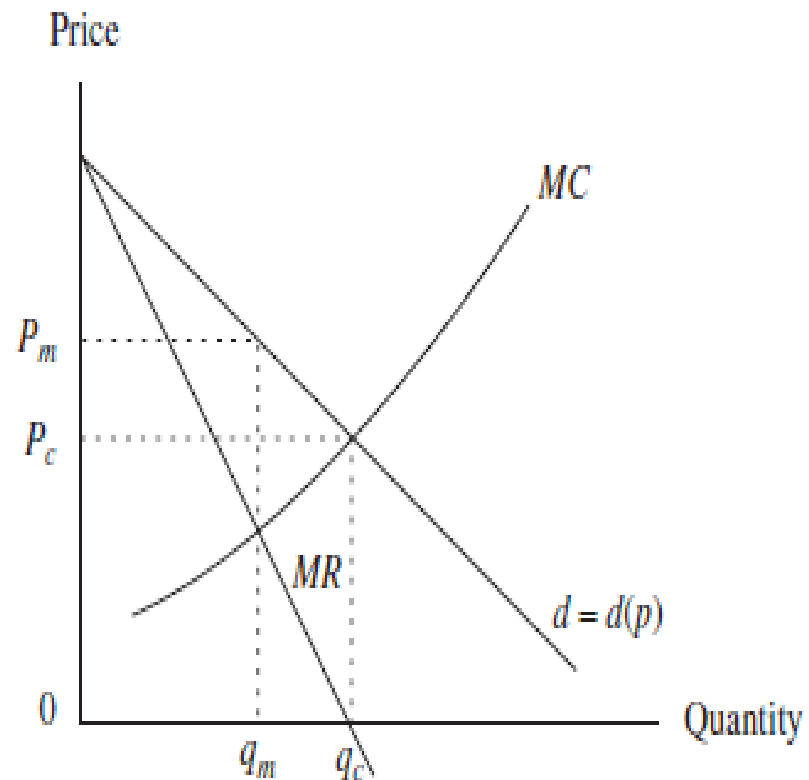
- A price of 400\$ is not a long term equilibrium because $P > AC$
- In the long run, these economic profits lead new firms to enter the market / existing ones to increase capacity
- Supply then increases and price adjusts until $P = MC = AC$

Equilibrium: Market power & Branding

- Firms enjoy market power when they can increase the price, without losing all customers to another firm
- We already saw the two extremes – perfect competition and monopoly
- Why do firms have market power?
 - Information asymmetries: customers don't know the difference
 - Product Differentiation (e.g. brands)
- How does market power relate to elasticity of demand? (example: patent on life-saving drug)
- How much should a firm with market power charge for its product? The competitive price? More? Less?

Equilibrium: Market power & Branding

- Firms maximize their profit when $MR=MC$.
- For a monopoly, this holds at the point which is lower than the demand curve (in the graph – at q_m)
- The monopoly thus produces less than a firm in perfect competition ($=q_c$)
- And charges a higher price



Perfect competition = no Market power

- **Conditions for a perfect competition (no single firm can impact the price)**
 - Sufficiently many buyers and sellers
 - No differentiation - each firm produces the same product (or products are perfect substitutes)
 - No information asymmetries regarding the price
 - No transaction costs
 - No barriers to enter and exit the market
- Demand is then perfectly elastic for each of the small firms that participate
- Recall: In a perfect competition, $MR=MC=P=D$
- In the long run, firms enter/exit until economic profits are zero.

The firm's problem (formally)

- Suppose the firm decides how much to produce - quantity q .
- Assume:
 - marginal costs increase in q (i.e., $c'(q) > 0$)
 - There are fixed costs F
 - The price behaves as follows: if the firm is a monopoly, then $p'(q) < 0$, but if the firm is in a perfectly competitive market, then $p'(q) = 0$ (i.e. a price taker)

- The firm's problem is

$$\max_q \Pi(q) = (p(q) - c(q)) * q - F$$

- The first order condition (FOC) is

$$\frac{\partial \Pi(q)}{\partial q} = (p'(q) - c'(q)) * q + p(q) - c(q) = 0$$

- With perfect competition: produce such that $p = c(q) + c'(q)q$
- But with a monopoly: produce such that $p(q) + p'(q)q = c(q) + c'(q)q$
- The LHS in both is the MR, whereas the RHS is the MC. Thus, MR=MC.

Elasticity and Market power: Example - SSNIP Test

- How is market power related to the price elasticity of demand?
- Elasticity of simultaneous and small (5-10%) price increases is used to define **relevant markets** in competition policy all over the world. Why does this make sense?
- Hint: How would you know if a market is “worth monopolizing”?

Other elasticities

- **Cross-price elasticity of demand**
 - $\% \text{Change in quantity demanded} / \% \text{change in price of another product}$
 - If it is positive – products are substitutes. If negative – complements.
- **Income elasticity of demand**
 - $\% \text{Change in quantity demanded} / \% \text{ change in income}$
 - May be positive (Luxury), negative (inferior) or zero (“sticky goods”)
- **Price elasticity of supply**
 - $\% \text{Change in quantity supplied} / \% \text{Change in price}$
 - Main determinants – how fast can a company change production
 - Availability of input materials (e.g. wood vs. diamonds)
 - Production complexity (e.g. reliance on low-skilled workers v. experts)
 - Ability to switch production line (e.g. can Apple use its factory to produce Macbooks)
 - Time horizon [e.g. supply is elastic in long-run, but less in short-run]
 - Production capacity [e.g. are there unused factories]

Exercises

Exercise 4: Changes in production costs

- Suppose that Coca-Cola produces its magical drink using only one input product – a syrup.
 - Assume perfect competition.
 - If the price of the *syrup* increases and no other changes occur, what should Coca-Cola do?
1. Increase the quantity of cola produced
 2. Decrease the quantity of cola produced
 3. Increase the price of cola
 4. Answers (1) and (3) are correct
 5. Answers (2) and (3) are correct

Exercise 4: Changes in production costs – solution

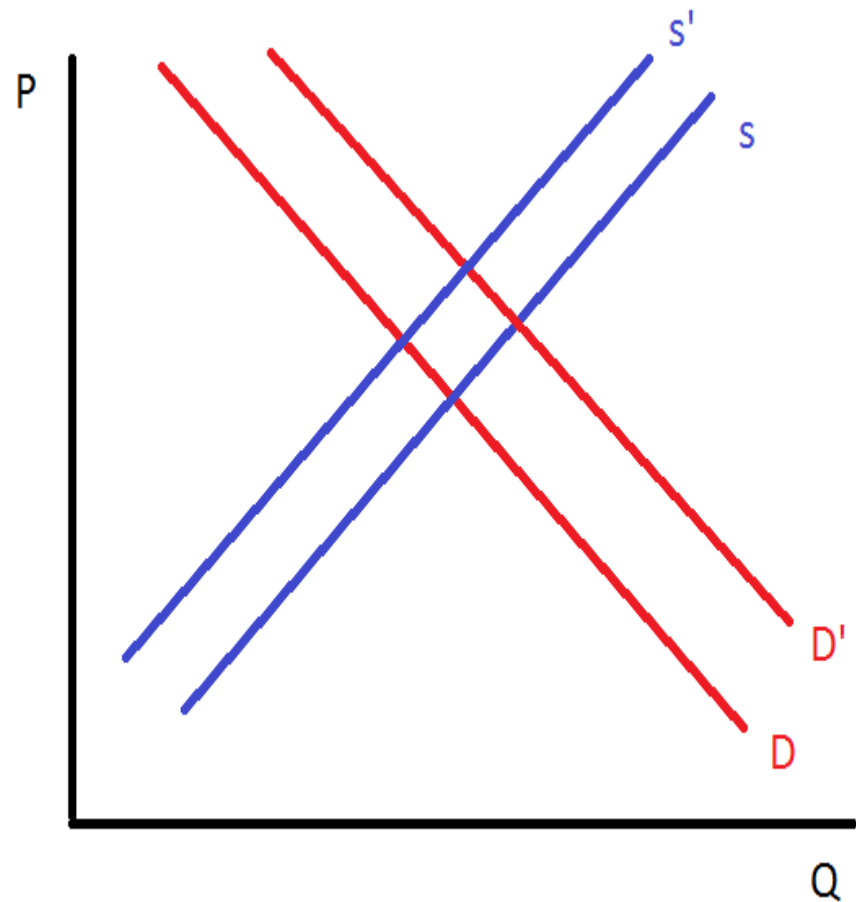
- An increase in the price of an input implies an increase in marginal costs
- This means that the supply will be decreased
- Since we assume perfect competition, Coca-Cola will lose all its customers if they try to raise the price
- Therefore they can only **decrease the quantity**

Exercise 5: Simultaneous Changes

- Product X is soled in a market with regular demand and supply curves.
- What happens if many firms leave the market and additionally the income of consumers increases? (choose the correct answer)
 - “It is possible the equilibrium quantity decreases”
 - “Firms that remain in the market will increase production”
 - “Overall revenues in the market will necessarily increase”
 - “The Price will increase “

Exercise 5: Simultaneous Changes - solution

- Firms leave the market $\rightarrow$ supply decreases from S to S'
- Income increases $\rightarrow$ demand increases from D to D'
- Equilibrium quantity may or may not change
- Revenues may or may not change
- Price increases



Exercise 6: Superstars

Earns €481,000 per week



Does not earn €400,000 per week



- Why does Leo Messi earn relatively so much?
 - Supply of superstar soccer players is very low, while the supply of competent academic teachers is much larger.
 - Demand for Messi's talents is very high since he can generate so much revenue for a firm.
 - Consumers enjoy Soccer I to the point that they are willing to spend lots of money and time attending games and watching commercials
 - All of the above

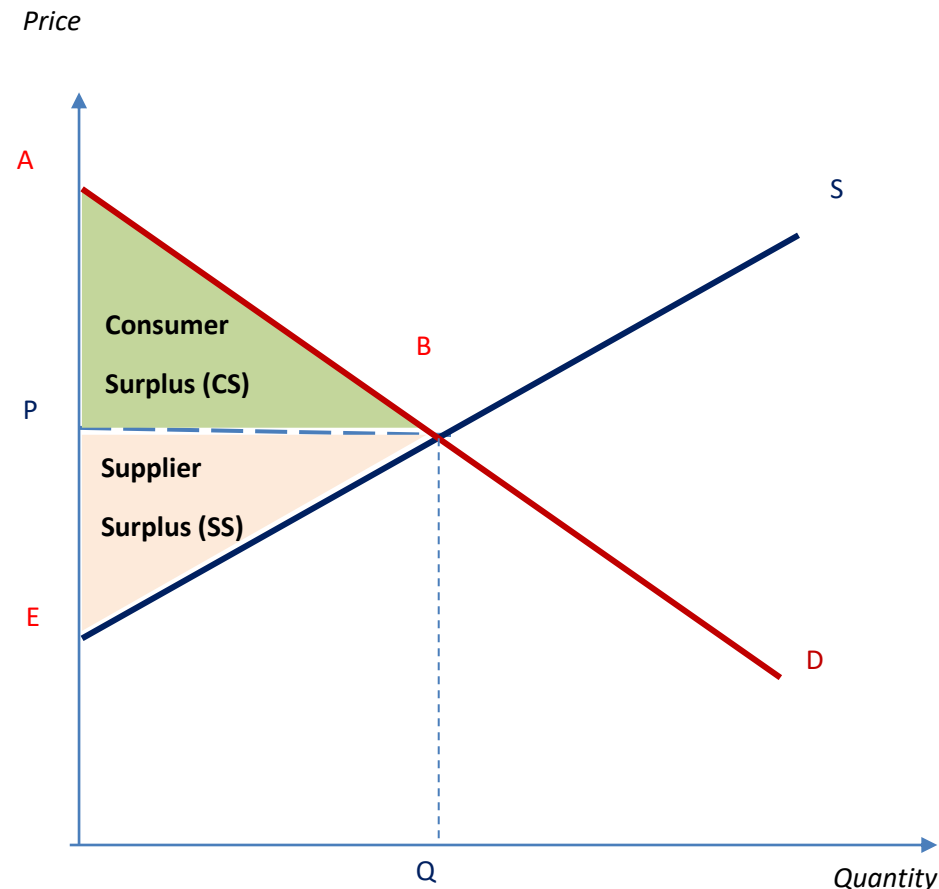
Surplus

Consumer surplus

- In equilibrium, the price is lower than the willingness to pay for lower quantities (otherwise, there is no trade)
- This implies a “surplus” to consumer: they were willing to pay more, but paid less
- Different definitions (all mean the same)
 - **Consumer surplus** is a measure of the **welfare** that people gain from consuming goods and services
 - Consumer surplus is defined as the difference between the total amount that consumers are **willing and able to pay** for a good or service (indicated by the demand curve) **and the total amount that they actually do pay** (i.e. the market price).
 - Consumer surplus is shown by the area under the demand curve and above the price.

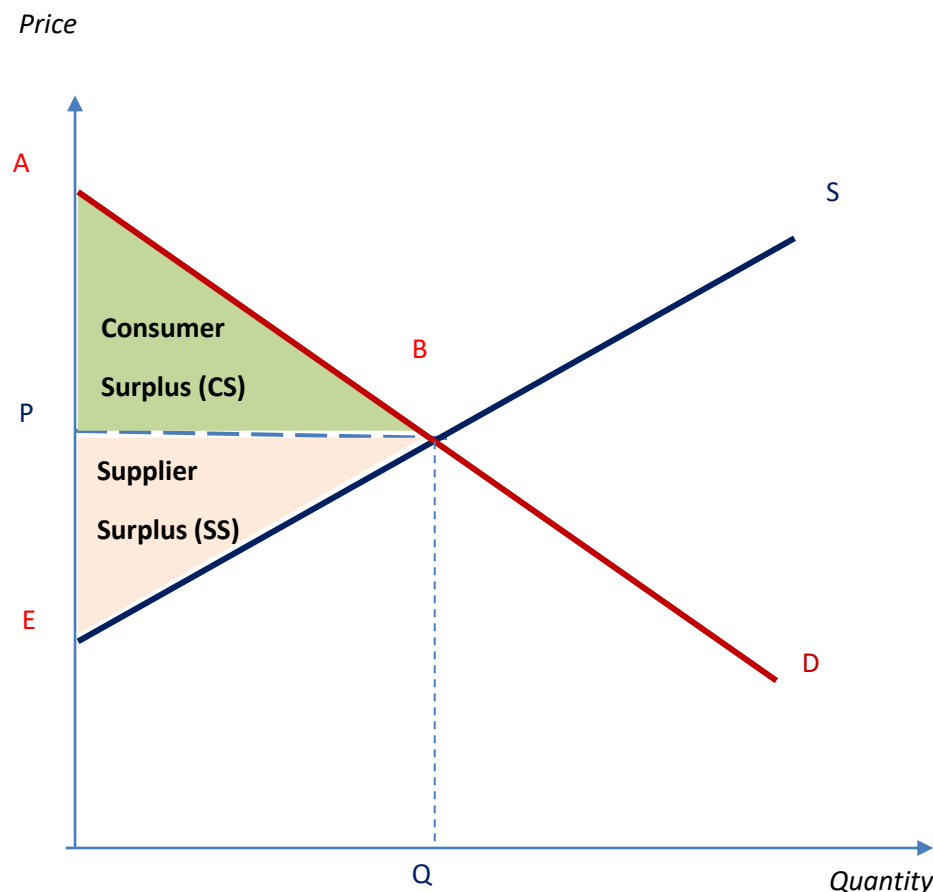
Consumer and supplier surplus

- Producers also have a surplus, created by selling in a price $>$ production cost
- The total surplus thus consists of consumer + supplier surplus
- To calculate surplus:
 - Consumer surplus: area of the green shaded triangle.
 - Supplier surplus: pink shaded area.
 - Think back to geometry class: formula for area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$.
 - In this case, the base of the triangle is the equilibrium quantity (Q)



Consumer surplus: numerical example

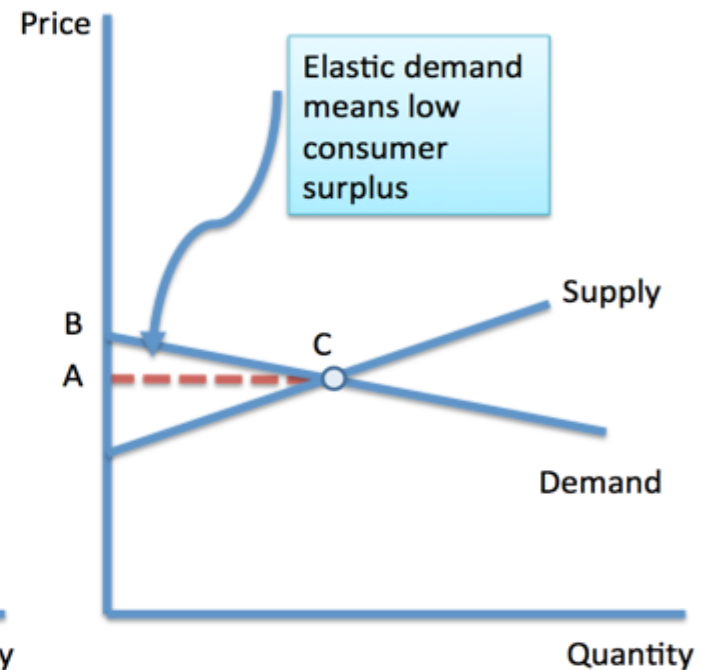
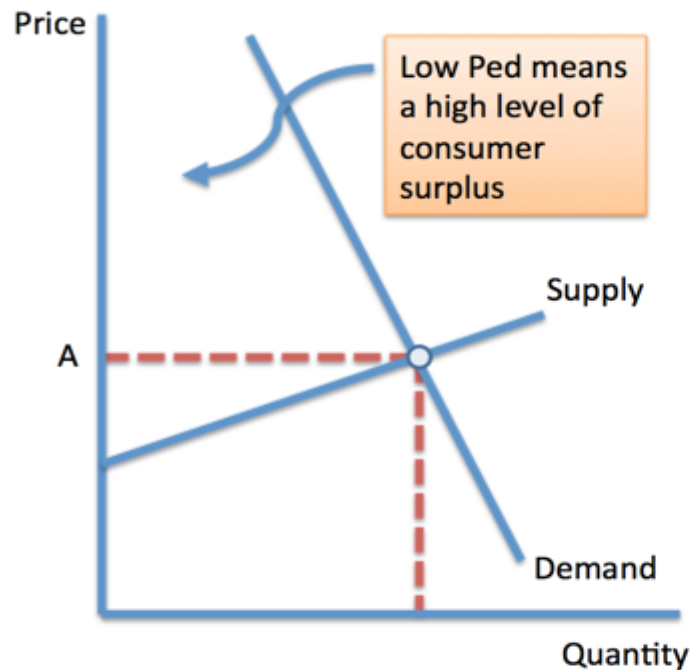
- Suppose: $P=8-2Q$ and $P=2$ in equilibrium
- Consumer surplus:
 - Find point “A” where $Q=0 \rightarrow P=8$
 - Find point “B”, where $P=2 \rightarrow 2=8-2Q \rightarrow Q=3$.
- Consumers will purchase three units of the good if its price is \$2.
- Triangle Area:
 $(AP \cdot PB)/2 = (8-2) \cdot 3/2 = 9$



Consumer surplus and elasticity

Consumer Surplus and Price Elasticity of Demand

When demand is inelastic, there is a greater consumer surplus because there are some buyers willing to pay a high price to continue consuming the product



Efficiency

Efficiency: Two Types

- The traditional target that economist want to achieve is **efficiency**
 - Other (more controversial) goals: equality, fairness, etc.
- There are different types of efficiency considered:
 - **Pareto-efficiency**: "state of allocation of resources from which it is impossible to reallocate so as to make any one individual or preference criterion better off without making at least one individual or preference criterion worse off."
 - **Kaldor-Hicks efficiency**: "A re-allocation is a Kaldor–Hicks improvement if those that are made better off could hypothetically compensate those that are made worse off and lead to a Pareto-improving outcome. The compensation does not actually have to occur (there is no presumption in favor of status-quo) and thus, a Kaldor–Hicks improvement can in fact leave some people worse off." [https://en.wikipedia.org/wiki/Kaldor-Hicks_efficiency]
- Usually when we say "Efficiency" we mean Kaldor-Hicks

Efficiency: Two Types

- Homer likes bacon and values it at 10 EUR. Bart also likes bacon, but values it only at 5 EUR.
- You have a slice of bacon, and can distribute it between Homer and Bart:
 - **Option 1:** give the bacon to Homer.
 - **Option 2:** give the bacon to Bart
 - **Option 3:** slice the bacon in two, and give half to each.
 - **Option 4:** slice the bacon in two, give half to Homer and through the rest to the garbage.
- Which option is a pareto improvement compared to the baseline (no one has bacon)?
- Which option is pareto-efficient?



Efficiency: Two Types

	Homer	Bart	Total
Baseline	0	0	0
Option 1 (bacon to Homer)	10	0	10
Option 2 (bacon to Bart)	0	5	5
Option 3 (half bacon to each)	5	2.5	7.5
Option 4 (half bacon to homer)	5	0	5

- Baseline is not pareto efficient – switching to any option improves welfare of one w/o hurting the other.
- All options are pareto *improvements* compared to the baseline.
- Options (1-3) are pareto efficient, but option 4 is not (we can switch to option 3, which improves the welfare of Homer w/o hurting Bart)
- Only Option 1 is Kaldor-Hicks efficient!

KALDOR-HICKS VS. JUSTICE

Distribution 1



Distribution 2



KALDOR-HICKS VS. JUSTICE

What is the efficient outcome?



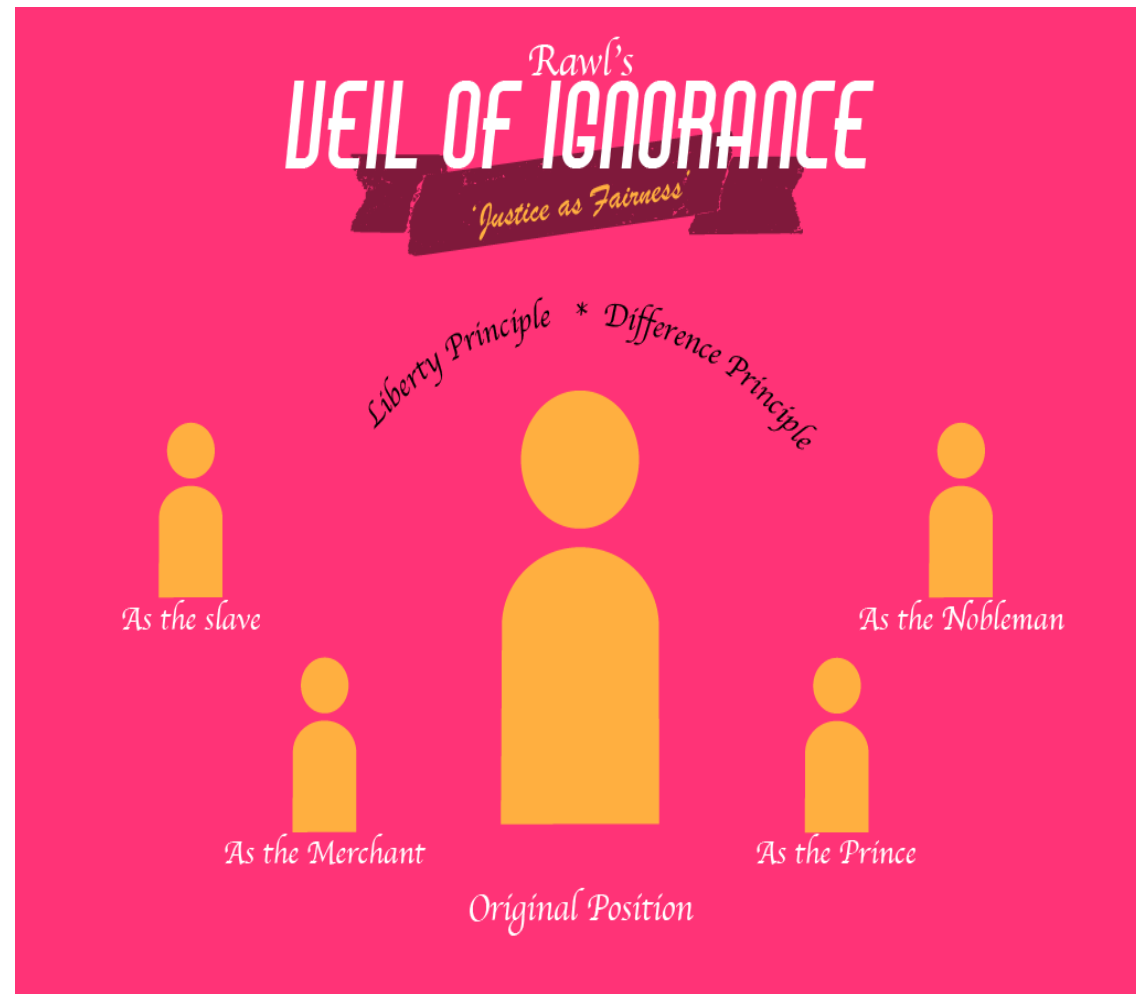
<https://youtu.be/H4BNbHBcnDI>

Efficiency: perspectives and approaches

- KH efficiency requires increasing the *size* of the “pie”, irrespective of distribution.
 - Recall: the “pie” is the social welfare = sum of utilities.
- This seems to contradict the approach of jurists, who are often interested in distributive justice (equality)
- How can we settle the two approaches?

Efficiency: perspectives and approaches

- KH is a **utilitarian** approach
- John **Rawls**:
 - “veil of ignorance”
 - SW increases only when the utility of the **least better off** person increases.
 - E.g. SW increases only when we give more to the poorest individual.



Efficiency: Settling law with economics

1. Economics provides tools to maximize **any kind of function**, not necessarily the sum of utilities.
 - E.g. Rawl's definition of social welfare
2. Utility functions can also be adjusted to relax the assumption of pure selfishness.
 - E.g. If I gain utility from helping the poor, then the usual maximization of my utility will indirectly have distribution effects.

How do we know what is really included in people's utility function?

General Equilibrium

General Equilibrium 1

- **Relationship between the labor market and the product market**
 - So far we looked mainly at the product market – how the supply and demand of a product determine the equilibrium price
 - But.. Since the wage of workers determines supply, there is an interaction between the product market and the market for workers
 - In the labor market, the supply side is the workers and the demand side is the employers.
 - If workers are willing to work for any wage, it means the supply is perfectly elastic
- **How is the wage of workers determined (in a nutshell)?**
 - The wage that each worker gets is the marginal output of the last worker

General Equilibrium 2

- **Example: suppose we have two types of plots: Plot A and Plot B.**
- Workers want to be paid a higher wage, but employers want to pay less
- Workers will sort themselves to where their marginal output is higher
- That marginal output would be their wage and determine the equilibrium wage
- E.g. with 1 worker: he would go to A and earn 1100. With 2 workers: second goes to B and earns 800, which would be the equilibrium wage.

Plot A

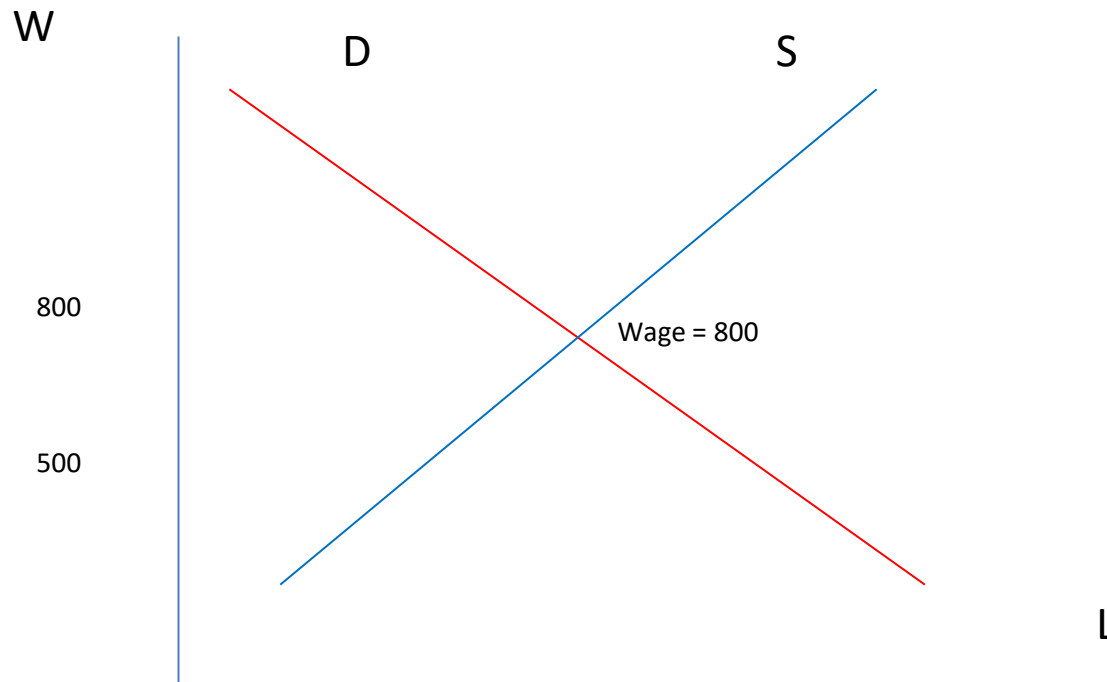
Number of workers	Total output	Marginal output
1	1100	1100
2	1750	650
3	2100	350
4	2300	200
5	2400	100

Plot B

Number of workers	Total output	Marginal output
1	800	800
2	1500	700
3	2100	600
4	2600	500
5	3000	400

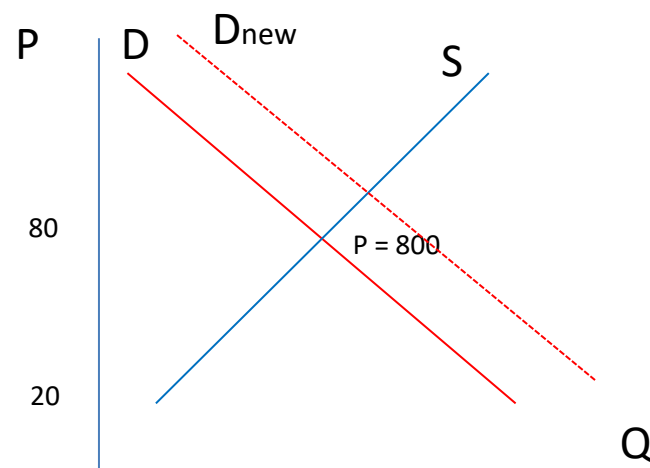
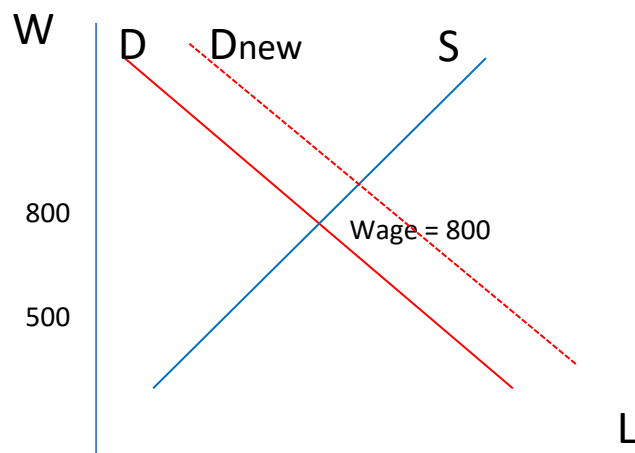
General Equilibrium 3

- Labor market then can be graphed exactly like the product market, where the Y-Axis is the wage and the X-Axis is labor



General Equilibrium 4

- **There is a relationship between the labor market and product market:**
- **Example:** positive shock to product demand $\rightarrow$ demand goes up $\rightarrow$ employers want to hire more workers to produce (close the excess demand) $\rightarrow$ wage goes up [$\rightarrow$ feedback: cost of production goes up $\rightarrow$ product supply goes down...]



General Equilibrium 5

- The intuition of feedback effects is more general than the example of product and labor markets
- In economics we differentiate between “partial” and “general” equilibrium
- Partial equilibrium = deals only with one market (e.g. the product market)
- General equilibrium = deals with the markets in the economy and their interaction. → implies feedback effect



Partial vs. general equilibrium - example

- Suppose a criminal considers committing a crime. What does he consider?
- **Benefit from crime:** e.g. how much money he gets from robbing a bank.
- **Cost of crime – the expected sanction:**
 - Probability of apprehension and conviction: how likely is he to get caught?
 - Sanction size: how much time will he have to serve in prison?
- He commits the crime if:

$$B > P * S$$

- In a partial equilibrium, when P or S increases, there is more crime deterrence
[This is roughly the famous Becker (1968) model]

Partial vs. general equilibrium - example cont'd

- But what if other markets respond? E.g. suppose that the legislator increases the penalty for rape (i.e. 'S' is now larger).
- Judges participate in another "market" which determines which punishment they give. If they are afraid to make a mistake, which implies sending someone innocent to a long time in prison, they might respond by convicting less.
- So there is feedback to the market for crime [Andreoni, 1991] and the P becomes lower.
- In some cases, the feedback effect may dominate – which is why a general equilibrium is important.

PRIMER ON GAME THEORY

ECONOMICS AND GAME THEORY

- Classical economics derives predictions using “game theory”
- Game theory is a framework for analyzing strategic interactions between “players”, where each player has a “payoff” and can choose between different “strategies”.
- Usual assumption is of “non-cooperative game theory”: people are selfish, so each person just focuses on his own payoff, which often hinders cooperation.

EXAMPLE: PRISONER'S DILEMMA

- Two robbers brought for interrogation at a police station. Each robber is put in a separate room. The officers can prosecute for theft, but don't have enough evidence to prove robbery unless someone confesses.
- Each suspect is given the following details:
 - If both confess, they are both sent to prison for 5 years
 - If no one confesses, they both get 1 year in prison for theft
 - If one confesses and the other doesn't, the "snitch" goes free and the other gets 10 prison years.
- If you are one of the robbers – what do you do?



PRISONER'S DILEMMA: PAYOFFS

Player 2

		Player 2	
		Confess	Don't confess
Player 1	Confess	$(-5, -5)$	$(0, -10)$
	Don't confess	$(-10, 0)$	$(-1, -1)$

- In each cell, the left number is the payoff of player 1 and the right number is the payoff of player 2
- Which outcome is jointly the best for the two players? **KH efficient!*
- If each player maximizes only his own payoff, what would be the (equilibrium) outcome?

PRISONER'S DILEMMA: DOMINANT STRATEGY

Player 2

		Player 2	
		Confess	Don't confess
Player 1	Confess	$(-5, -5)$	$(0, -10)$
	Don't confess	$(-10, 0)$	$(-1, -1)$

- To solve the game we have to make the following thought exercise:
 “suppose my opponent chooses action X, what should I choose then? Now, suppose he chooses Y instead, what should I choose then? If the answer is the same – we have a **“dominant strategy”**”

WEAKLY DOMINANT VS. STRICTLY DOMINANT

Player 2

		Player 2	
		Confess	Don't confess
Player 1	Confess	$(-5, -5)$	$(0, -10)$
	Don't confess	$(-10, 0)$	$(-1, -1)$ $(0, 0)$

- Suppose that the game was slightly different, such that if both criminals don't confess, the payoff is 0 rather than -1.
- In this case, choosing “confess” yields either a better result ($-5 > -10$), if the other player doesn't confess, or the same result ($0 = 0$), if the other player also confesses. We then say that confessing is a “**weakly**” dominant strategy.
- → Tip: remember from math that if $x \geq y$ we sometimes say “x is weakly greater than y” – it's the same thing.

WEAKLY DOMINANT VS. STRICTLY DOMINANT

Does player 1 have a strictly dominant strategy, weakly dominant strategy, or neither?
(player 1 chooses rows)

	A	B
A	1,1	0,0
B	0,0	0,0

	A	B
A	9,1	0,0
B	8,0	-2,-1

	A	B
A	1,2	1,1
B	0,0	2,0

	A	B
A	1,1	1,1
B	1,1	1,1

WEAKLY DOMINANT VS. STRICTLY DOMINANT

Does player 1 have a strictly dominant strategy, weakly dominant strategy, or neither?
(player 1 chooses rows)

weakly

	A	B
A	1,1	0,0
B	0,0	0,0

	A	B
A	9,1	0,0
B	8,0	-2,-1

strictly

neither

	A	B
A	1,2	1,1
B	0,0	2,0

	A	B
A	1,1	1,1
B	1,1	1,1

neither

WEAKLY DOMINANT VS. STRICTLY DOMINANT

What about player 2 (who chooses columns)?

	A	B
A	1,1	0,0
B	0,0	0,0

	A	B
A	9,1	0,0
B	8,0	-2,-1

	A	B
A	1,2	1,1
B	0,0	2,0

	A	B
A	1,1	1,1
B	1,1	1,1

WEAKLY DOMINANT VS. STRICTLY DOMINANT

What about player 2 (who chooses columns)?

weakly

	A	B
A	1,1	0,0
B	0,0	0,0

	A	B
A	9,1	0,0
B	8,0	-2,-1

strictly

weakly

	A	B
A	1,2	1,1
B	0,0	2,0

	A	B
A	1,1	1,1
B	1,1	1,1

neither

PRISONER'S DILEMMA: EFFICIENCY

	Confess	Don't confess
Confess	$(-5, -5)$	$(0, -10)$
Don't confess	$(-10, 0)$	$(-1, -1)$

- The equilibrium outcome (confess, confess) is inefficient.
- There is a better outcome $(-1, -1)$, where payoffs increase without hurting anyone → Pareto improvement.
- But each player plays his dominant strategy, so we don't get there.

NASH EQUILIBRIUM

The concept that determines the equilibrium outcome is the “Nash Equilibrium”, named after Nobel Prize Winner John Nash

This concepts is more general than dominant strategies (dominant strategies are a special case of a NE)



John Forbes Nash (1928 - 2015)
Nobel laureate in economics sciences 1994

THE KEY EQUILIBRIUM CONCEPT IS “NASH EQUILIBRIUM”: ALL PLAYERS SELECT BEST RESPONSES.

Nash Equilibrium (NE)

In a Nash equilibrium, each player selects a strategy which is the best response to what the other players have chosen; i.e.: each player acts optimally, given the other players' choices.

		Advertising Game	
		Player 2	
		Don't Advertise	Advertise
Player 1	Don't Advertise	(10,10)	(5,13)
	Advertise	(13,5)	(8,8)

Any equilibrium in dominant strategies is also a Nash Equilibrium, but there are also equilibria without dominant strategies

TO IDENTIFY A NASH EQUILIBRIUM, WE HAVE TO ANALYZE EACH OUTCOME INDIVIDUALLY.

How do we find a Nash equilibrium?

- In principle, we have to check for each outcome (i.e., for each combination of strategic choices) whether or not each player acts optimally; if a player has a profitable deviation, the outcome is not a Nash equilibrium.
- In other words – we ask for each player “can this player profit by unilaterally deviating”? If the answer is no, it’s a Nash Equilibrium.

Games with discrete payoffs

		Player 2	
		A	B
Player 1	a	(1,10)	(1,13)
	b	(3,4)	(2,3)

We will not cover games with continuous payoffs here

(generally the way to solve these is finding an optimal action using an explicit profit function and taking the derivative to find the first order conditions)

NASH EQUILIBRIUM: AN EXAMPLE

- Suppose that Player A and Player B each have three possible moves.
- Is there a dominant strategy?
- What is the Nash Equilibrium?

	B1	B2	B3
A1	10,10	0,6	2,2
A2	15,0	5,5	4,4
A3	3,5	7,8	6,6

NASH EQUILIBRIUM: AN EXAMPLE

- → Start from somewhere and keep checking if there is a profitable deviation, keeping the choice of the other player fixed.
- Example: Start from (A1,B1). Player A will deviate to A2. Player B will deviate to B2. Player A will then deviate to A3. (A3, B2) is an equilibrium → No one has an incentive to deviate further.

	B1	B2	B3
A1	10,10	0,6	2,2
A2	15,0	5,5	4,4
A3	3,5	7,8	6,6

WE NOW WANT TO ALLOW FOR THE CASE WHERE THE TIMING OF STRATEGIC CHOICES MATTERS.

Example: Market Entry Game

Situation

- There are two firms which can offer very high speed internet access, a telecommunications incumbent (A) and a TV cable company (B).
- Setting up the infrastructure requires huge investments of 10 each.
- The telecommunications incumbent has a first mover advantage: it can enter the market, “sink” the investment and will then be able to supply the market.
- If B does not follow, A will get (gross of investment) profits of 12, leaving a total profit of 2.
- If B follows, each firms’ (gross) profits are 9 (and net profits are -1).
- If A does not enter but B enters, B gets (gross) profits of 12 if it enters.

SIMULTANEOUS VERSION OF THE GAME.

The Market Entry Game

- Suppose that the game is played simultaneously.
- The game can then be represented by the familiar strategic form.
- The game has two Nash equilibria.
- Are these two Nash equilibria also equilibria in the sequential game?
- If one firm enters, it is optimal for the other firm to stay out. But given that the other firm stays out, it is optimal for the first firm to enter.
- Therefore, the answer is yes!
- But is the Nash equilibrium in which firm B enters and firm A stays out a plausible one?

Strategic Form Representation

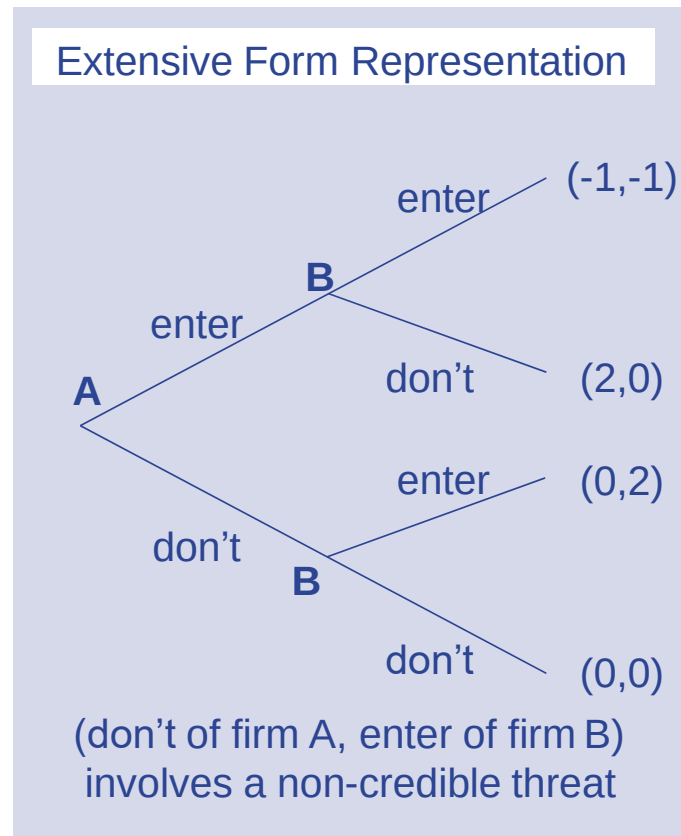
		B	
		enter	don't
A	enter	$(-1, -1)$	$(2, 0)$
	don't	$(0, 2)$	$(0, 0)$

This game has two NE

FOR DYNAMIC GAMES, AN "EXTENSIVE FORM" REPRESENTATION IS HELPFUL

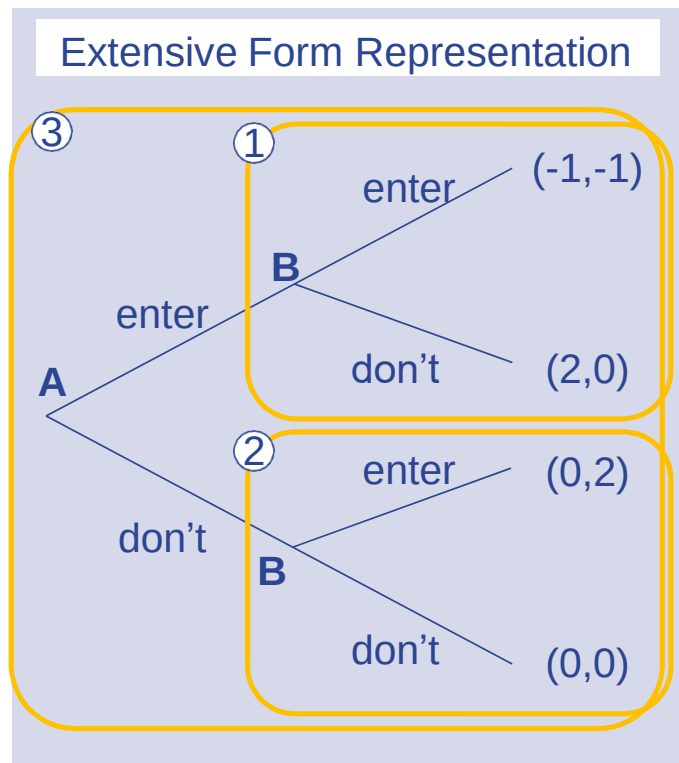
Extensive Form

- Problem with the equilibrium (don't, enter): It depends on the threat of B to enter if firm A has chosen to enter in the first stage.
- This threat is non-credible since firm B prefers to stay out given that firm A has entered.
- However, on the equilibrium path, i.e., when firm A stays out, firm B does not need to carry out its threat.
- Hence, (don't, enter) is a Nash equilibrium.
- Can we refine the Nash equilibrium concept to rule out such implausible equilibria?



THE WAY TO SOLVE SEQUENTIAL GAMES IS TO IDENTIFY SUBGAME PERFECT EQUILIBRIA.

Subgame Perfect Equilibrium



1) In finite games of complete information.

- Each decision node constitutes a “subgame”.
- Our market entry game has 3 subgames.

Subgame Perfect Equilibrium

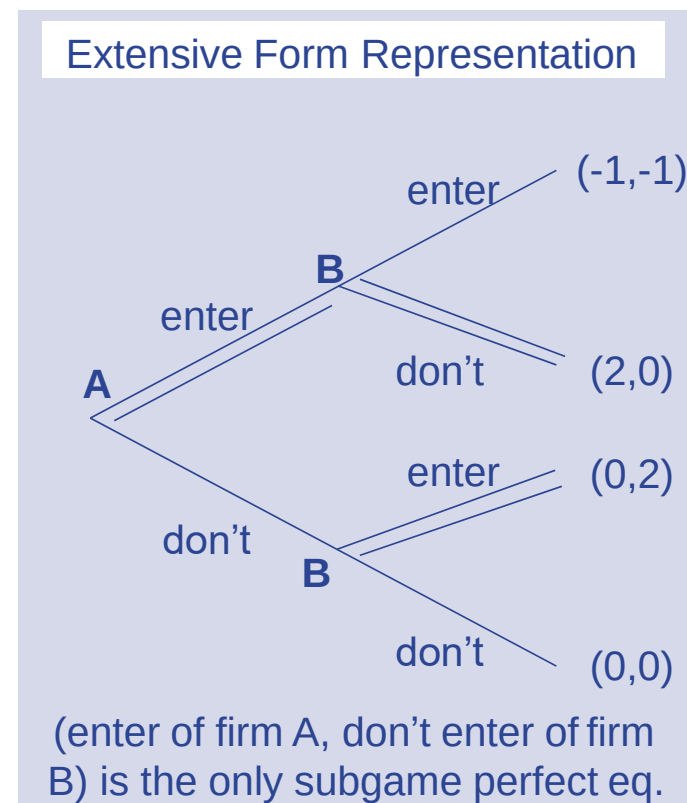
In a subgame perfect equilibrium (SPE), a strategy induces a NE in every subgame.

- Often¹⁾, we can find the SPE by applying “backward induction”.
- Let us now apply backward induction.

BACKWARD INDUCTION PROCEDURE

Extensive Form

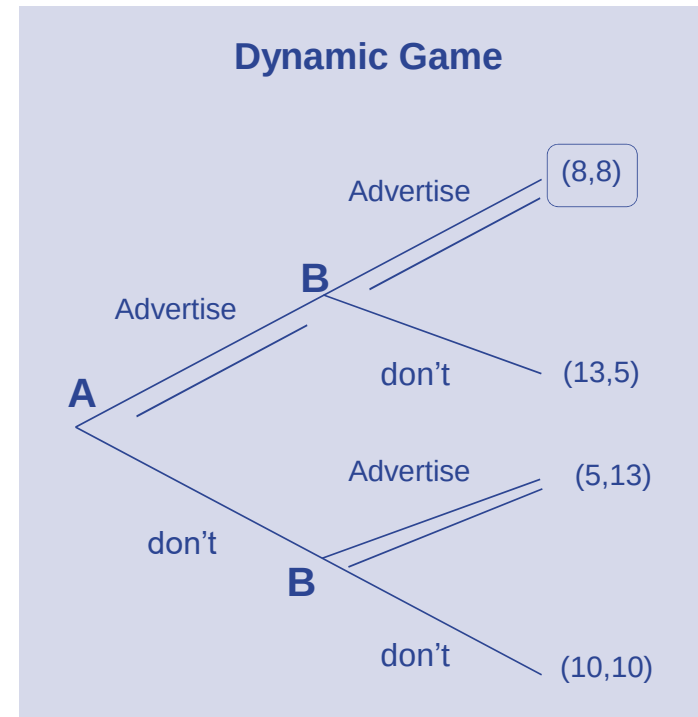
- In subgame 1, (don't) is the optimal choice of firm B.
- In subgame 2, (enter) is the optimal choice of firm B.
- Since firm A knows this, it knows that the resulting payoff from (enter) is 2 and the resulting payoff from (don't) is 0.
- Therefore, in subgame 3, (enter) is the optimal choice of firm A.
- The Nash equilibrium (don't of A, enter of B) is not subgame perfect because it relies on the non-credible threat that firm B enters in the second stage.



DOES IT MATTER WHETHER A GAME IS DYNAMIC?

Static game

		Player 2	
		Don't Advertise	Advertise
Player 1	Don't Advertise	(10,10)	(5,13)
	Advertise	(13,5)	(8,8)



WHEN DOES IT MATTER IF THE GAME IS DYNAMIC?

- Unlike the Market entry game, where the SPE concept allows to rule out an (implausible) Nash Equilibrium, in the prisoner's dilemma it makes no difference at all whether the game is dynamic.
- The reason is dominant strategies – if I do the same thing irrespective of what the other does, it doesn't matter if he goes first or I go first.

Part 1: Principles of Economics

1. Basics

2. Necessary
Conditions for
Functioning Markets

3. Market Failure &
Market intervention

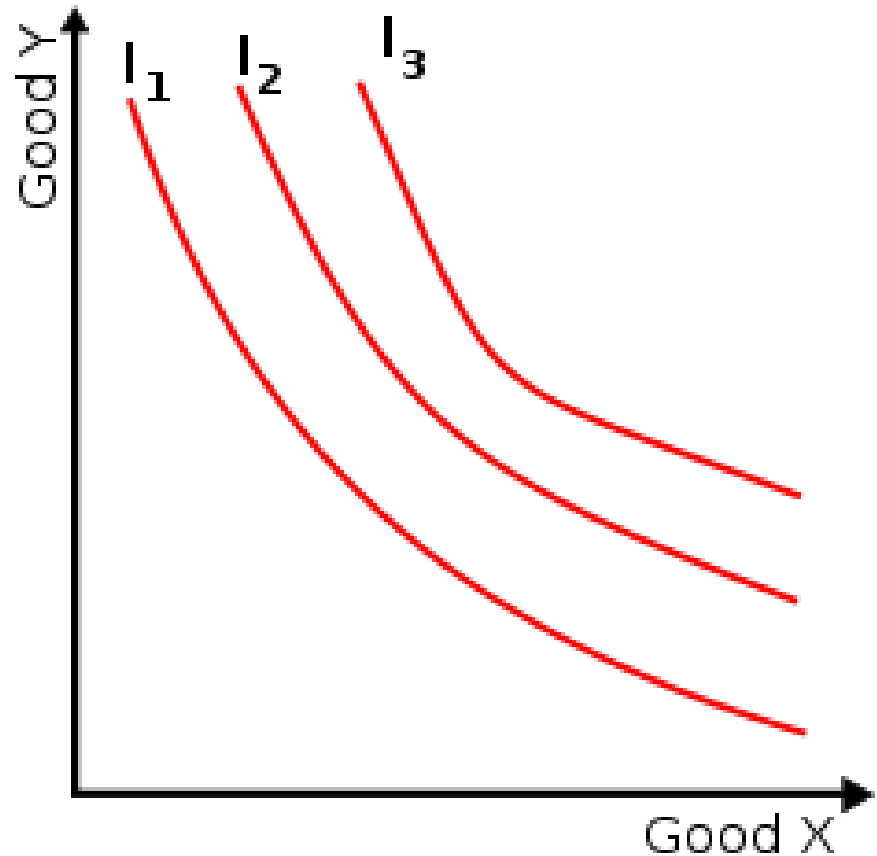
4. International Trade

5. Political Economy



Indifference curves

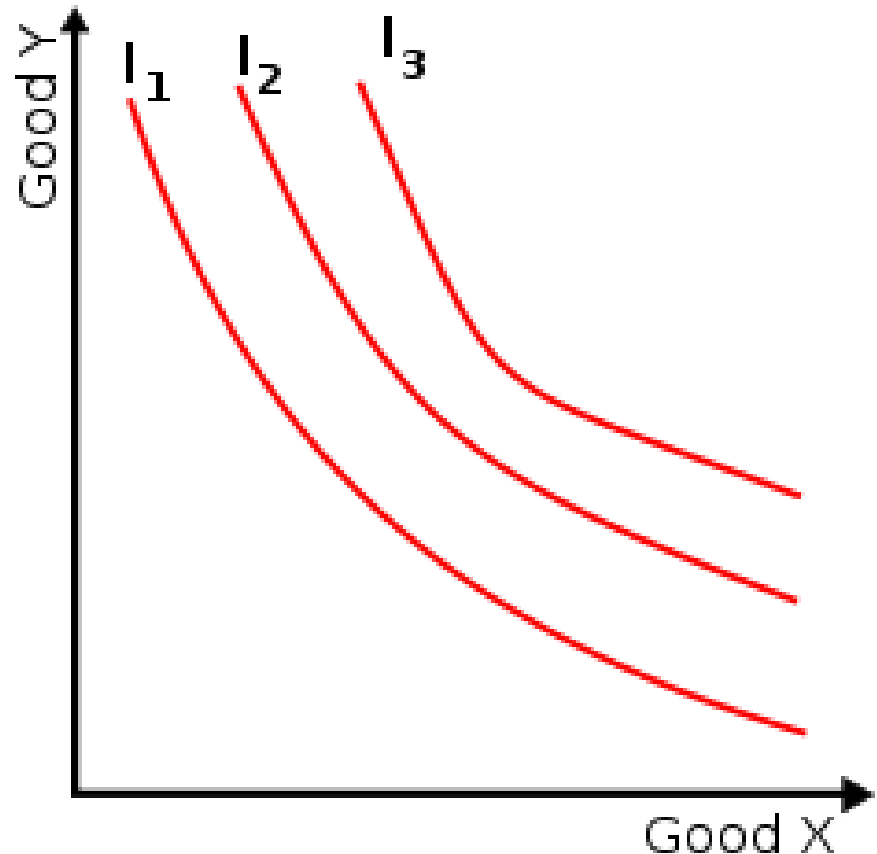
- Suppose a consumer can hold a bundle of two goods: “X” and “Y”
- We want to describe his preferences for different combinations
 - E.g. Is 10x and 20y better than 15x and 15y?
- We can do so with an “indifference curve”
 - Note: we’ve seen something similar with the production frontier



Source: Wikipedia

Indifference curves

- The indifference curve is a line going through all combinations that the consumer is *indifferent* between
 - =That give the same utility.
- There are *infinitely many* indifference curves, each capturing different combinations
 - E.g. in the figure, I_1 starts from, say, $10Y + 1X$ and ends in $1Y + 10X$; whereas, I_2 starts from, say, $10Y + 5x$ and ends in $3Y + 10X$.
- Why are indifference curves convex?



Source: Wikipedia

Indifference curves

- Convex indifference curves express the same idea as the law of demand: consumers are willing to “pay” (=give up another product) less and less for one additional product.
- Formally, this is known as the “Marginal rate of substitution”:

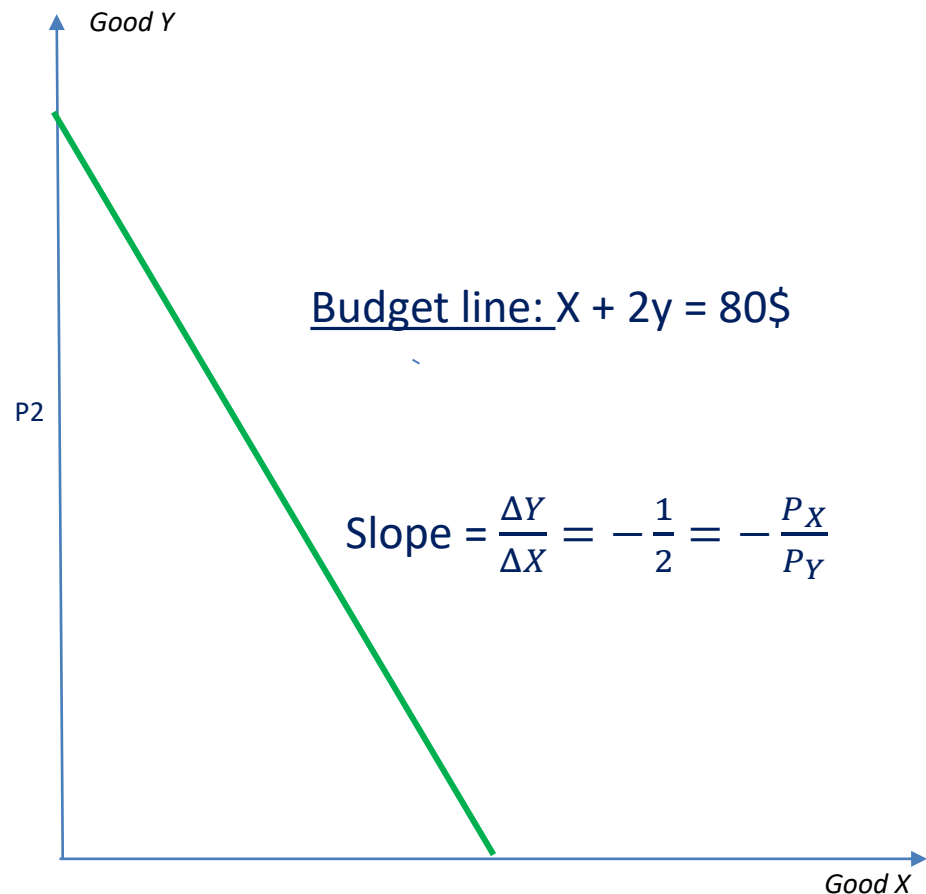
$$MRS = -\frac{\Delta Y}{\Delta X}$$

- → We will get a better mathematical intuition for this later on.

Source: Wikipedia

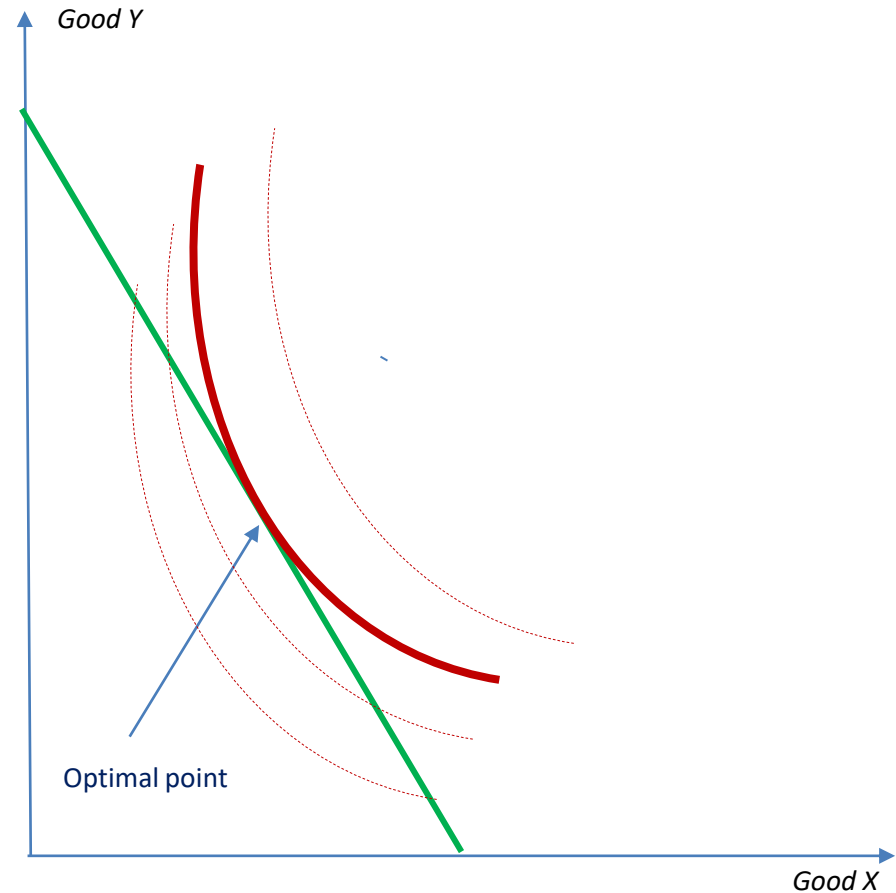
Budget Lines

- “A budget line describes the combinations of goods that can be purchased given the consumer’s income and the prices of the goods” (Pindyk & Rubinfeld, p. 105)
- The slope is equal to the ratio of prices
 - This expresses the consumer’s ability to trade-off the bundle of produces given the market prices
- In the example: the price of X is 1\$, the price of Y is 2\$, and the consumer’s budget is 80\$.



Budget Lines

- The consumer chooses a point on the indifference curve, subject to the budget constraint
- The optimal point must lie on the highest indifference curve that just touches on the budget (is tangent to it)
- At the optimal point, the slope of the indifference curve is exactly equal to the slope of the budget line, which is also the MRS
- Consumer chooses bundle so that $MRS = -\frac{P_X}{P_Y}$



Suppose people can trade in a pure exchange economy.

Pure Exchange Economy

- We look at the simplest case with two goods:
 - Two different consumers, no production, exogenously given quantities
 - Robinson is stranded on an island
 - He lives on the west coast of the island; due to the humidity on the west coast, there are plenty of bananas growing, but little coconuts
 - He can harvest 2 units of coconuts and 8 units of bananas; that is all he has
 - One day (actually, a Friday), he realizes that another person lives on the island – Friday
 - Friday lives on the east coast, where he can better grow coconuts
 - Thus, Friday has 8 units of coconuts and 2 units of bananas
 - Both islanders would prefer a bit more balanced diet – how can they improve the allocation of the goods available on the island, i.e.: what is the optimal allocation of the 10 units of bananas and the 10 units of coconuts?
-

How to analyze the example of Robinson & Friday?

- We use the **Edgeworth Box** to illustrate the total amounts of both bananas and coconuts and all the feasible ways to allocate them between Robinson and Friday
- We assume that the southwest corner represents the origin for Robinson while Friday's origin is the northeast corner of the **Edgeworth Box**
- The **Indifference curves** of Robinson and Friday indicate all the different bundles of bananas and coconuts that yield the same level of utility for each of them



How to (efficiently) allocate the fixed quantities of bananas and coconuts between Robinson and Friday?

Recall: The meaning of an indifference curve

Intuition behind an Indifference Curve

- Interpretation of an indifference curve:
All points on an indifference curve give the same utility to a consumer
- An indifference curve which is further outwards provides a higher utility to a consumer
- Slope of the indifference curve:
How many bananas is Robison willing to give up to obtain one more coconut?

An indifference curve: Formal derivation

- Formally: $U^R(x_B^R, x_C^R) = \text{constant}$

- $$\frac{\partial U^R(x_B^R, x_C^R)}{\partial x_B^R} dx_B^R + \frac{\partial U^R(x_B^R, x_C^R)}{\partial x_C^R} dx_C^R = 0$$

- $$\frac{dx_B^R}{dx_C^R} = - \frac{\frac{\partial U^R(x_B^R, x_C^R)}{\partial x_C^R}}{\frac{\partial U^R(x_B^R, x_C^R)}{\partial x_B^R}}$$

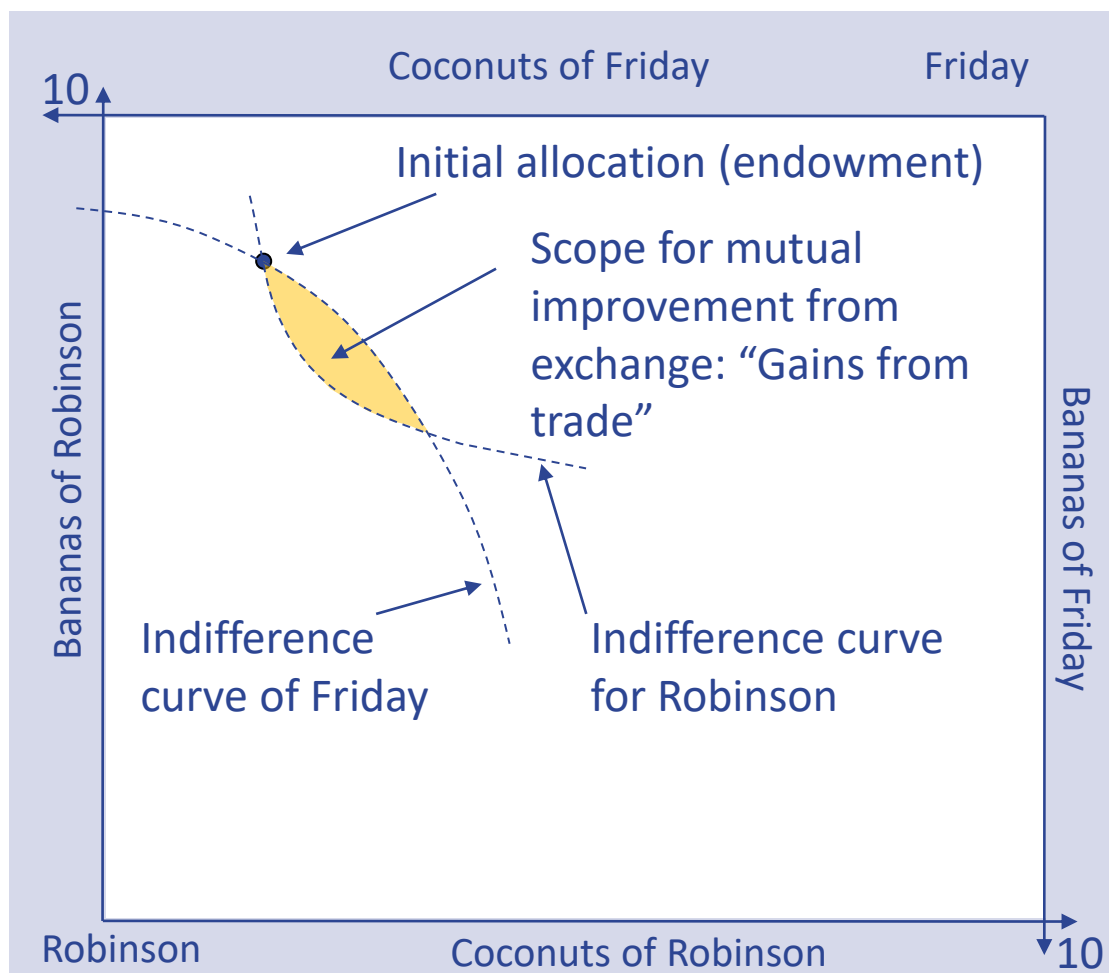
- The right-hand side of the last expression is the **Marginal Rate of Substitution**.
- It is the formal representation of how many units of bananas Robison is willing to give up to get one more unit of coconuts.

An “Edgeworth Box” depicts all possible allocations in the exchange economy.

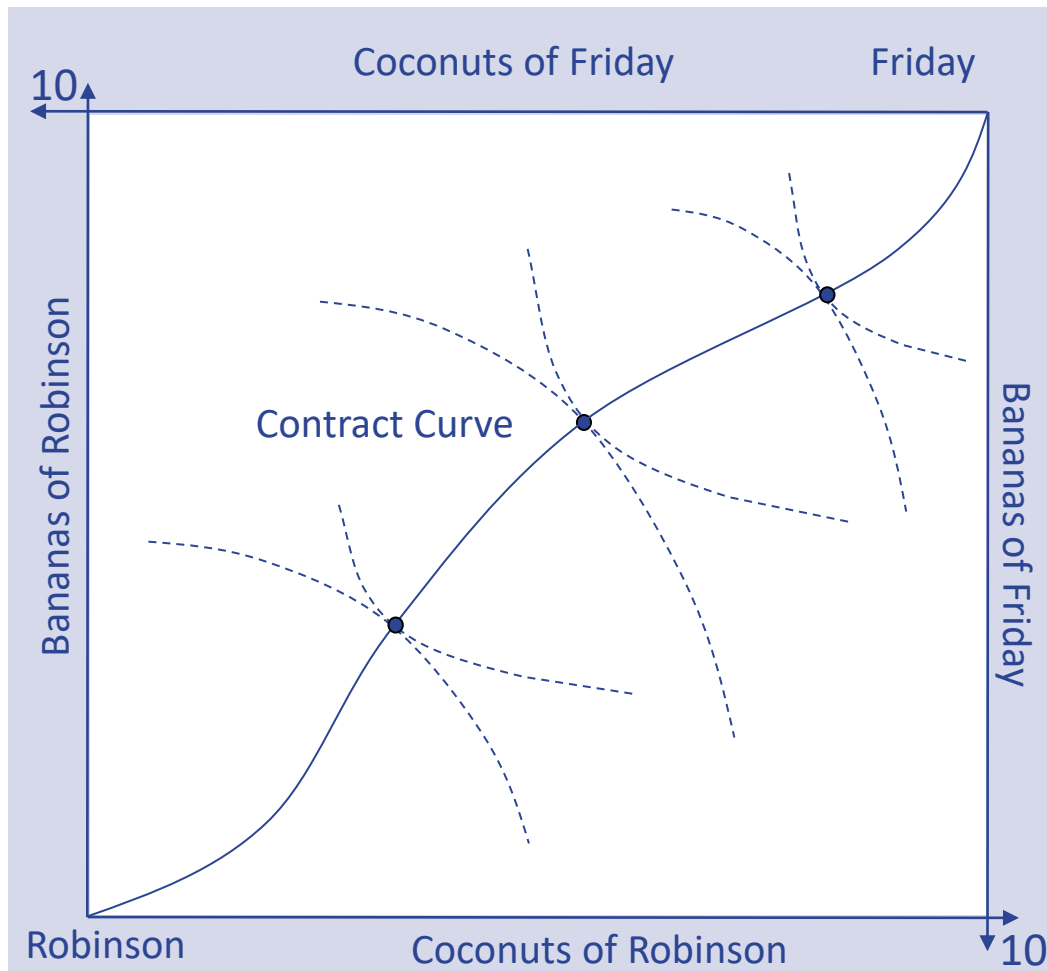
Pareto Improvement

- Our island economy is endowed with 10 units of each good
- By “chance”, we have some initial allocation (2,8) and (8,2)
- Robinson’s and Friday’s indifference curves are depicted with reference to their origins
 - Robinson’s utility increases in the north-east direction (shifting his indifference curve toward Friday’s origin) and Friday’s utility increases in the south-west direction (shifting his indifference curve toward Robinson’s origin)
 - Starting from the initial allocation, Robinson and Friday can realize gains from trade by exchanging goods and realize an allocation indicated by the yellow area in the graphic on the next slide
- This would be a “**Pareto**” **improvement**: both are better off!

An “Edgeworth Box” depicts all possible allocations in the exchange economy.



Pareto efficiency means that all mutual gains from trade have been realized.



Pareto Efficiency

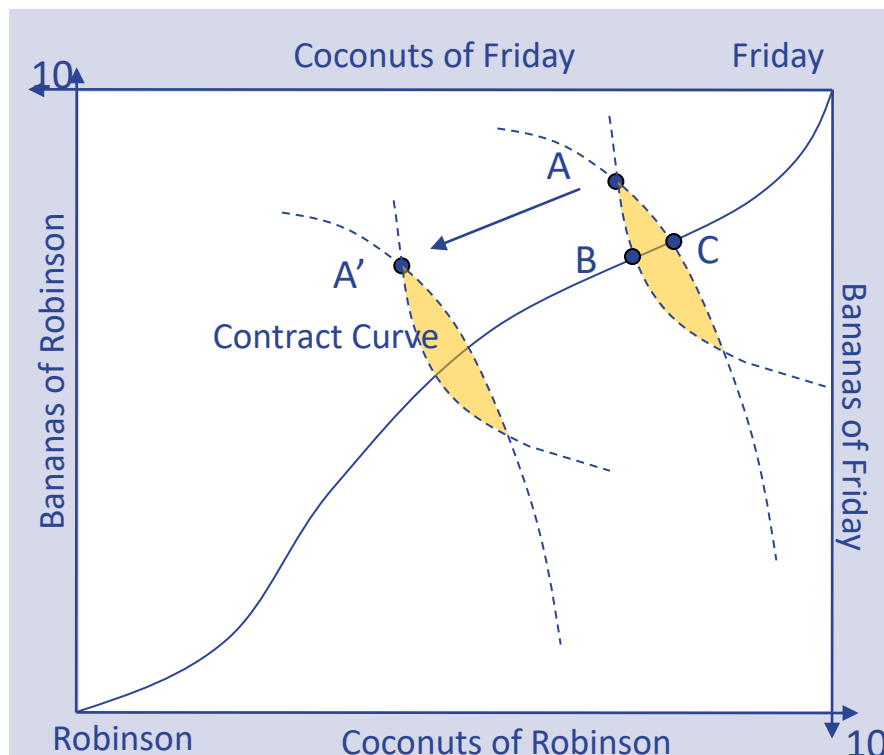
- Where indifference curves are tangent there is no further scope for mutually beneficial trade
- These points are called “contract curve”
- Points on the contract curve are Pareto efficient

Pareto Efficiency

An allocation is Pareto efficient, if no one can be made better off without making someone else worse off.

Pareto efficiency is clearly desirable; but it says nothing about the fairness of allocations

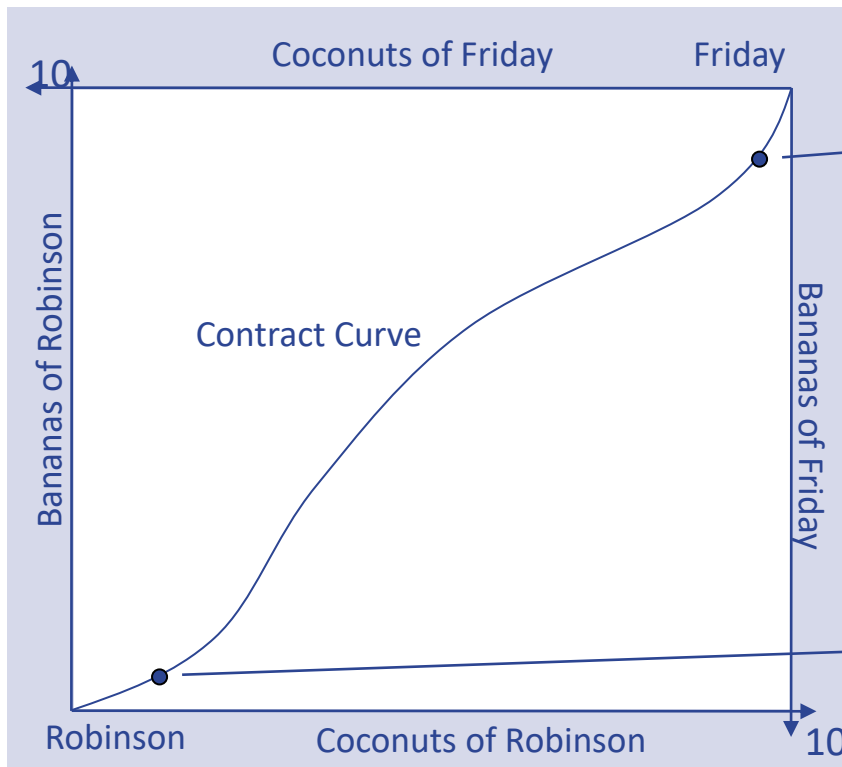
Distribution and Pareto Efficiency



- Realizing A is clearly inefficient since we know we can make at least one subject better off without harming the other.
- However, subjects clearly have diverging interests, which Pareto efficient allocation (B or C) shall be realized.
- The initial allocation determines which allocations are acceptable
- To achieve an efficient and more equitable allocation, one could redistribute initial endowments (A')

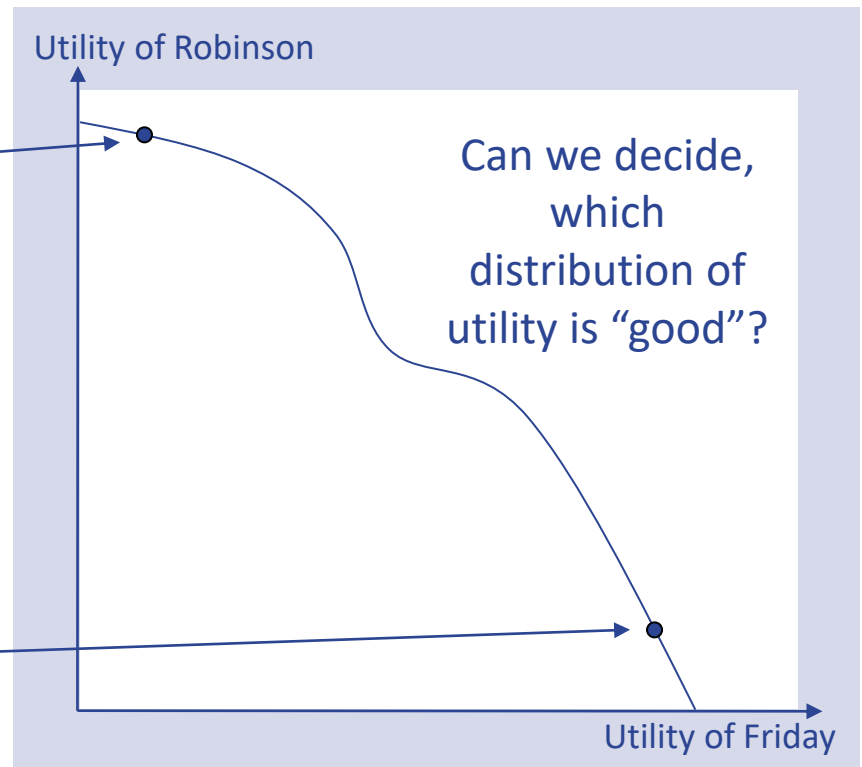
From the contract curve, we can construct the “utility frontier”.

Contract Curve



All Pareto-efficient allocations

Utility Frontier



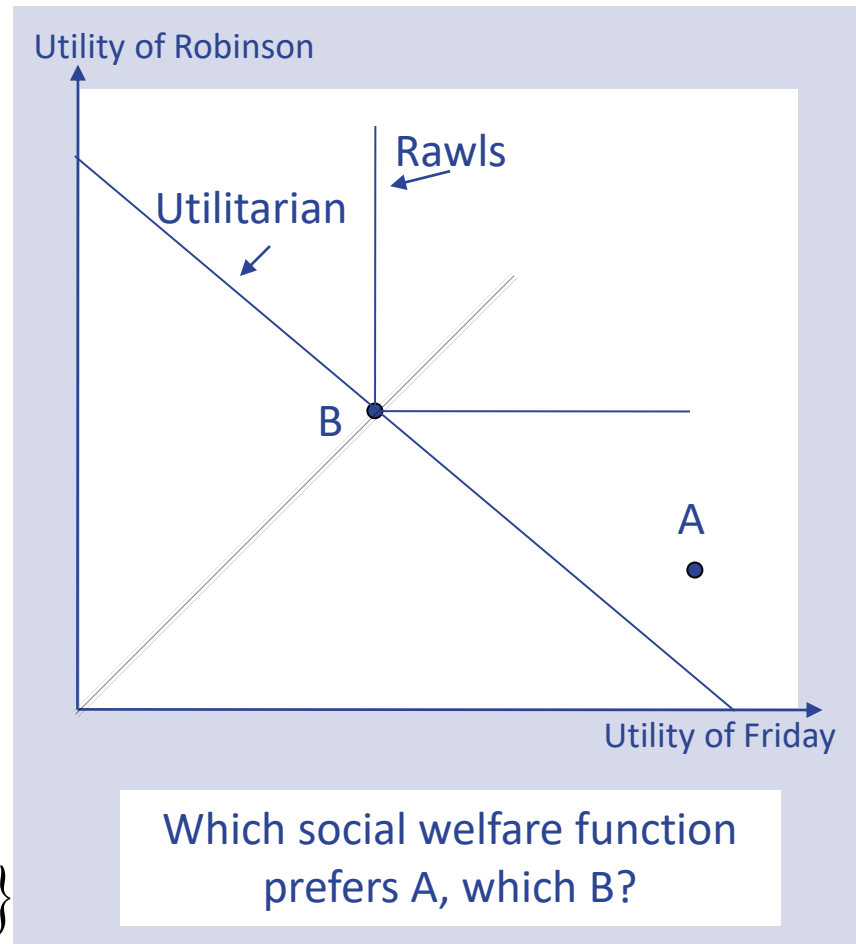
Distribution of utility of Pareto-efficient allocations

Welfare functions are tools to discuss different distributions of utility.

Welfare Functions

- Economics can say little about which distribution is “good”, apart from: avoid inefficiencies! (i.e., realize Pareto efficient outcomes)
- Utility as an ordinal concept can not be compared between individuals
- Economics can not provide a normative judgment which of two efficient allocations is better
- But: Social welfare functions can provide a tool to discuss different views on distribution of individual welfare; two examples:
 - Utilitarian: $W = u_{Robinson} + u_{Friday}$
 - Rawls: $W = \min\{u_{Robinson}, u_{Friday}\}$

Iso-Welfare Lines



A competitive equilibrium is characterized by market clearing and price taking behavior.

Definition of a competitive equilibrium

A competitive equilibrium is characterized by a set of prices such that markets clear (i.e., there is neither excess demand nor excess supply of any good) and that, given these prices, each subject maximizes its objective function (e.g., consumers maximize their utility, given their budget constraint).



Two important properties

- Market clearing
- Price taking behavior
 - Interpretation: No subject believes that she can influence the level of prices, i.e., we assume absence of market power
 - Example where this is violated: a monopolist sets the price

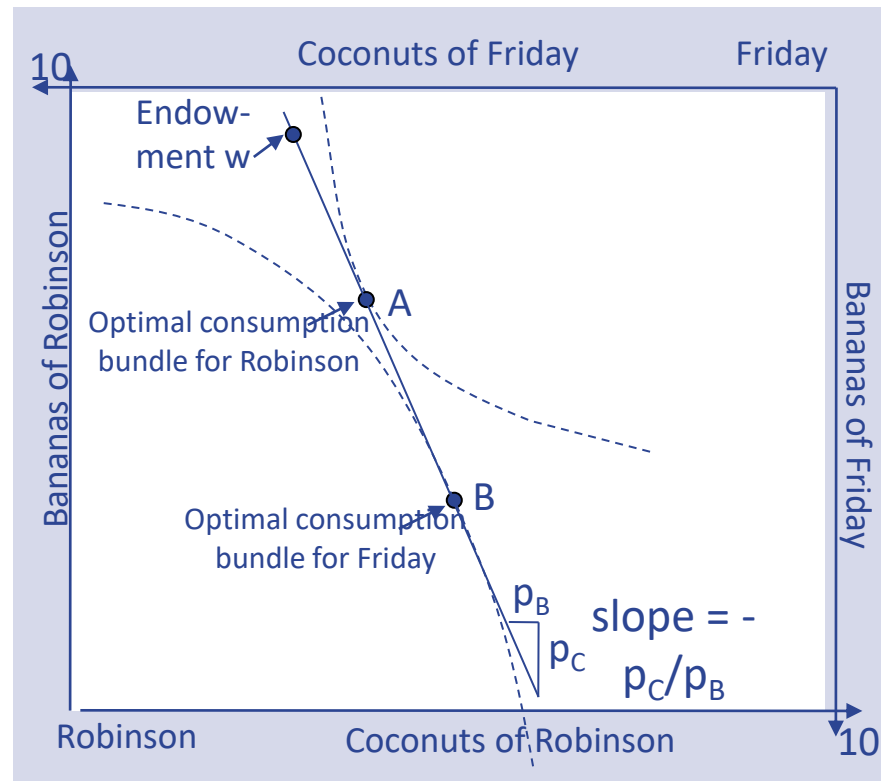
A competitive equilibrium is characterized by market clearing and price taking behavior.

- Assume now that there is an existing price for coconuts p_C and bananas p_B
- Given these prices, Robinson and Friday must choose the bundle of coconuts and bananas that is best for them in the set of feasible allocations (depicted by A and B in the following graphic)
- The prices p_C and p_B determine Robinson's and Friday's budget line
- Given price ratio p_C/p_B , all bundles of coconuts and bananas on the budget line are exactly as expensive as the initial endowment bundle (depicted by w in the following graphic)

Can Robinson and Friday reach their optimal consumption bundles by trading coconuts and bananas with each other?

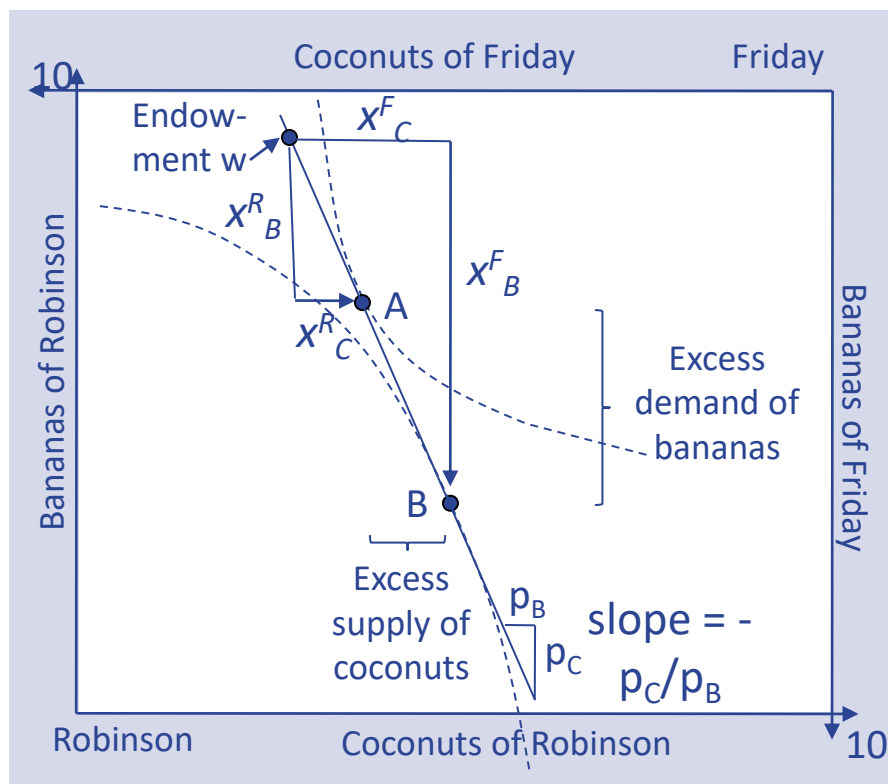
A competitive equilibrium is characterized by market clearing and price taking behavior.

Excess Demand / Excess Supply



Consumers optimize consumption bundles subject to given prices – thus, there can be excess demand and excess supply of goods.

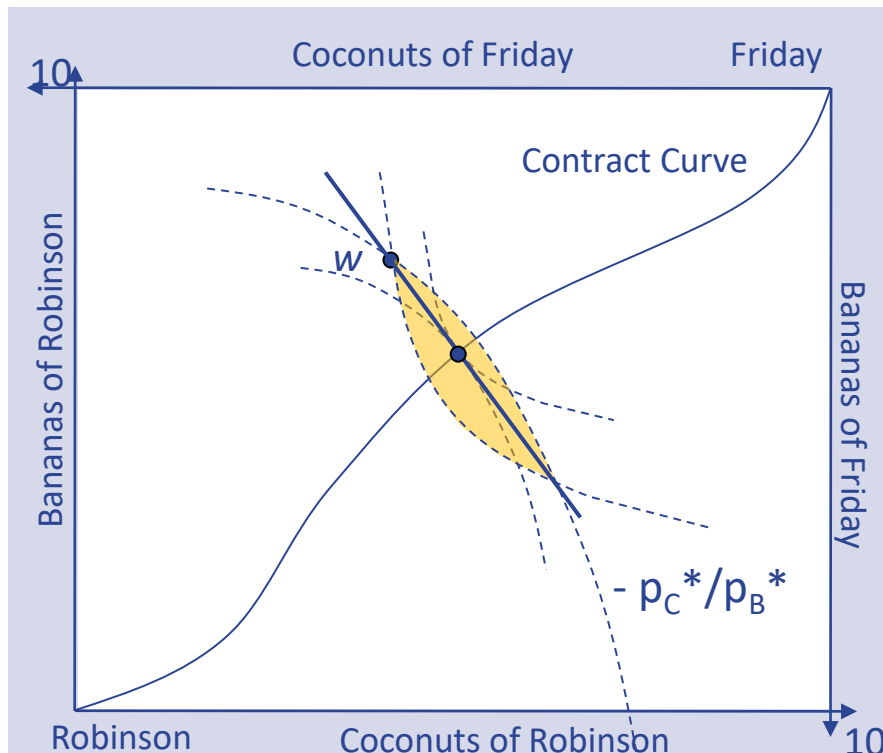
Excess Demand / Excess Supply



- At given prices p_B and p_C and endowment w , Robinson would like to sell x_B^R bananas to get x_C^R coconuts, and Friday would like to sell x_C^F coconuts and buy x_B^F bananas
- **Problem.** At the given prices, the optimal consumption bundles lead to an imbalance between demand and supply
- Excess supply for coconuts $= x_C^R + (-x_C^F)$
- Excess demand for bananas $= x_B^F + (-x_B^R)$

With the “correct” prices we can implement an equilibrium which is also Pareto-efficient.

Competitive equilibrium prices



- Solution to the problem: revise prices to balance demand and supply for coconuts and bananas
- Assume prices p_C^* and p_B^*
- At these prices, the subjects trade such that excess demand and excess supply are zero
- This is where the indifference curves are tangent
- Thus: the competitive equilibrium is Pareto-efficient!

The first theorem of welfare economics formalizes the idea of an “invisible hand”.

First Fundamental Theorem of Welfare Economics

A competitive equilibrium is (Pareto) efficient.

Why?

In our exchange economy we have that:

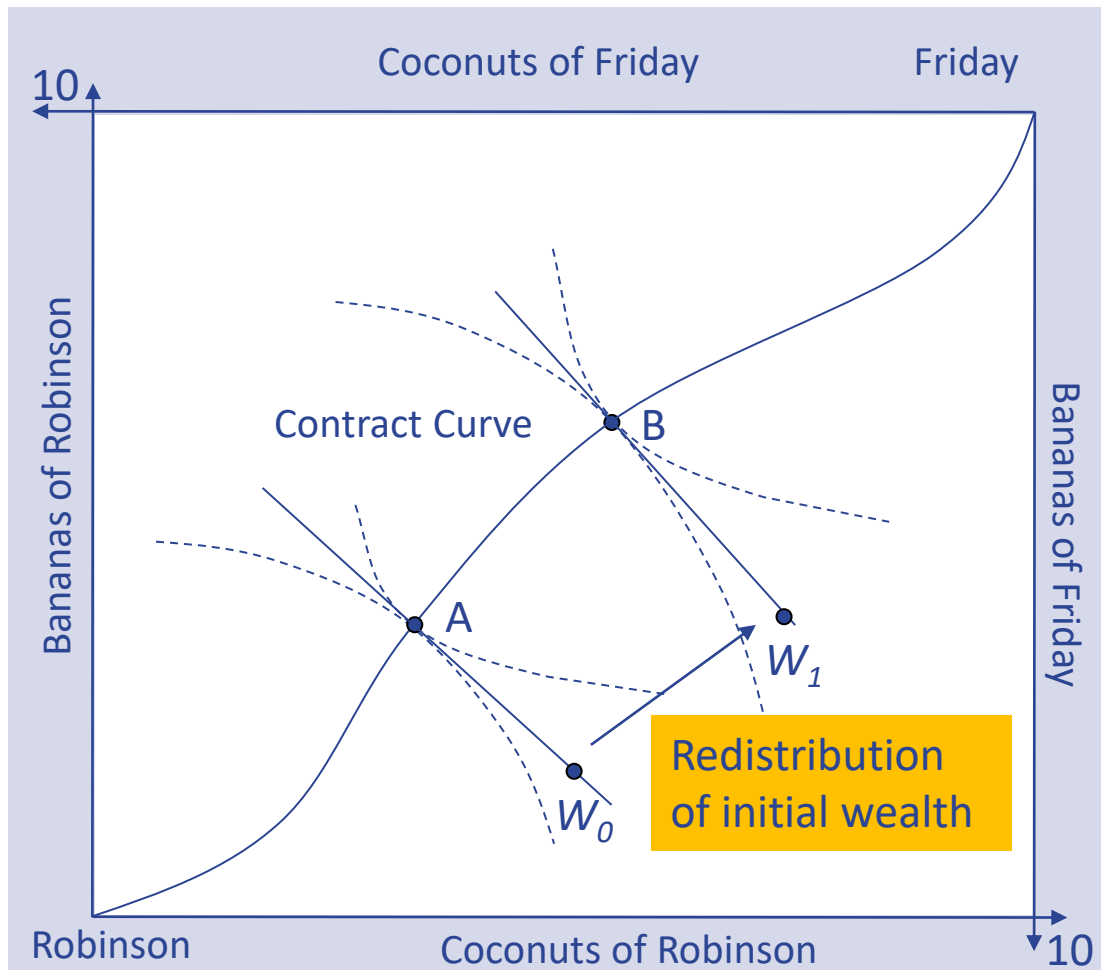
- In equilibrium, for each consumer the marginal rate of substitution equals the relative prices
- Since prices are the same for all consumers, this implies that the marginal rates of substitution of all consumers are identical
- If the marginal rates of substitution are identical, no consumer can be made better off without making some other worse off – the definition of Pareto efficiency.

The first welfare theorem is fundamental for our understanding of market economies.

Interpretation of 1. Welfare Theorem

- The first welfare theorem is important, since it provides us with a **“mechanism” that ensures Pareto-efficiency: competitive markets.**
 - The mechanism is simple in the sense that it requires **very little information**: each individual needs to know only the own preferences and the market prices – that’s it.
 - In our toy economy with only two individuals this might not seem to be a big deal (two individuals could easily get together and exploit all gains from trade), but the first welfare theorem **applies also if there are millions of individuals.**
 - This is exactly the formalization of **Adam Smith’s** idea of the market acting as an **“invisible hand”**: each individual acts only in its own interest but when interacting in markets, a “socially” desirable (i.e., Pareto-efficient) outcome occurs.
 - The first welfare theorem applies for almost all kinds of preferences and for all kinds of technologies, i.e., it requires only **weak assumptions on preferences and none on technologies.**
 - **Key assumptions** are:
 1. Markets are competitive, i.e., **absence of market power.**
 2. **No externalities.**
 3. **Symmetric information** (“complete markets”).
-

Each allocation can be efficiently implemented using a market mechanism.



2. Welfare Theorem

- If we want to implement a certain distribution of utility levels, we can do so by redistributing initial wealth and then let market forces work
- If we prefer B to A, we can just reallocate from W_0 to W_1
- The outcome will be efficient due to the first welfare theorem
- We need not give up efficiency in order to achieve a certain distributional aim.

The second welfare theorem states that we can separate distribution and allocative efficiency.

Interpretation of 2. Welfare Theorem

- Strictly speaking, the second welfare theorem says that we can implement every desired outcome as a competitive equilibrium by redistributing initial wealth.
- This must not be taken literally: A social planner would need to know the consumers' preferences in order to identify the correct redistribution; and: if the social planner already knew the desired allocation: Why does he not implement it directly?
- The importance of the second theorem comes from its general intuition:
 - If we want to attain a different distribution, we need not reallocate final consumption goods, e.g., by giving them away for free or subsidized to certain consumers (this would “distort” consumption decisions)
 - Rather, we should redistribute initial wealth.
 - Thus, we do not (in principle) have to give up on the efficiency target in order to achieve redistribution.

Part 1: Principles of Economics

1. Basics
2. Necessary
Conditions for
Functioning Markets
3. Market Failure &
Market intervention
4. International Trade
5. Political Economy



Market intervention – why?

- If the market functions properly, we will “magically” reach the efficient outcome of Demand=Supply and the competitive equilibrium, which maximizes welfare is reached.
- Adam Smith’s “The invisible hand” – by letting people maximize their own profit, the market is socially efficient
- But we know that governments do intervene in markets.
- Why?

Market failures occur when there are frictions

- Reasons for market failure
 - Market power – e.g. a monopoly.
 - Externalities
 - Information Asymmetry
 - Behavioral market failure?
 - Recall the assumption of homo(o)economicus!
- We will consider each of these reasons and then look at the effect of intervention



<https://daveashelman.wordpress.com/2015/11/25/small-town-bringdowns-and-economic-failure/>



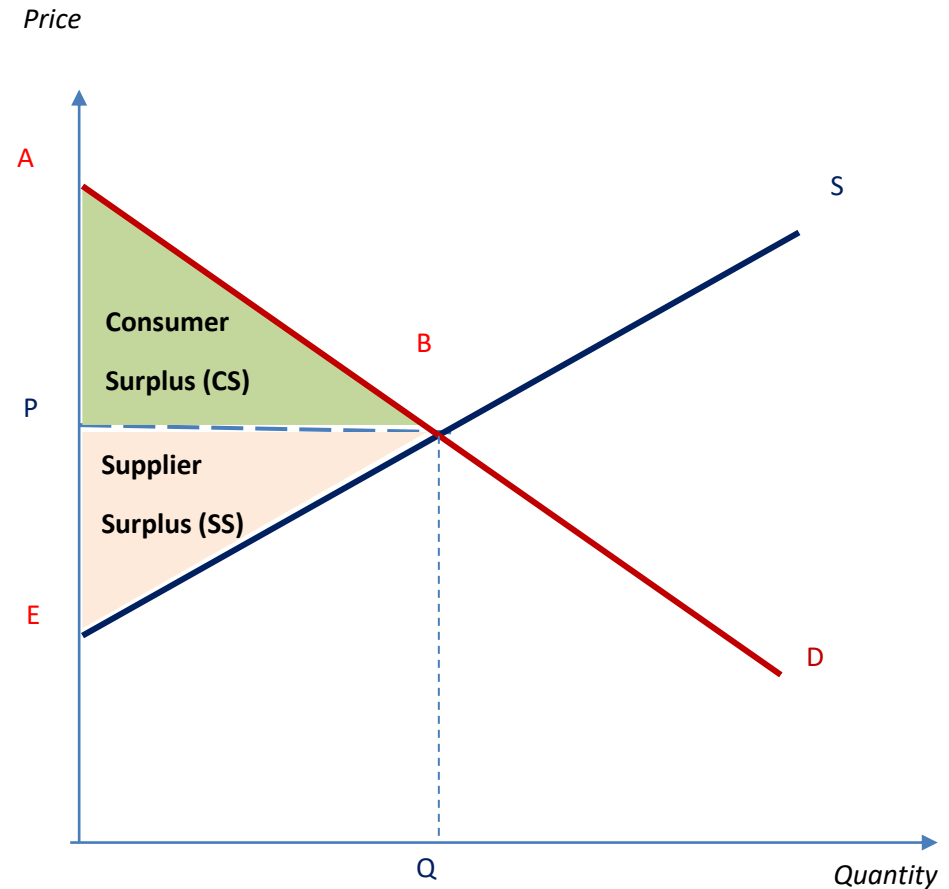
PHOTO COURTESY OF CABLE.COM

http://textbook.stpauls.br/Microeconomics/page_157.htm

Market Failure: Monopoly

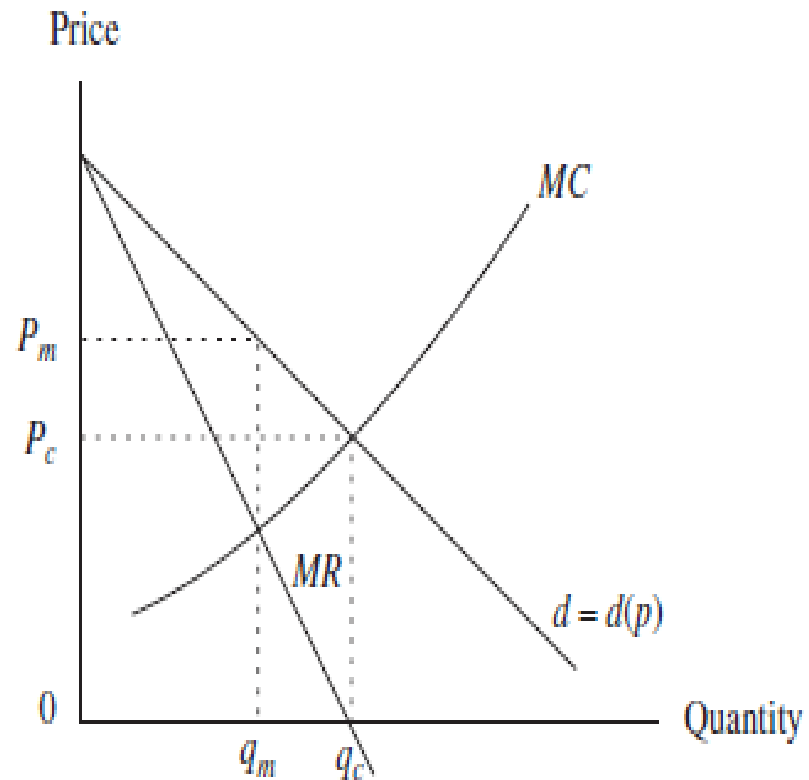
Market Failure: Monopoly

- Recall that (Kaldor-Hicks) efficiency is achieved when the **total surplus** (consumer + supplier) is maximized.
- This happens when the market is at its **competitive equilibrium**
- Are monopolies efficient?



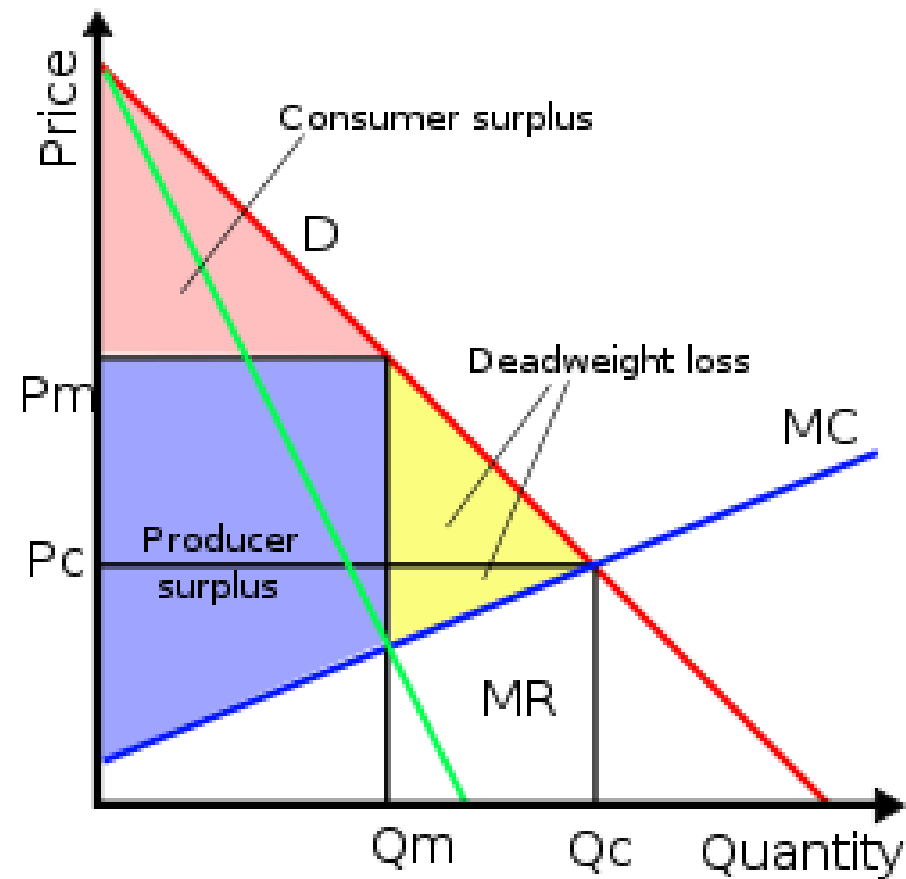
MF: Monopoly

- Recall: $MR < D$ when there is a monopoly.
- Since a monopoly chooses $MR = MC$, the *quantity* would be *lower* (q_m in the diagram) and the *price* will be *higher* (p_m in the diagram)
- This is inefficient: the supplier surplus is larger, but the overall surplus is smaller



MF: Monopoly

- The yellow area is a “deadweight loss”, due to the monopoly
- Consumer surplus with a monopoly (pink area) is smaller than without it
- Supplier surplus is larger with a monopoly



Source: <https://apecon2.wikispaces.com/Consumer+%26+Producer+Surplus+in+a+Monopoly>

MF: Monopoly

If monopolies are so inefficient, why do governments allow monopolies to exist?

Ex-ante vs. Ex-post

- Suppose the government proposes a prison sentence to people who don't wear a mask (because of Covid-19)
- Is this efficient?
- How does this relate to monopoly?



MF: Monopoly

- Monopolies are inefficient once they exist (=ex-post perspective)
- But what would happen if we forbid monopolies from being formed (=ex-ante)?
 - → Low incentive to enter the market (problematic when R&D investments are high)
 - → This is why we give *patents*.
- This is a typical tradeoff in (Law&)Economics: we often can't get ex-ante and ex-post efficiency at the same time.

MF: Monopoly

- Alternative governmental interventions:
 - Taxes & Subsidies
 - Price Regulation
 - Participating as market player (buying & selling)
- Before looking at monopolies, we will consider what intervention does in a competitive market



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Do you know why I'm always
worried when the professor is
drawing a graph?

Because he's definitely
plotting something!

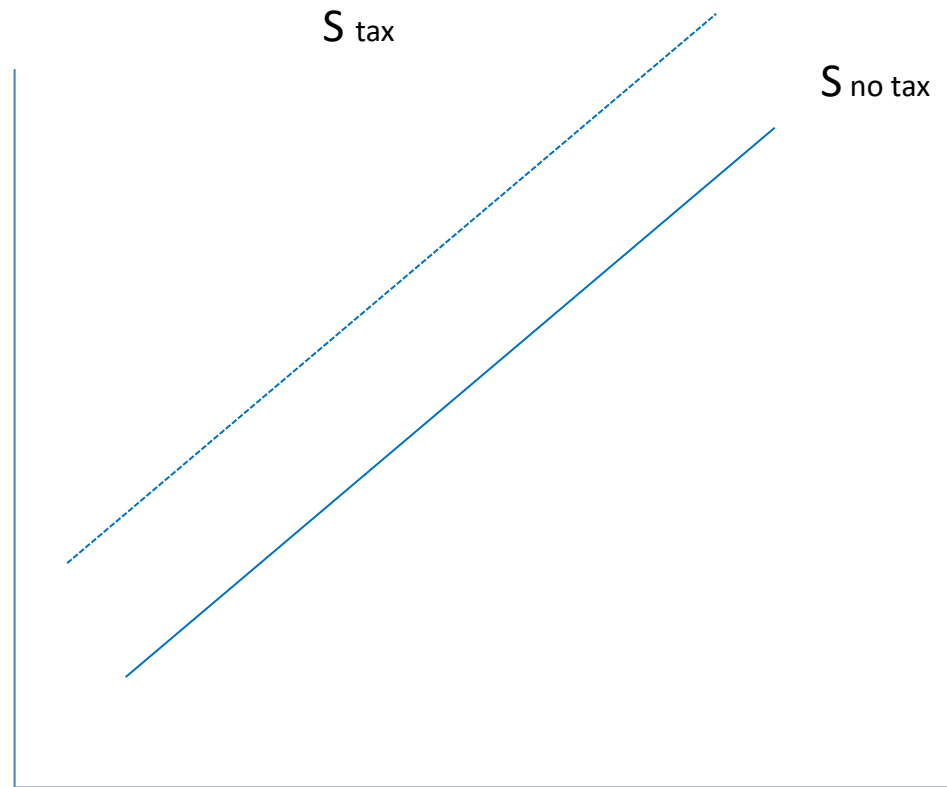
Taxes

Note the difference between a *lump sum* tax and a per-unit tax:

Lump-sum:

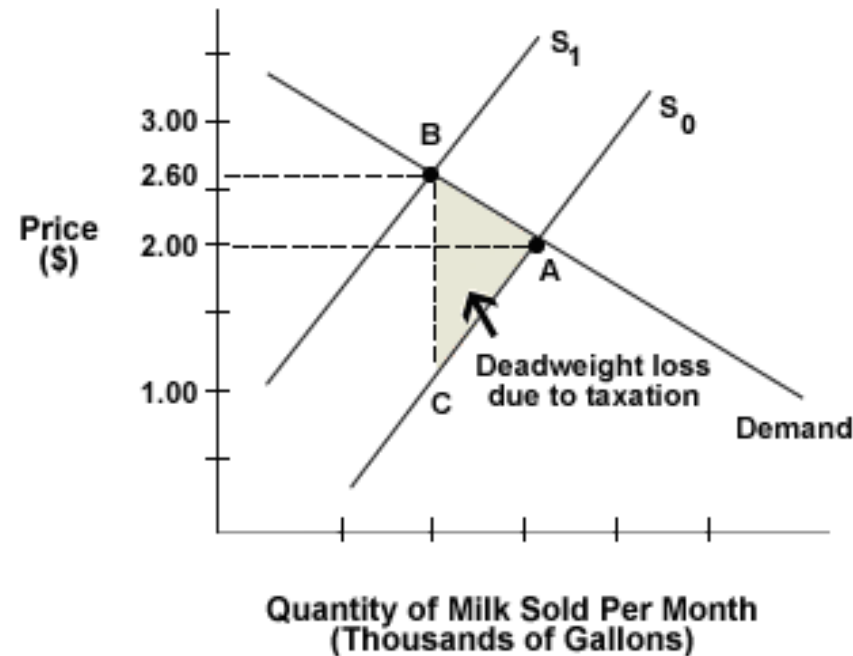
- Does not affect the marginal cost (it is a fixed costs)
- But it may cause firms to exit the market

Per-unit: See graph.



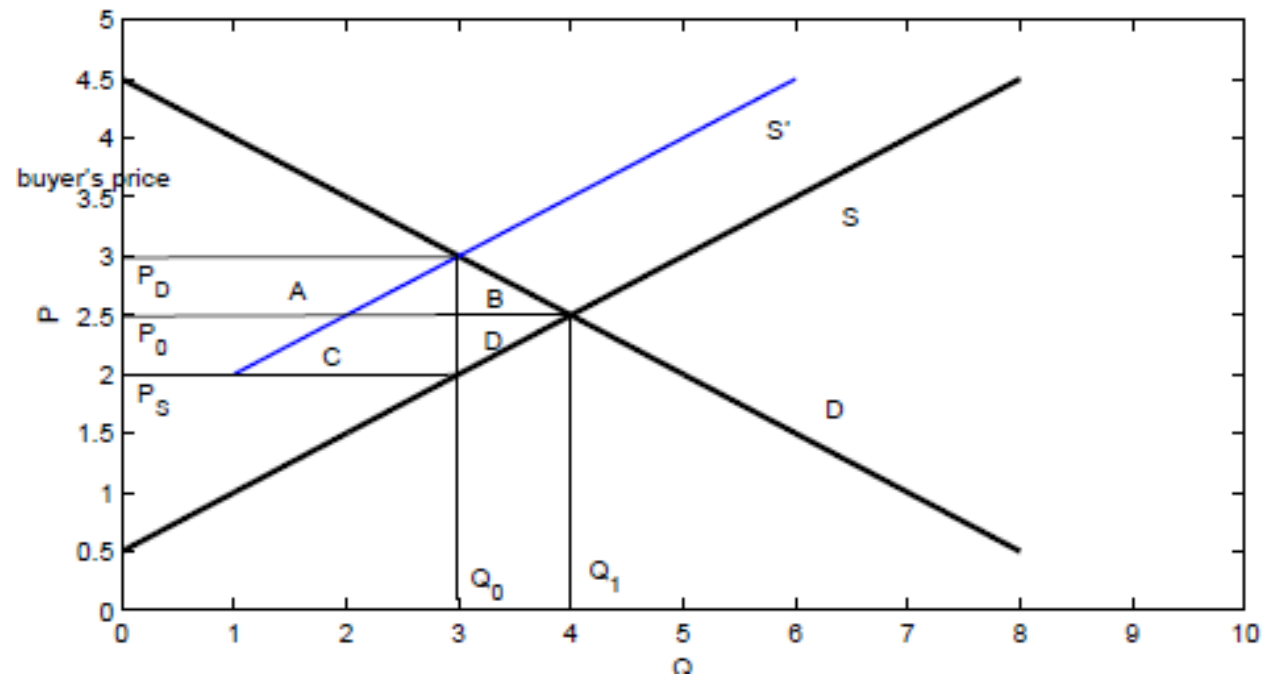
Taxes - example

- Government imposes a tax of 1\$ per gallon of milk
- Price before tax: 2.
- Price after tax: 2.6.
- Supply shifts up. Consumers pay 2.6. Seller gets 1.6.
 - → Consumers paid part of the tax (0.6).
- Outcome: a deadweight-loss (“DWL”). Why is there a loss?



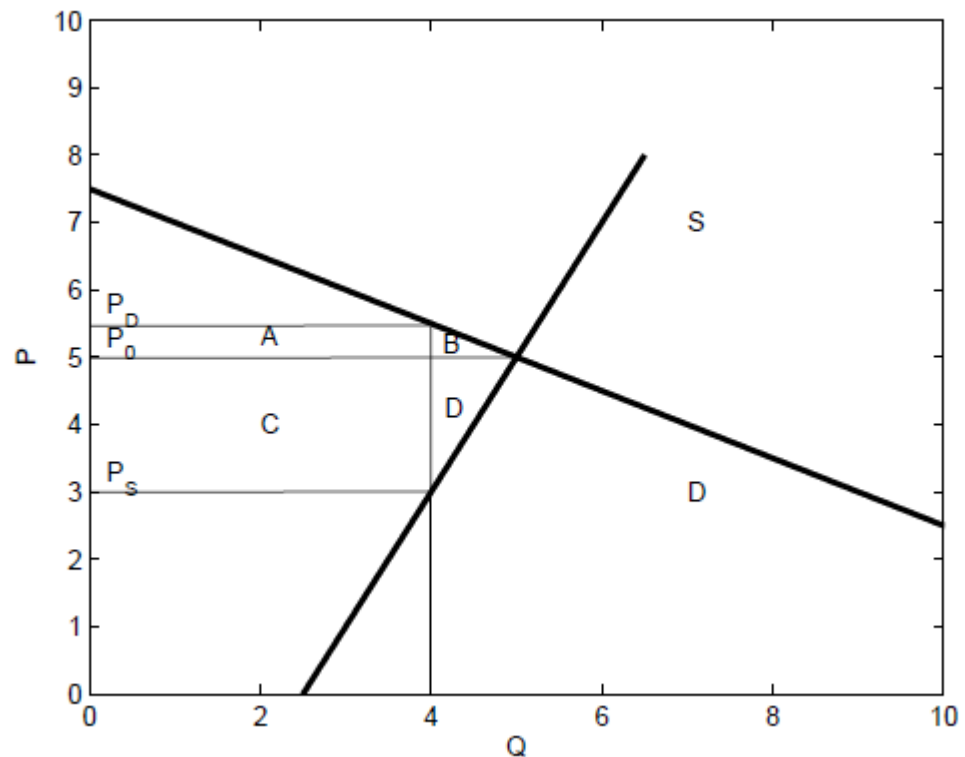
Taxes - example

- What about the government's revenue from taxes?
- $\Delta CS = -(A+B)$
- $\Delta SS = -(C+D)$
- $\Delta Gov = A+C$
- $DWL = B+D$
- Gov. revenues do not fully cover the loss



Taxes – effect of elasticity

- If the supply is relatively inelastic, suppliers bear most burden
- Recall: $\Delta SS = -(C+D)$
- The opposite happens if the demand is inelastic (=buyers bear burden)
 - Which is usually the case when there is market power

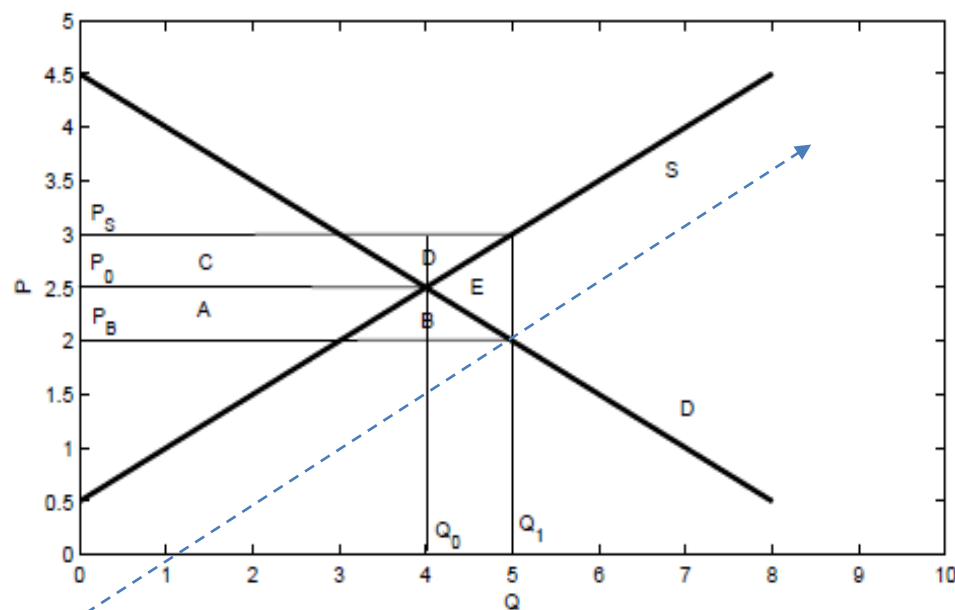


Taxes – Monopoly

- What would happen if we tax a monopoly?
- It is possible to draw a (little bit complex) graph showing what happens, but we will avoid this because the intuition is simple.
- A monopoly already produces *less* than the efficient quantity.
- Thus, imposing a tax will cause the monopoly to produce even less and the price will increase even more.
- The deadweight loss might also increase, whenever the revenues from the tax are smaller than the loss due to the price increase (which is the typical case).

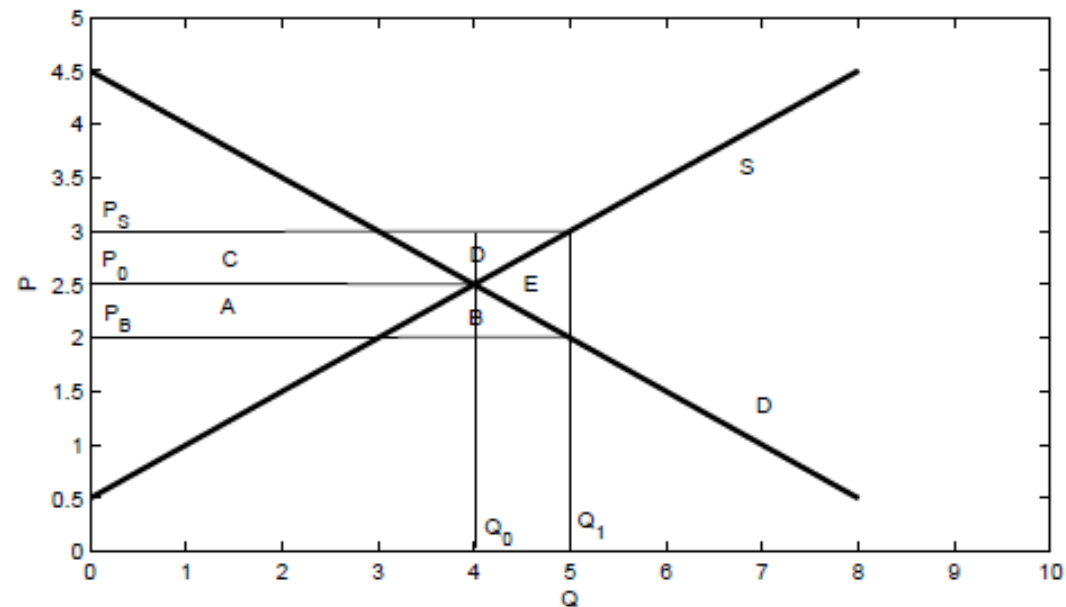
Subsidy

- Again we look first at competitive markets.
- A subsidy “sounds good” but may be inefficient
- When a subsidy occurs, the price that the seller sees is higher than the price that the buyers sees
- E.g. if the subsidy is 1\$ per unit, buyers pay p_B but sellers get $p_S = p_B + 1$.



Subsidy 2

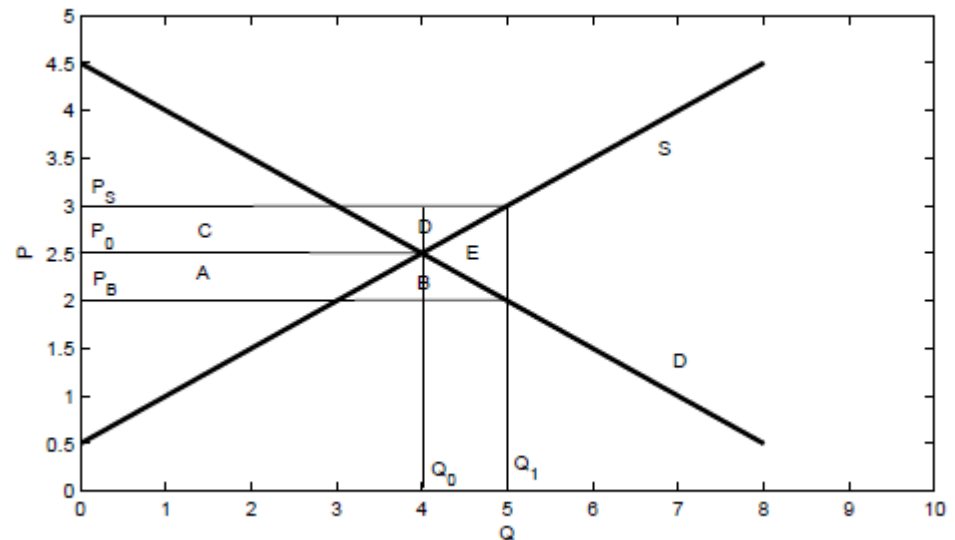
- Consumer surplus and Supplier surplus both increase:
- $\Delta CS = A + B$
- $\Delta SS = C + D$
- $\Delta Gov = -(A + B + C + D + E)$
- Expenditures do not fully cover benefit to the market
- Dead Weight Loss = E



*To see why C+D are relevant for the SS, see <https://economics.stackexchange.com/questions/12716/consumer-and-producer-surplus>

Subsidy 3 – effect of elasticity

- The effect of elasticity is a mirror image of the effect on tax
- If the supply is inelastic, sellers gain more.
- If the demand is more inelastic, buyers gain more.



Subsidy - Monopolies

Clarification

- Assume constant MC for simplicity (otherwise the graph gets messy).
- Before subsidy: $DWL = C$
- After subsidy:

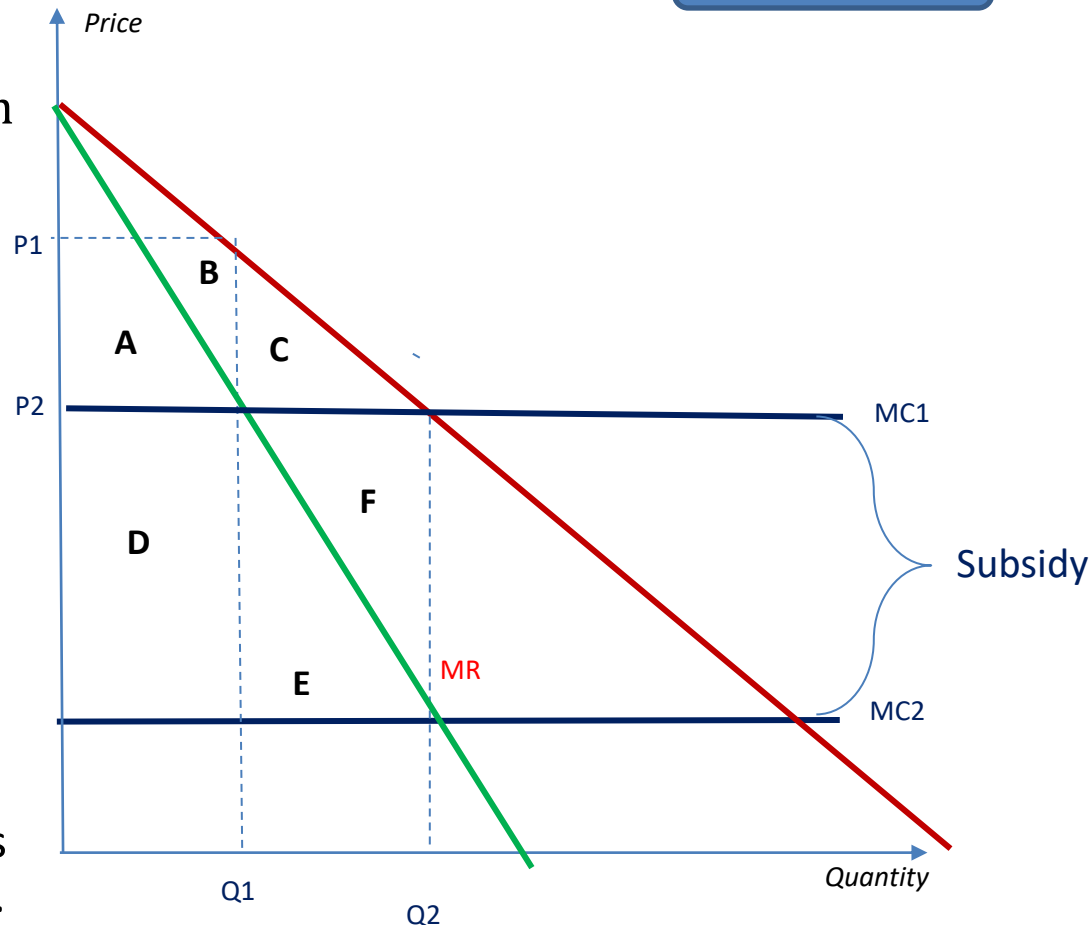
$$\Delta Gov = -D - E - F$$

$$\Delta CS = A + B + C$$

$$\Delta SS = D + E + F - A - B$$

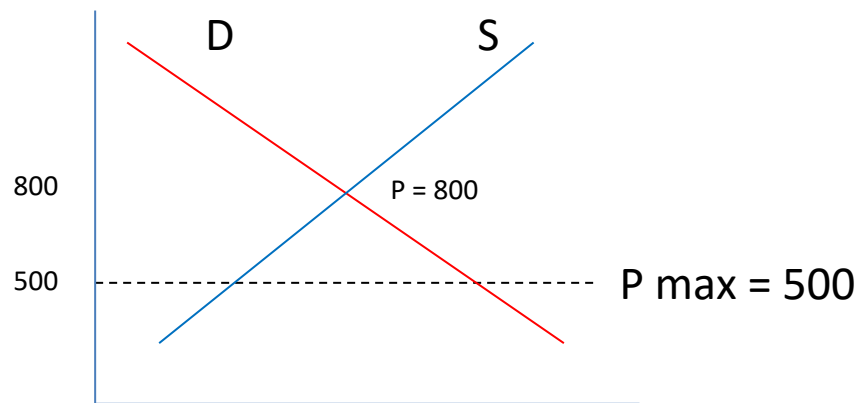
$$\text{Total } \Delta = C$$

- Note: Price falls from P_1 to P_2 , but the decrease is *less* than the subsidy (i.e. the monopoly keeps some of the benefits of subsidy).



Price Regulation 1

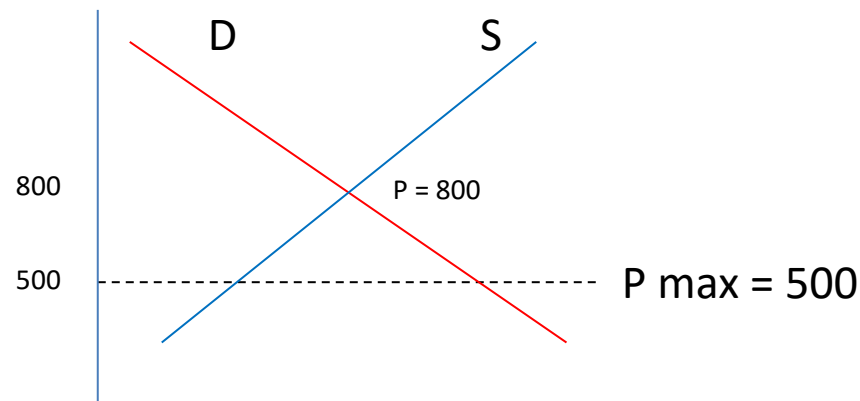
- Again, we first consider competitive markets.
- Suppose that the government thinks that landlords charge too much money for their apartments (e.g. 800€ per month, for 1-bedroom apartment in Hamburg)
- The Government sets a maximum price of 500 € a month for apartments.
What will happen?



- Does it matter whether the maximal price is strictly enforced or not?

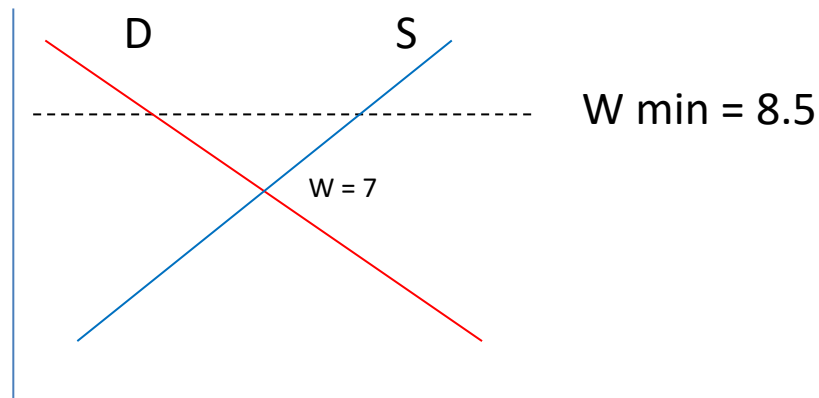
Price Regulation 2

- At the maximum price $P_{max} = 500$ there is excess demand $\rightarrow$ some landlords will not agree to rent.
- Thus, less apartments will be available (or quality decreases), people would have to stand in line to get an apartment.
- If the price is not strictly enforced, landlords may demand side-payments (“I’ll rent it for this price, but you have to buy the Sofa separately”).



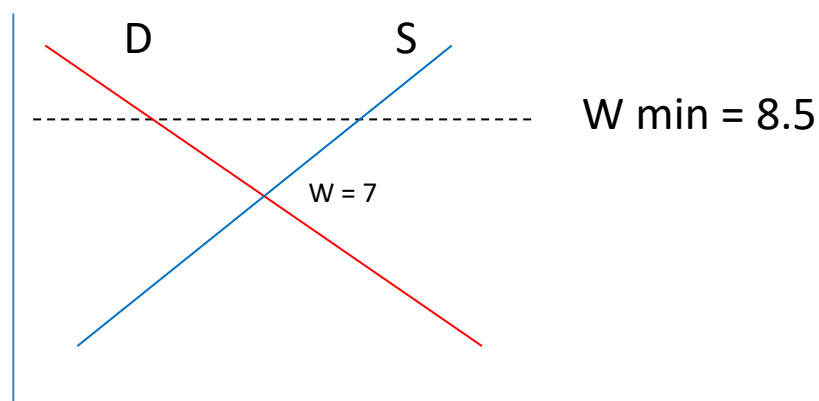
Price Regulation 3

- Different example with the same intuition: minimum wage
- Recall:
 - Demand=employer seeks workers
 - Supply = workers willing to work.
- With a minimum wage, there is excess supply.
- Workers must stand in line = unemployment
- If not strictly enforced, worker pay side-payments to work (e.g. a free “praktikum”)



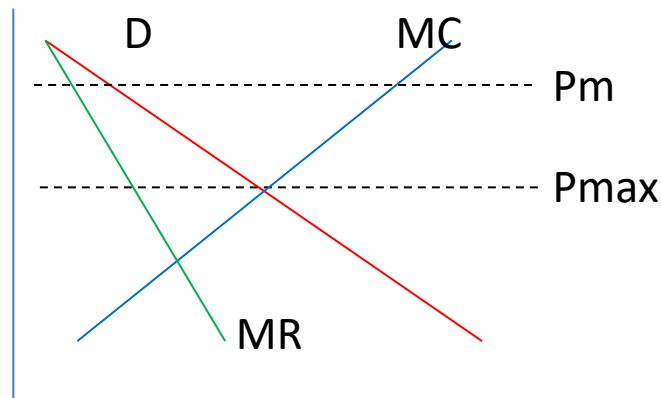
Price Regulation 4

- Hence, theorists usually think that imposing a minimum wage would cause unemployment, at least for low-skilled workers:
 - Demand for workers is lower at the minimum wage
 - Low-skilled workers are substitutable by other inputs (demand is elastic)
 - Only low-skilled workers are directly influenced by the max price (high skilled-workers get more anyway)
- Empirical evidence mostly supports the theory (with some exceptions that find no effect).



Price Regulation & Monopoly

- Recall: A monopoly chooses $MR=MC$.
- If the maximum price is above the monopoly price, it (of course) has no impact.
- Otherwise, If $MR < P_{max}$, the intervention works: The monopoly reduces the price while producing the same quantity.



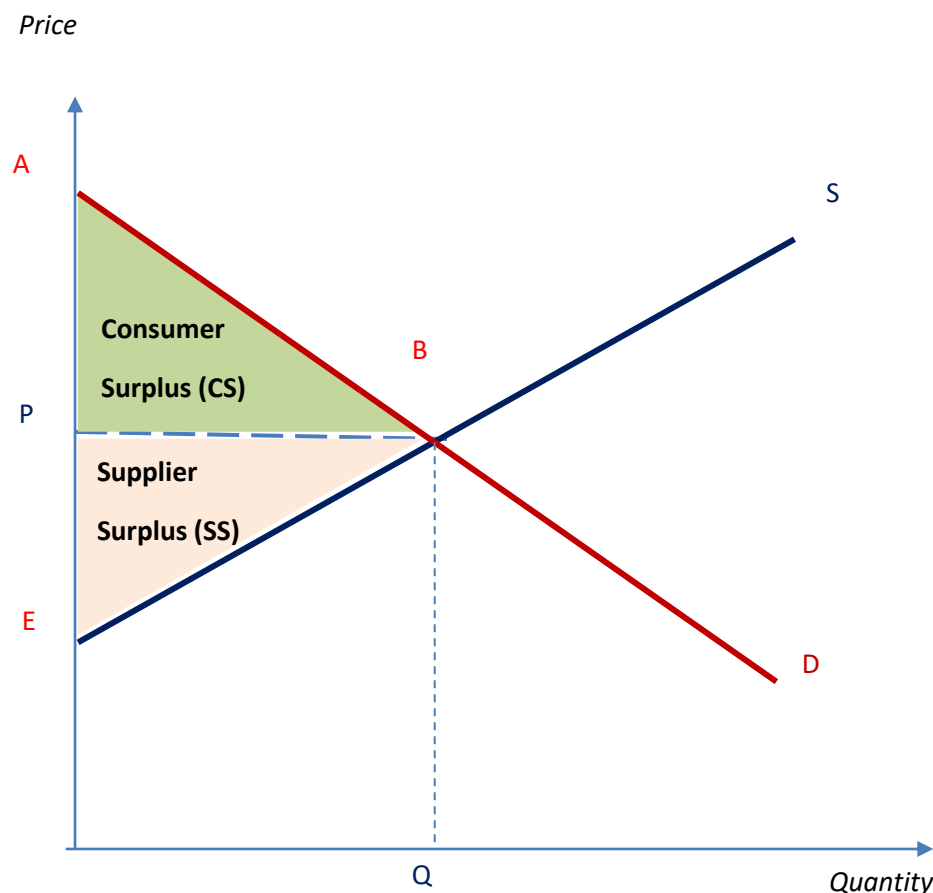
Interventions: Monopolies vs. Perfect competition

- We saw that interventions generally reduce welfare (cause a dead weight loss) when there is perfect competition.
- Conversely, the interventions might work when there is a monopoly
- But: one must be careful which policy to choose

Market Failure: Externalities

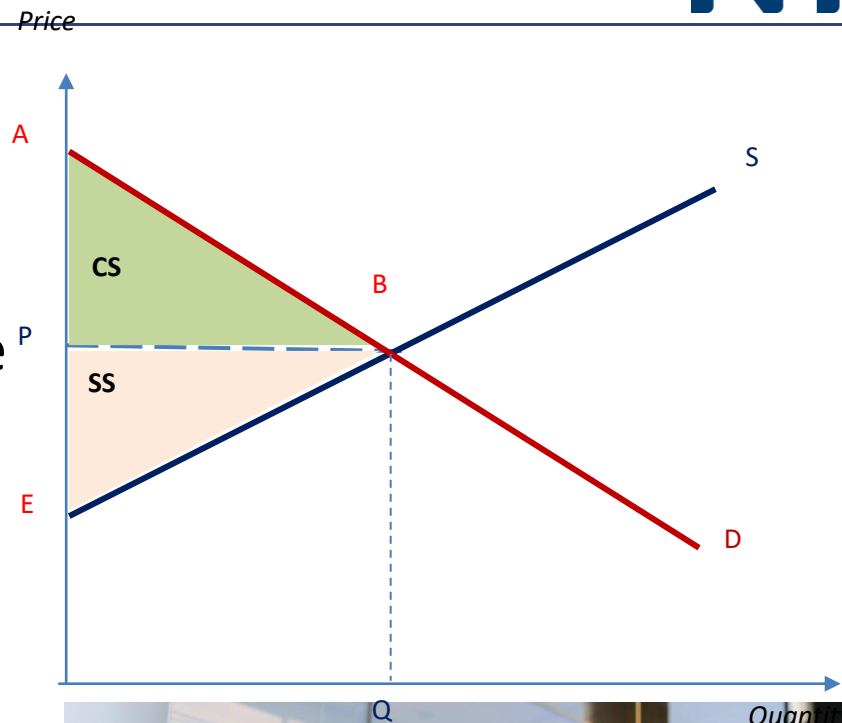
MF: Externalities

- A competitive market ensures that the total surplus (=consumers + producers) is maximized.
- But does it ensure that **social welfare** (=sum of utilities of *all* individuals) is maximized?



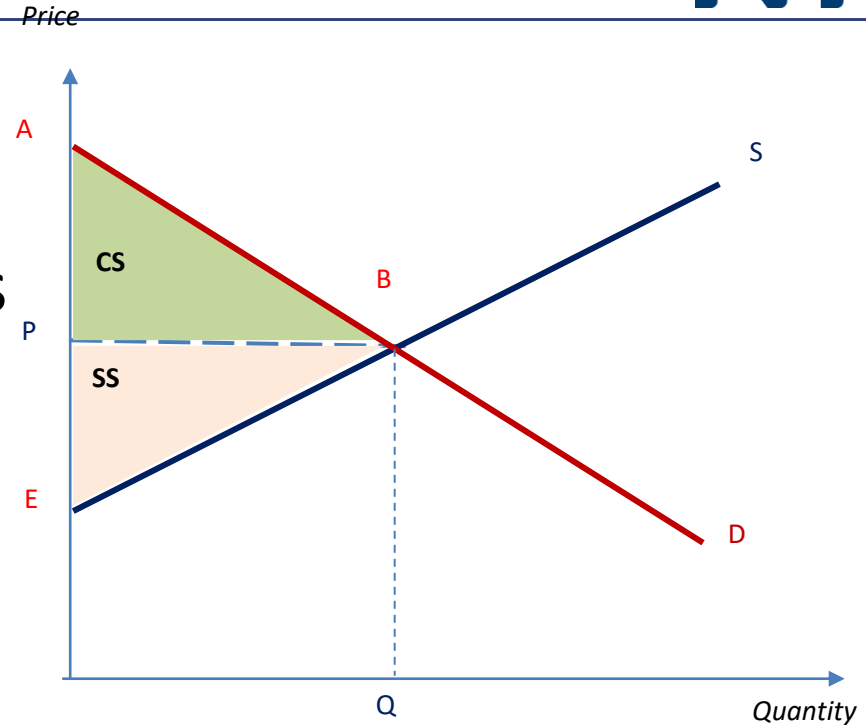
MF: Positive Externalities

- Suppose that the price of a masks is $p=5$ in the competitive equilibrium.
 - $\rightarrow$ Total surplus is maximized.
- Some buyers are not willing to buy for this price
 - Usually but that's ok, right?
- Utility of non-buyers might be maximized when more masks are bought, but this isn't priced in the market
- $\rightarrow$ Buying masks has a positive externality on others



MF: Negative Externalities

- Suppose that drug addicts are willing to pay 10 EUR for a new drug.
- Buyers who consume the drug have a 10% probability of becoming violent and killing others.
- This risk isn't internalized by the participants in the market
- → Negative externality



Interventions in case of externalities

- Taxes for negative externalities
- Subsidies for positive externalities
- Price regulation
- Participating in the market
 - E.g. Government supplies the “public good” (will be discussed more later).
 - Examples?

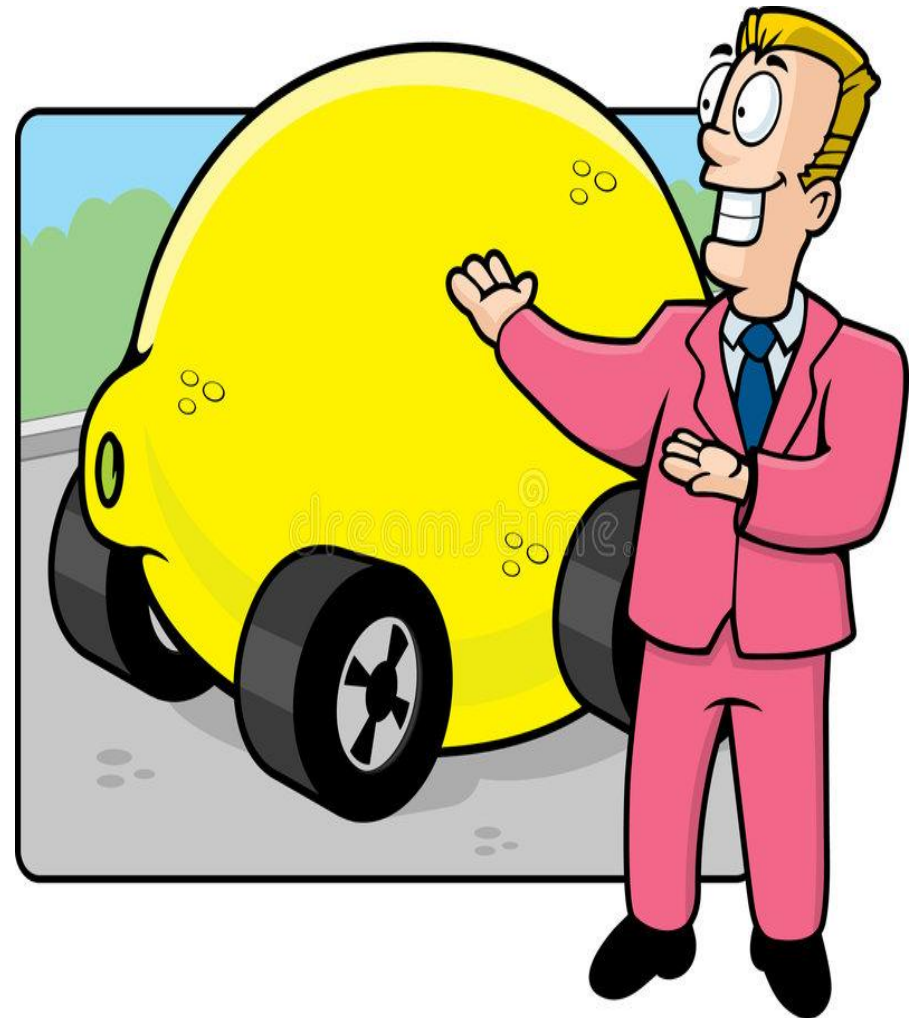
Market Failure: Information Asymmetry

Adverse Selection

“The Market for Lemons”

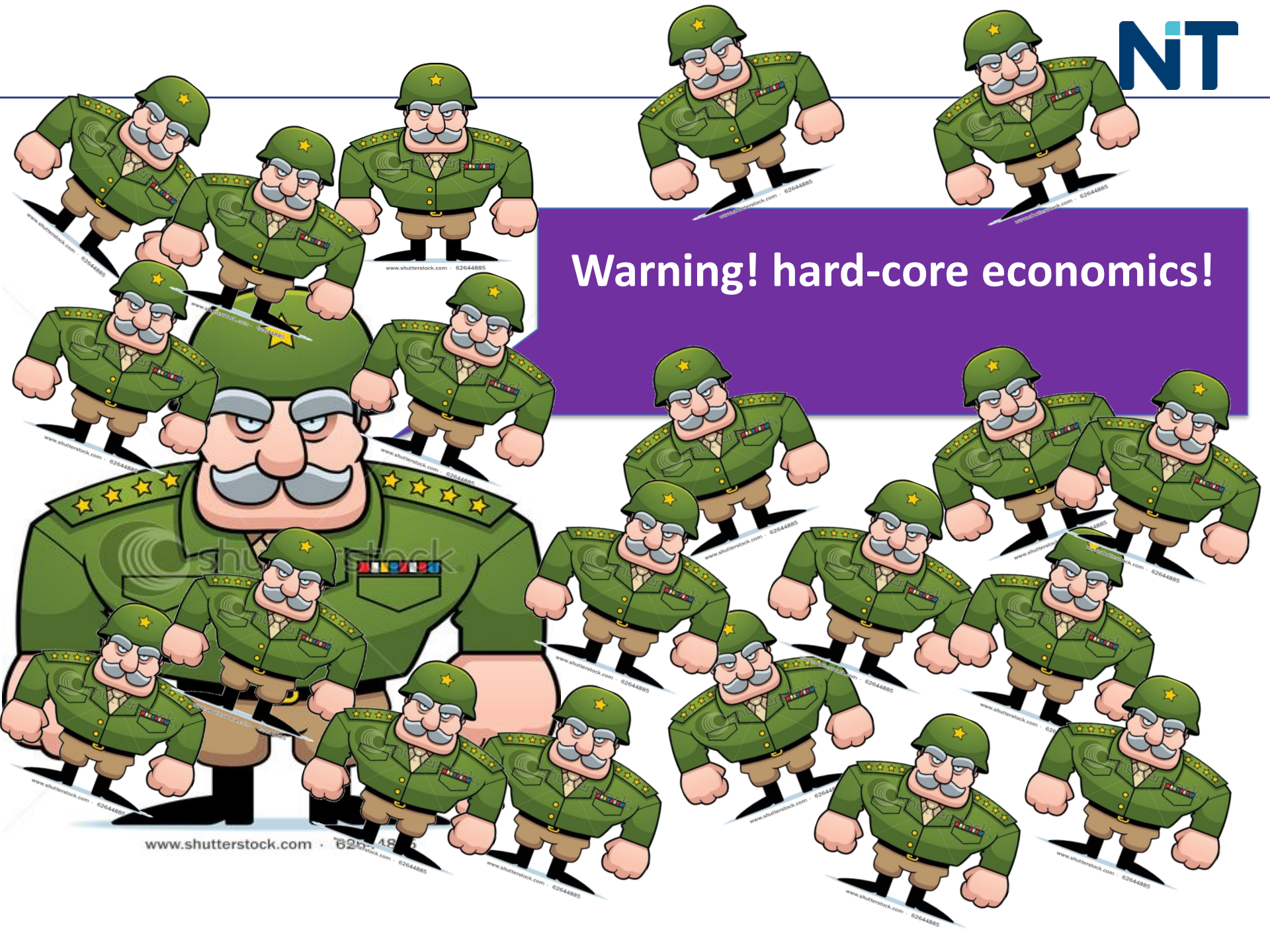
(George Akerlof, Nobel Prize winner)

*if he wants to sell that car, do
I really want to buy it?*



Adverse Selection: Example

- A used cars dealer can sell three different types of cars:
 - “High type” (H): worth 12 to a buyer, but seller incurs a cost of 9 when selling.
 - “Medium type” (M): worth 6 to buyer, but seller incurs a cost of 4 when selling.
 - “Low type” (L): worth 1 to buyer; seller incurs cost of 0 when selling.
- Seller has an equal amount ($1/3$) of each type.
- Buyer can’t infer the type of car ex-ante, and also can’t prove it ex-post (or a proof is too costly).
- Seller makes a “take it or leave it” offer.
- What is the market outcome?



Warning! hard-core economics!

Adverse Selection: Example

Objective Functions

- **Buyer** wants to maximize expected gains from trade: $U^B = E[v_i] - p$, where $i = \{H, M, L\}$, and p is the price demanded by the seller.

Type	Probability	Value v	Cost c
H	1/3	12	9
M	1/3	6	4
L	1/3	1	0

- **Seller** wants to maximize the profits from trade: $U^S = p - c$.

Possible Outcomes

- All types are sold
 - $S_1 = \{H, M, L\}$
- Only a subset of sellers is active
 - $S_2 = \{M, L\}$
 - $S_3 = \{L\}$
 - $S_4 = \{H, M\}$.
 - $S_5 = \{H, L\}$
- We want to check, which of these outcomes can be an equilibrium and what will be the resulting price if the seller makes a “take-it-or-leave-it-offer”.

Adverse selection

S_1 : Could it be that all types of cars are offered?

- Maximum price the buyer is willing to pay is

$$E(v) = \frac{1}{3} * 12 + \frac{1}{3} * 6 + \frac{1}{3} * 1 = 6\frac{1}{3}$$

- This is strictly less than 9, i.e., what a high car dealer needs to break even -> **thus, this cannot occur in equilibrium**

S_2 : Could it be that only M & L are active?

- Maximum price the buyer is willing to pay is

$$E(v) = \frac{1}{2} * 6 + \frac{1}{2} * 1 = 3\frac{1}{2}$$

- This is strictly less than 4, i.e., what a medium car dealer needs to break even -> **thus, this can also not occur in equilibrium.**

S_3 : Could it be that only L types are active?

- Maximum price the buyer is willing to pay is

$$E(v) = 1$$

- This is enough for the L-types to break even -> **thus this can occur in equilibrium.**

Adverse selection

S_4 : Could it be that only H & M are active?

~~HM~~

- Maximum price the buyer is willing to pay is

$$E(v) = \frac{1}{2} * 12 + \frac{1}{2} * 6 = 9$$

- Although this is exactly 9 (what a high car dealer needs to break even), it can't hold as an equilibrium. Why? Because for this price, it's also profitable to sell low cars! -> **thus, this cannot occur in equilibrium**

S_5 : Could it be that only H & L are active?

- Maximum price the buyer is willing to pay is

$$E(v) = \frac{1}{2} * 12 + \frac{1}{2} * 1 = 6.5$$

- This is strictly less than 9, i.e., what a high car dealer needs to break even -> **thus, this cannot occur in equilibrium**

Adverse Selection: Summary

- Because of asymmetric information, the buyer is only willing to pay the price for an average car. But for this price, the dealer will not sell good cars.
- Then, the “updated” average car is an average between medium and low. But the seller won’t sell medium cars for this price.
- Result: only low cars are sold.
- This process is called “adverse selection”.
- What would happen if selling bad cars would cost the seller 2 (instead of 0)? → Then, even low cars won’t be sold = *complete market breakdown*.

Adverse selection: solution?

- There are many possible solution for this problem, e.g.
 - Pay an expert (a garage) to examine the car before buying (so that type is certified) - e.g. TÜV in Germany.
 - Only works when the type *is* verifiable.
 - Give a guarantee: the parties sign a contract, stipulating that in case the product turns out to be bad, the seller will pay for replacing/fixing it.
- What is the purpose of guarantees?
 - **Guarantees as insurances**
 - Guarantees as an incentive for product quality
 - Guarantees as quality signals

Guarantee: Insurance function

- pay an insurance premium to have a guaranteed product quality, avoid the loss from the defect
 - Typical case: risk averse buyer vs. risk neutral seller
- What does „risk aversion“ mean? Suppose I offer you two choices:
 - Choice 1: get 10,000 Euros.
 - Choice 2: a lottery ticket, which gives 100 Million euros with probability of 0.01% and 0 otherwise.
- Both options have the same expected value, but most people prefer the safe amount → Risk aversion.
- As a result, people are willing to pay a „risk premium“: extra money to reduce/eliminate the risk.

Guarantee: Insurance function - Example

Assumptions

- Product is valued by the consumer at 100, but with 10% probability is defective (in which case it's worth 0). Expected value is thus 90.
- After purchase it's revealed whether the product is defected.
 - Slightly different from the cars example – here it can be verified ex-post.
- Suppose the seller offers the following prices:

100, with guarantee

90, without guarantee

Choice of a **risk-averse** consumer?

- *Buyer will not be willing to pay 90, because the utility from a risky product is less than 90 due to risk aversion.*
- *Buyer will be willing to pay 100, because the expected value becomes 100 and there is no more risk.*

Guarantee - product quality incentive

- Built on top of insurance function.
- A seller who is liable for damages due to the guarantee will have an incentive to produce better cars
- Seller will invest into quality (design or quality control) as long as the savings from reduced losses are larger than the investment costs → Efficient care

Guarantee - product quality incentive – example

Two car producers, A and B.

Production costs: 20.000 € per car.

Probability of defect: 10%

Loss (compensation to buyer): 10.000 €

B decides to improve the quality of his cars by investing 500 €/car

→ This reduces probability of defect to 1%

Expected costs of A: $20.000 \text{ €} + 10.000 \text{ €} \times 0.1 = 21.000 \text{ €/car}$

Expected costs of B: $20.500 \text{ €} + 10.000 \text{ €} \times 0.01 = 20.600 \text{ €/car}$

= despite higher production costs, total costs per car of B are lower

→ Guarantee incentivizes the extra efficient investment

Guarantee - Signalling function (Spence (1973); Grossman (1981))

- Another solution to the adverse selection problem in economics is *signalling*: the informed party (the seller) takes a costly action in order to „separate himself“ from the „bad type“.
- A seller can signal high quality of his product by an extensive guarantee.
- If this guarantee is too-expensive when selling a medium or low car, than the seller can successfully signal that he is selling a high car.

Guarantee - Signalling function - Example

Assume 2 types of sellers, selling a car worth 100 to buyer.

- H produces high-quality cars, which have a 10% probability of a defect (Expected value is 90).
- L produces low-quality cars, which have a 70% probability for a defect (Expected value is 30).
- Each type of seller is equally likely, and costs of production are zero for both.
- Without guarantee, the buyer will be willing to pay:

$$E(v) = 0.5 * 90 + 0.5 * 30 = 60.$$

Both sellers can give a guarantee, but doing so is more costly for the Low type.

**Assume everyone is risk neutral*

Guarantee - Signalling function - Example

Is there an equilibrium where the buyer pays 90 (value of high-type car) if he gets a guarantee and 30 (value from low-type car) otherwise?

Cost of guarantee G is 10 for H ($G_H = 10$), but 70 for L ($G_L = 70$).

H gives a guarantee, because

$$\underbrace{90 - G_H}_{\text{Profit with guarantee}} > \underbrace{30 - 0}_{\text{Profit w/o a guarantee}}$$

$90 - 10 = 80$ $= 30$

L doesn't give a guarantee because:

$$\underbrace{90 - G_L}_{\text{Profit with guarantee}} < \underbrace{30 - 0}_{\text{Profit w/o a guarantee}}$$

$90 - 70 = 20$ $= 30$

Can the low type copy the signal?

A guarantee needs to be a credible signal.

In the example, the guarantee works as a signal because imitating the high type is too costly for the low type.

→ Whether imitating works depends on the parameters

→ Example: what would happen if the buyer would value a high car at 150 in case of no-defect (keeping all else equal)?

Can the low type copy the signal?

Updated values: H cars are 135 ($= 0.9 * 150$) ; L cars are still 30.

Expected value w/o guarantee is $E(v) = 0.5 * 135 + 0.5 * 30 = 92.5$.

Can it occur that cars w/ guarantee are sold for 135 and w/o guarantee are sold for 30?

H would then give a guarantee, because $\underbrace{135 - G_H}_{\text{Profit with guarantee}} > \underbrace{30 - 0}_{\text{Profit w/o a guarantee}}$
 $135 - 10 = 125$ $= 30$

But now, L also gives a guarantee, because:

$$\underbrace{135 - G_L}_{\text{Profit with guarantee}} > \underbrace{30 - 0}_{\text{Profit w/o a guarantee}}$$

$135 - 70 = 65$ $= 30$

Can the low type copy the signal?

- Why does the signal not work in the second example?
 - Being perceived as the high type is now more profitable
 - → High incentive to imitate the signal
- What happens after the low type tries to imitate the high type in the second example?
 - Buyer can again not differentiate → pays average price.
 - High type responds by not giving a guarantee
 - Low type responds by also not giving a guarantee...
 -[we will not solve this fully until the end]

What about the buyer's actions?

- "Moral hazard":
 - Typical problem in principal-agent models
 - Instead of “unobservable types” (e.g. car sellers) we have unobservable actions, e.g. by the buyer after purchase.
- Example: Buyers may use the product either carefully or carelessly (e.g. abstain from regular maintenance, or over-use)
- Those who were not careful, may try to hide it and then demand compensation in case of harm.

Screening Solution

How can we solve the moral hazard problem?

→ Regular option: pooling equilibrium (“one price”)

- A single price that reflects low-risk buyers paying too much and high-risk buyers paying too little

- Cross-subsidisation

- Fear that low risk buyers will not buy the product (as the price is too high), i.e. *adverse selection problem*.

→ Solution: offer two contracts: (1) full guarantee + high price and (2) limited guarantee + low price, to separate the types.

→ Careful drivers will buy the limited guarantee.

Law-mandated guarantee - example

- *So far we have talked about a seller's individual decision to offer a guarantee.... But what happens if it is mandated by law?*
- *Example:*
 - Directive 99/44/EC (will be replaced in 2022 by directive **2019/771**)
 - Applies to b2c transactions.
 - **Legal guarantee:** mandatory minimum limitation periods of two years for new goods and one year for second-hand goods
 - **Commercial guarantee:** sellers can offer better terms than the required minimum, and such offers are binding.

*Should legal guarantees be mandatory? *how much can be signalled on top?*

Part 1: Principles of Economics

1. Basics
2. Necessary
Conditions for
Functioning Markets
3. Market Failure &
Market intervention
4. International Trade
5. Political Economy



International Trade: Comparative Advantage

International Trade

- Suppose Robinson & Friday each establish a state at their edge of the island.
- The new states can trade, just as the individuals could.
- But what happens if the following changes happen?
 - (The states of) Robinson & Friday learn how to grow bananas and coconuts themselves (i.e. they can produce).
 - More people arrive at the beach and join the states, so that the decision-making process is more complex.

International Trade

- We first tackle the complication due to the ability to produce.
- When discussing production problems, we saw an example where there were two types of workers that had different opportunity costs.
 - Our conclusion was to assign each worker to the task that he performs cheaper
 - → “Comparative Advantage”
- The same concept can be extended to multiple countries that have different specializations

International Trade

- Robinson is better in producing both bananas and coconuts
- But in terms of *opportunity costs*, It is cheaper for Robinson to produce coconuts, and cheaper for Friday to produce bananas.
- What does efficiency dictates?

Production capacity in 1 day

	Bananas	Coconut
Robinson	2	6
Friday	1	5

Marginal costs

	Cost of producing a Banana	Cost of producing a coconut
Robinson	3C	1/3 B
Friday	5C	1/5 B

International Trade

- Each state should focus on it can produce cheaper. Then, they should trade
 - This way, resources aren't wasted
- What is the price range in which they will trade?

Production capacity in 1 day

	Bananas	Coconut
Robinson	2	6
Friday	1	5

Marginal costs

	Cost of producing a Banana	Cost of producing a coconut
Robinson	3C	1/3 B
Friday	5C	1/5 B

International Trade

Robinson: “if it’s only me, then to get 1 coconut I have to give up 1/3 of a banana. I am thus willing to buy a coconut if it is cheaper than that”

Friday: “For me, producing coconuts costs only 1/5 of a banana. So I’m happy to trade!”

Trade if:

$$\frac{1}{5}B < p_c < \frac{1}{3}B$$

$$3C < p_B < 5C$$

Production capacity in 1 day

	Bananas	Coconut
Robinson	2	6
Friday	1	5

Marginal costs

	Cost of producing a Banana	Cost of producing a coconut
Robinson	3C	1/3 B
Friday	5C	1/5 B

International Trade

Ricardo's Comparative Advantage Theorem (1817)

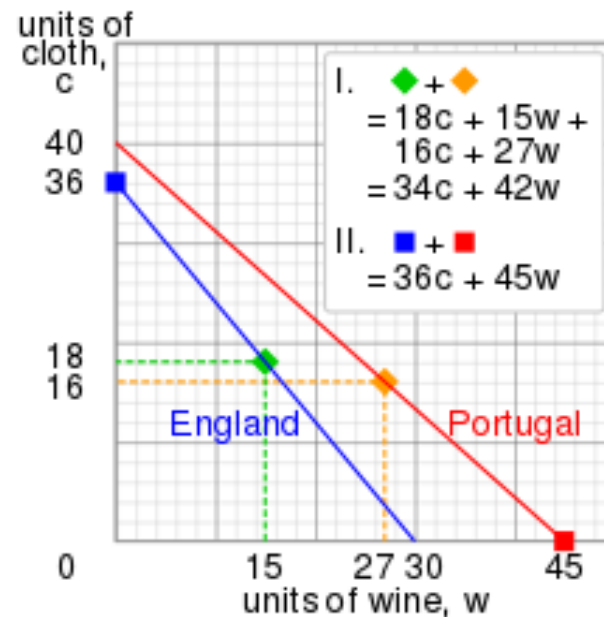
- David Ricardo wanted to explain why a country that is more efficient in producing two products might still want to trade with another country.

Hours of work needed to produce the product	Cloth	Wine
England	100	120
Portugal	90	80

Marginal opportunity costs	Cost to produce Cloth	Cost to produce Wine
England	5/6 W	6/5 C
Portugal	8/9 W	9/8 C

*Note: the numbers in the first table represent costs. Thus, a higher number means less productive.

England has a comparative advantage for producing cloth
Portugal has a comparative advantage for producing wine



Source: https://en.wikipedia.org/wiki/Comparative_advantage

Example: Suppose each country can invest 3,600 hours.

Case I. Each country produces a mixture of Cloth and wine.

Case II. Each country produces according to the comparative advantage.

International Trade

Suppose that the only input is labor, so that the cost of production is just the wage (w).

How are the prices decided?

- In *perfect competition* and free trade, everyone produces at $p = \text{unit cost}$.
- The relative prices of the two goods are then equal to the relative unit costs:

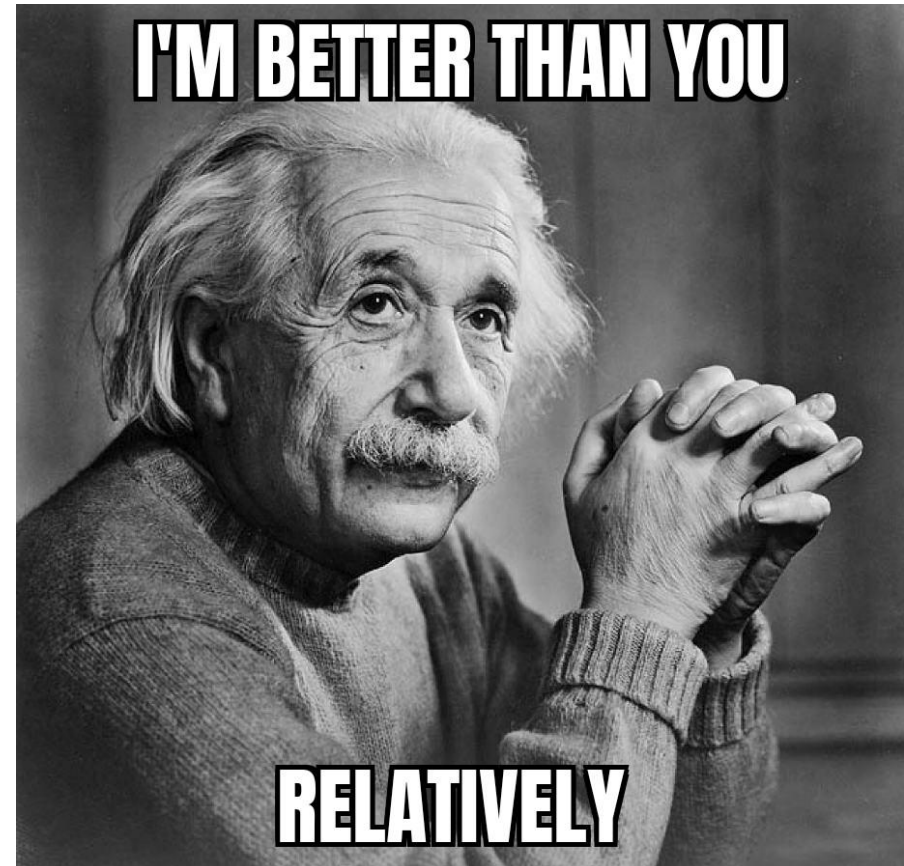
$$P_i = \frac{p_i^x}{p_i^y} = \frac{w_i H_i^x}{w_i H_i^y}, i \in \{A, B\}$$

	Country A	Country B
<u>Product X</u>		
Hourly wage	10	5
Hours worked	1	4
Unit costs	10	20
<u>Product Y</u>		
Hourly wage	10	5
Hours worked	2	6
Unit costs	20	30
Prices		
P_i	$\frac{10}{20} = \frac{1}{2}$	$\frac{20}{30} = \frac{2}{3}$

Good X is exported from country A to country B if and only if $P_A < P_B$.
Flexible *exchange rates* are the simplest transmission mechanism.

Comparative advantages depend on....

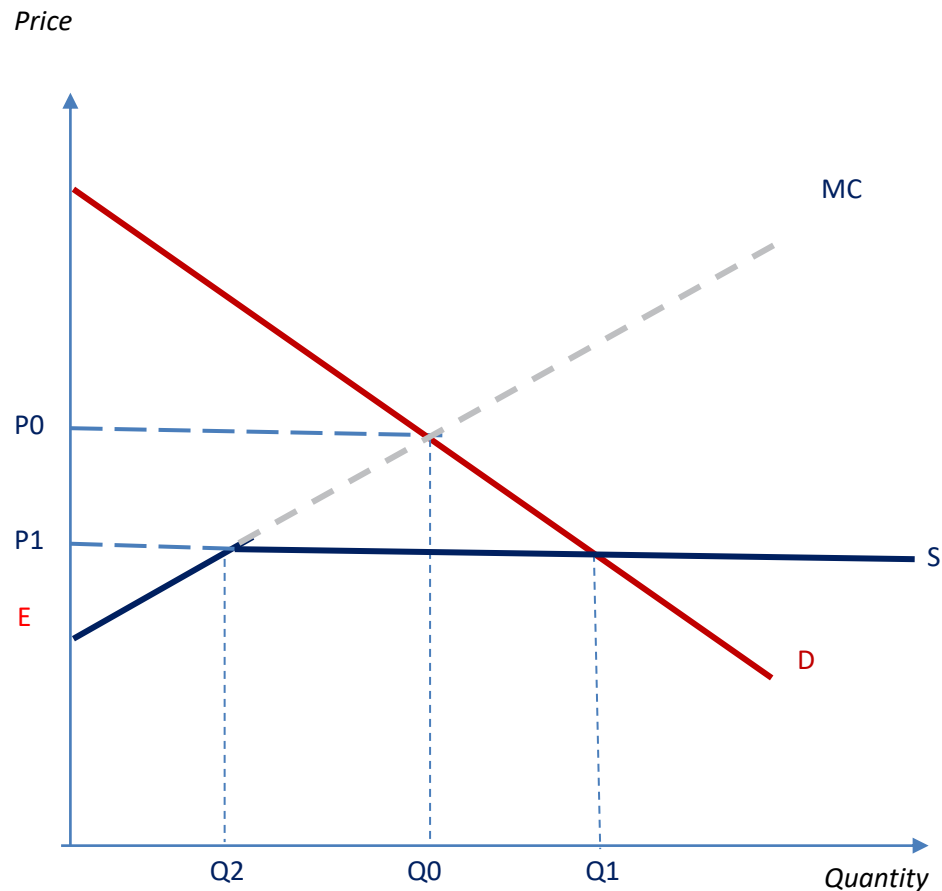
- Natural resources
- Relative scarcity of factors (factor endowments)
- Preferences
- Capabilities/Human Capital
- Technology
- Institutions
- Wage differentials between industries (net of productivity)



International Trade: Import & Export

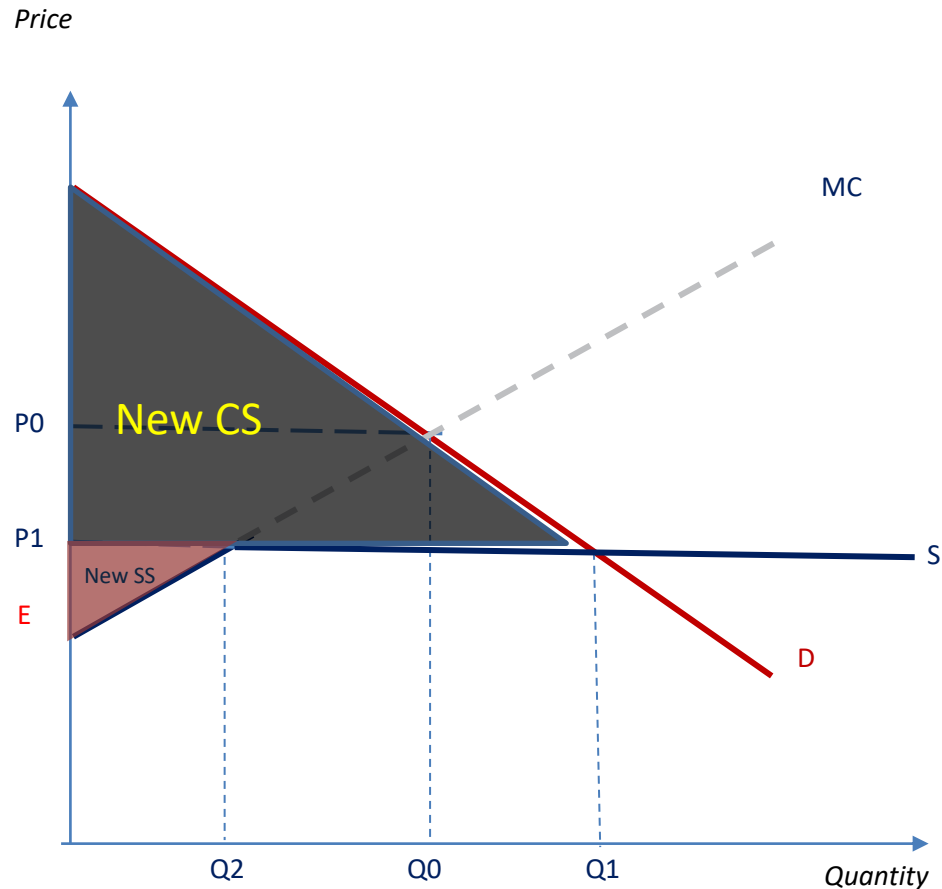
Equilibrium with free trade: Import

- Suppose the state can import at the world market price, p_1 .
 - *Assume that the state is too small to affect the world price.
- What happens then?
 - The local supplier produces Q_2
 - The quantity $Q_1 - Q_2$ is imported.
 - The equilibrium quantity is Q_1 .
 - The equilibrium price is P_1 .
- What happens to the surplus?



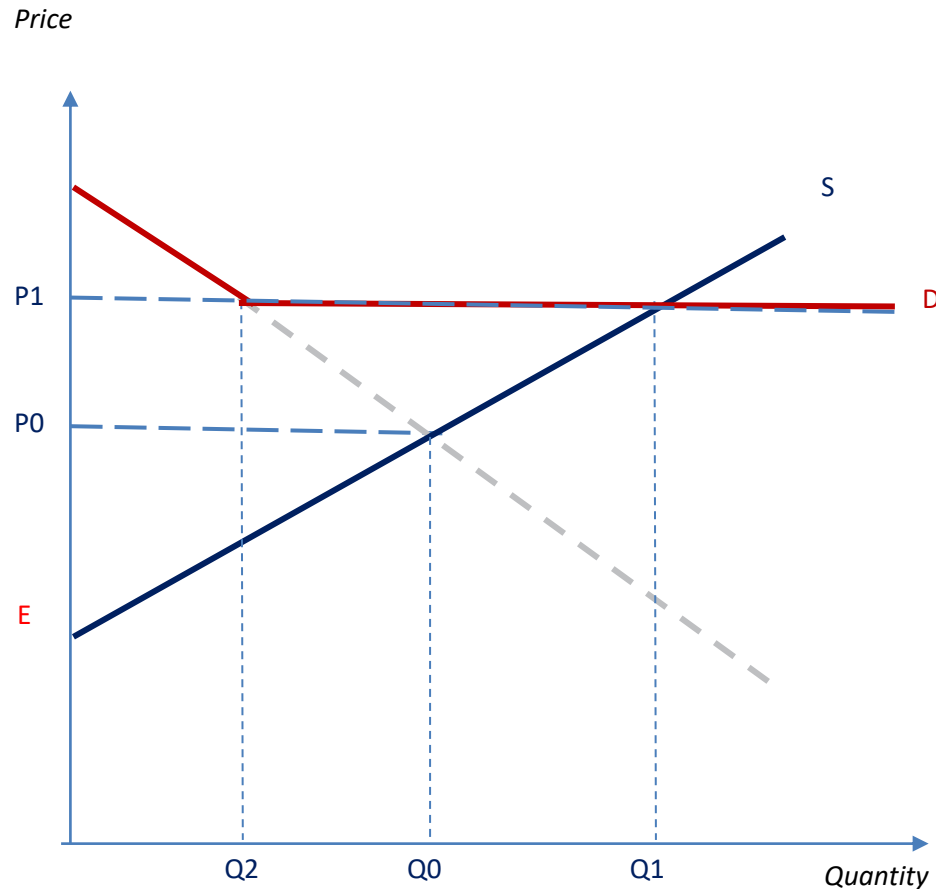
Equilibrium with free trade: Import

- Consumer surplus increases
- Local supplier's surplus decreases
 - If the supplier is able to reduce its MC (e.g. by technological improvement) he can get a higher surplus).
- Total surplus increased!
 - Kaldor-Hicks improvement



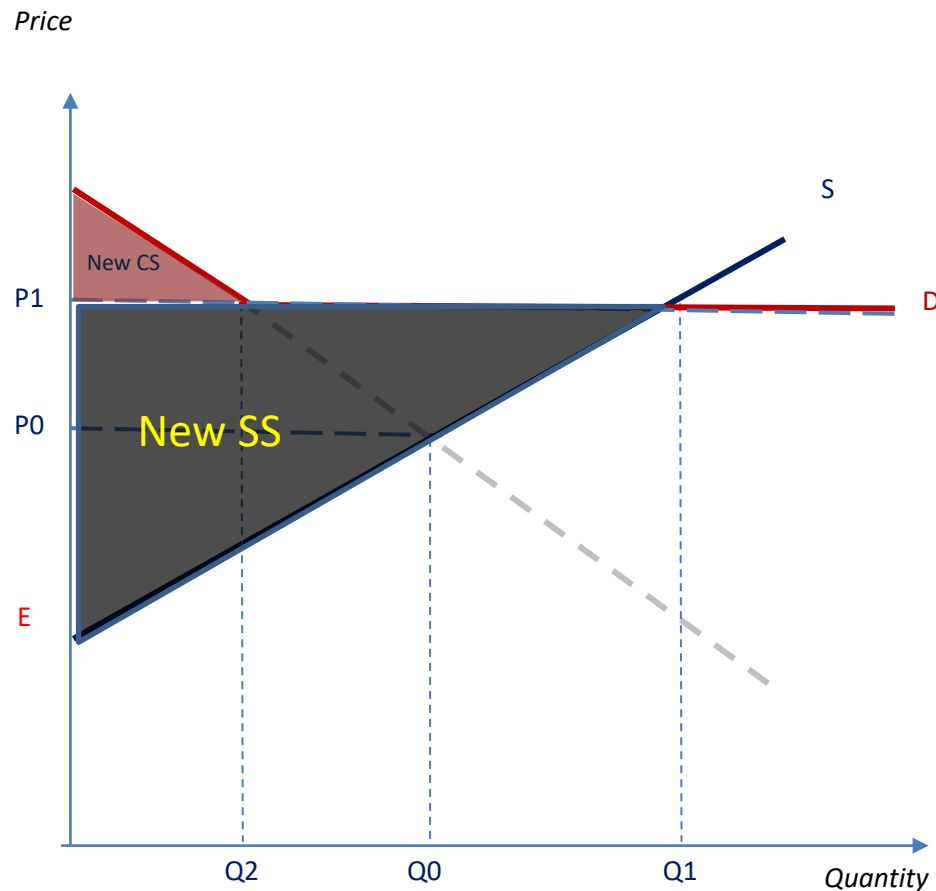
Equilibrium with free trade: Export

- Suppose the state can export at the world price, p_1 . What happens then?
 - The local customers buy Q_2 .
 - The quantity $Q_1 - Q_2$ is exported.
 - The equilibrium quantity is Q_1 .
 - The equilibrium price is P_1 .
- What happens to the surplus?



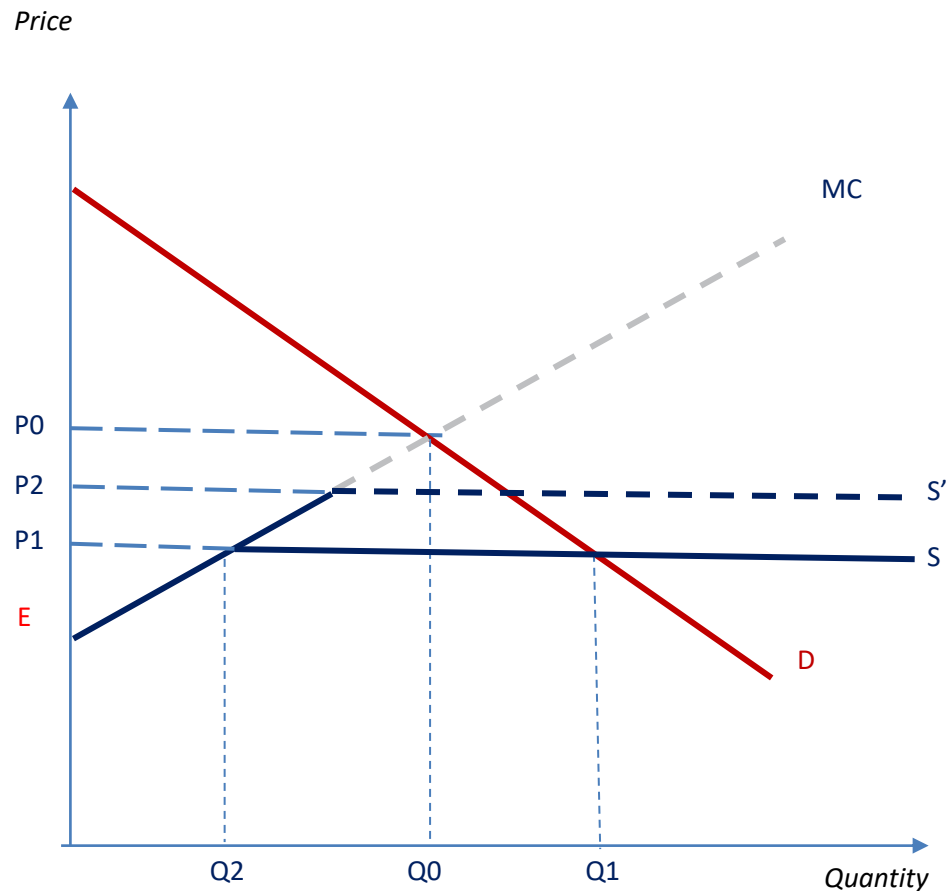
Equilibrium with free trade: Export

- Supplier's surplus increases
- Local consumers' surplus decreases
- Total surplus increased
 - Kaldor-Hicks improvement



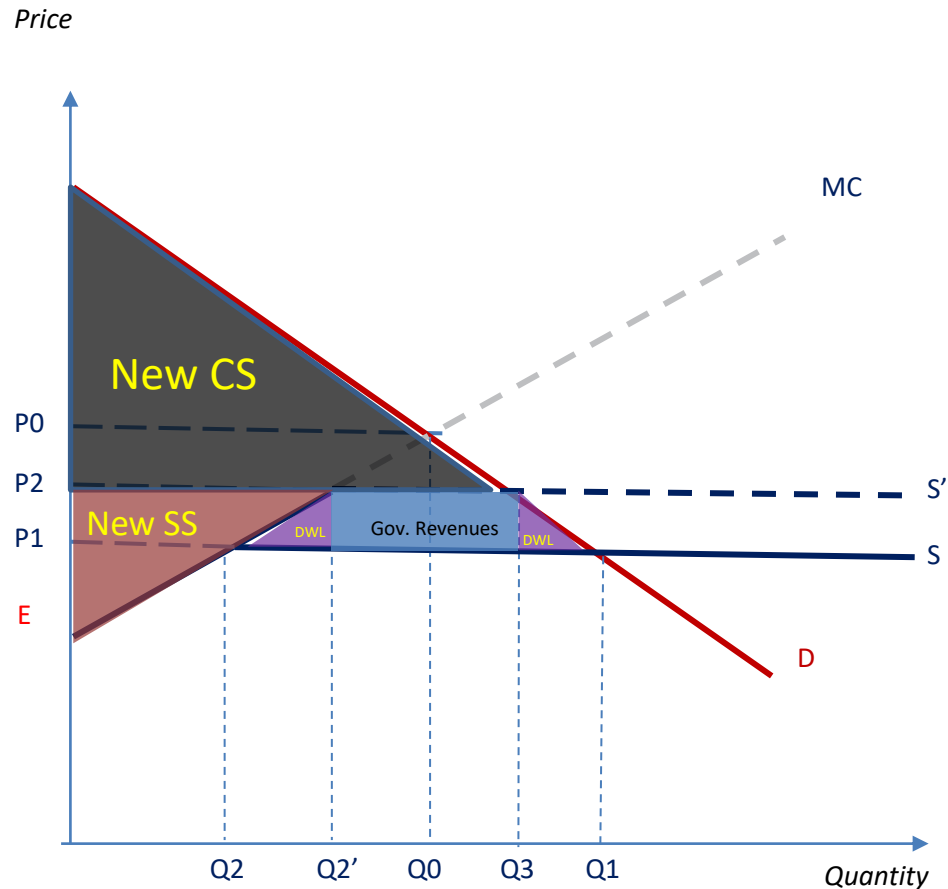
Governmental Intervention

- Suppose the government imposes a unit-tax for importing (=tariff). What happens?
 - Supply curve shifts up to s'
 - Price increases from p_1 to p_2 .
 - CS decreases
 - Local SS increases
- → Who does the tariff protect?
- → and what about the revenues from tax?



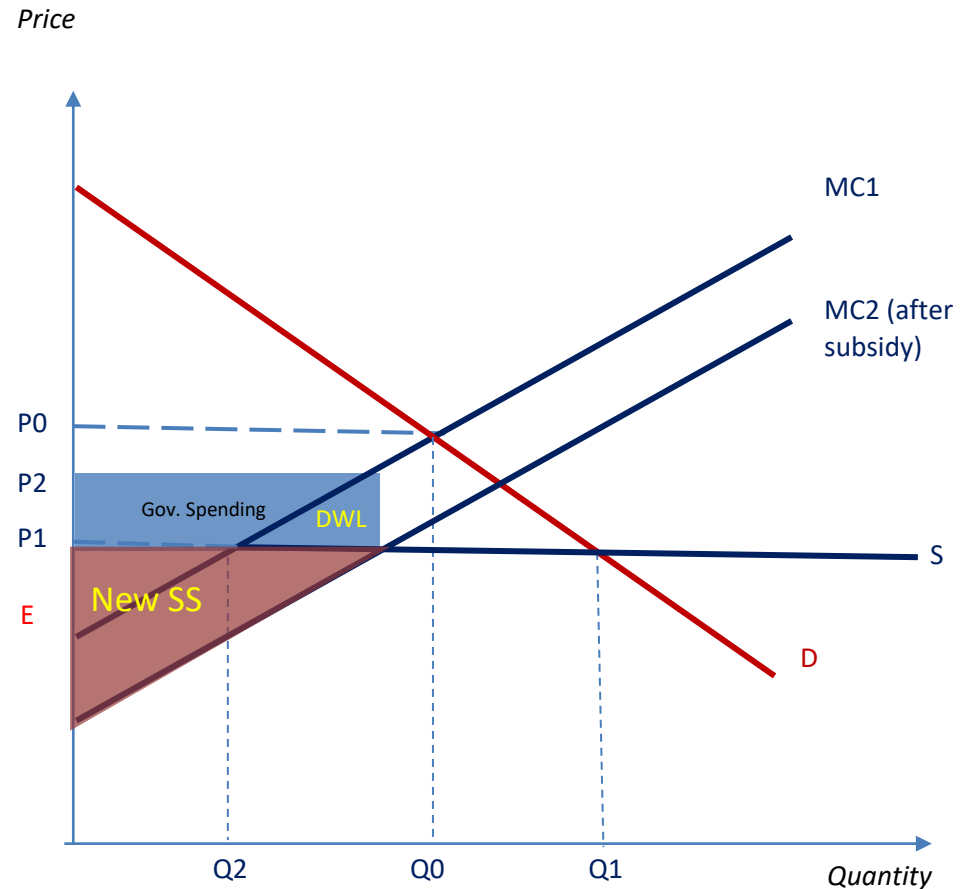
Governmental Intervention

- Suppose the government imposes a unit-tax for importing (=tariff). What happens?
 - Supply curve shifts up to s'
 - Price increases from p_1 to p_2 .
 - Local supplier produces more (Q_2' instead of Q_2)
 - Quantity sold decreases from Q_1 to Q_3
 - CS decreases (compared to import and no tariff)
 - Local SS increases
 - There are Governmental revenues
- → A tariff is an intervention that protects local suppliers
- But there is DWL due to the intervention!
 - As was the case in a single market w/o trade.



Governmental intervention (importing country)

- If the government gives a (small) subsidy instead of a tariff:
 - Suppliers gain
 - Equilibrium Quantity & CS don't change
 - Gov. spends money to subsidize inefficient producers = DWL.



International Trade: Game Theory

International Trade & Prisoner's dilemma

- Remember the prisoner's dilemma?
- What does prisoner's dilemma have to do with international trade???
- Suppose Robinson and Friday each must decide whether to impose a tariff on trade
 - The tariff protects the local (inefficient!) supplier and hurts the foreign supplier
 - For both local (=no deadweight loss) and overall efficiency (=let the supplier with the comparative advantage export) , it would be better if no tariffs were imposed.
 - BUT: both states are in a prisoner's-dilemma-like situation

Prisoner's dilemma – tariffs

	tariff	Free trade
tariff	2,2	5,1
Free Trade	1,5	4,4

- Germany & Switzerland can produce both “Labskaus” and Rum-Kokos Chocolate.
- If they have free trade, they get a benefit of 4.
- If one imposes a tariff, the profit is even larger
 - They protect the local supplier
 - The consumers can still buy the imported good
 - The local product will still be exported without limits, bringing in more income.
- But if both impose tariffs they just cause a DWL with no extra benefit from trade.
- → *Prisoner's Dilemma!*



Source: <https://cookidoo.at/recipes/recipe/de-AT/r45824>



Source: <https://www.amazon.co.uk/Casali-Rum-Kokos-Dragee-Classic-300/dp/B003WVP1NU>

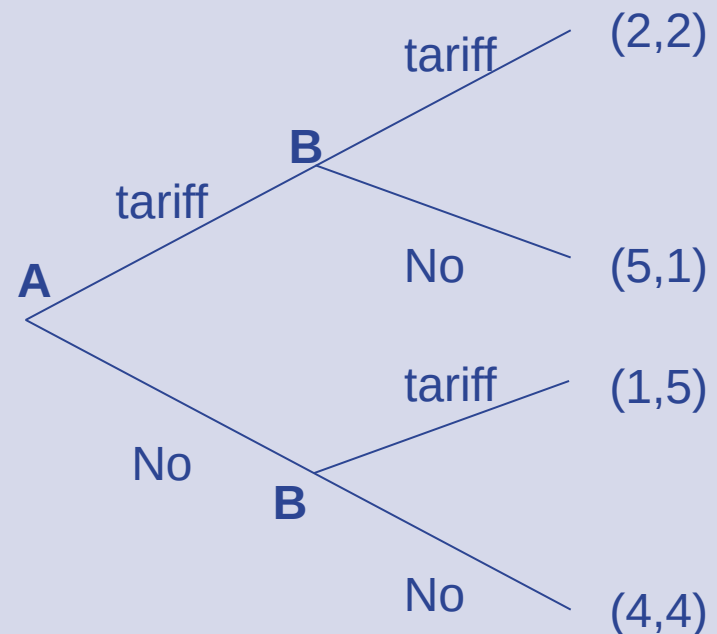
Prisoner's dilemma – tariffs

- Why do we see free trade nonetheless?
 - Governmental intervention: The countries sign a treaty which forces a legal restriction on tariffs
 - But how and why are these agreement stable?
 - Repeated Game – the prisoner's dilemma isn't played just once, it is *infinitely* repeated
 - Cooperative-Game theory: A different stream in game-theory, focusing on outcomes: if there is a surplus to be made from trade, the parties will realize this and agree on splitting the profits.

Dynamic (or repeated) Games

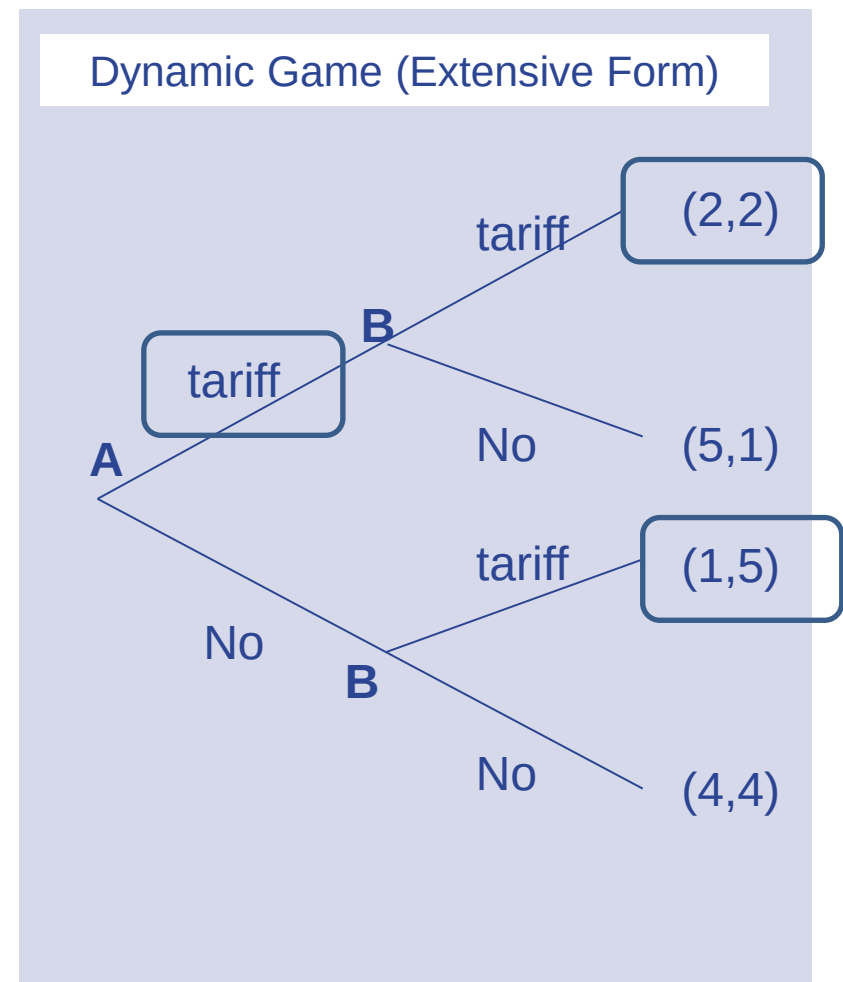
- Suppose the tariff game is played just twice (or any number of finite times). What is the equilibrium?
- In a dynamic game, we use *backward induction* and solve the result for each “subgame”
- Recall: In a one-shot game the equilibrium is that both charge a tariff.

Dynamic Game (Extensive Form)



Dynamic (or repeated) Games

- Solving backwards:
 - Last game is a one-shot game (both charge tariff)
 - So there is no reason to change the behavior in the first stage (nothing to be gained)
- The Subgame Perfect Nash Equilibrium (SPNE) is (“tariff, tariff”)
- Why does a repeated game matter at all then?



Dynamic (or repeated) Games

Discussion

Summary:

With finite horizon cooperation cannot emerge!

Result is not very convincing:

- If the time horizon is large, we would probably expect that countries behave cooperatively for some time and non-cooperatively towards the end. (This is observed in laboratory experiments.)
- Problem: last period effect.

Question:

Can the last period effect be avoided when allowing for infinitely many periods?

Grim Trigger Strategy

Cooperative strategy for country i , $i=1,2$.

In period $t=1$:

Start by charging *no tariff*

In periods $t=2,3,\dots$:

Keep charging no tariff as long as both countries have done so in all previous periods.



Cooperation phase

If one country has deviated, charge a tariff (=Nash equilibrium) forever.



Punishment phase

Strategy is called grim trigger strategy because deviation from cooperative behavior triggers punishment and countries stick to non-cooperative behavior forever. (Note that the punishment constitutes a Nash equilibrium.)

Warning: math ahead...



“Teach your kids to do Math
rather than do Meth”

(Roe Sarel)

www.shutterstock.com · 62644885

Cooperation

Profit with cooperation

We will now determine under which conditions, the grim trigger strategy of the previous slide constitutes a (subgame perfect) Nash equilibrium.

Since there is no terminal period, we cannot apply backward induction.

Way to proceed: Suppose country 2 follows the grim trigger strategy. What is the best response for country 1?

Payoff of country 1 when sticking to the grim trigger strategy is $\Pi^C = 4$ (the profit from cooperation, so that there's free trade, is 4)

$$\pi_1^C = \Pi^C(1 + \delta + \delta^2 + \delta^3 \dots) = \Pi^C * \frac{1}{1 - \delta} = \frac{4}{1 - \delta}$$

*Note: in parentheses is the sum of a geometric series

where δ is a discount factor. Usually: $\delta < 1$ (i.e. money now is better than money later)

Cooperation

Profit with defection

If country 1 defects (=charges tariff) at all, it's better to do so in period 1..

The profit from defection when country 2 *does not* defect is $\Pi^D = 5$.

The profit from defection when country 2 also charges a tariff (=punishing country 1) is $\Pi^P = 2$.

Profit from this strategy of no-cooperation is then

$$\pi_1^{NC} = \Pi^D + \Pi^P(\delta + \delta_1 + \dots) = \Pi^D + \Pi^P \frac{\delta}{1 - \delta} = 5 + 2 * \frac{\delta}{1 - \delta}$$

Cooperation

Profit with defection

Country 1 prefers cooperation to defection if $\pi_1^C > \pi_1^{NC}$, i.e. if $\frac{4}{1-\delta} > 5 + \frac{2\delta}{1-\delta}$
(Country 2 has the same preference, because we assume that they are symmetrical.)

Which gives us, in our example: $\frac{1}{3} \leq \delta \leq 1$

Intuition: The countries must care enough about the future (the discount factor is not too low) in order to fear the consequences of a “punishment”.

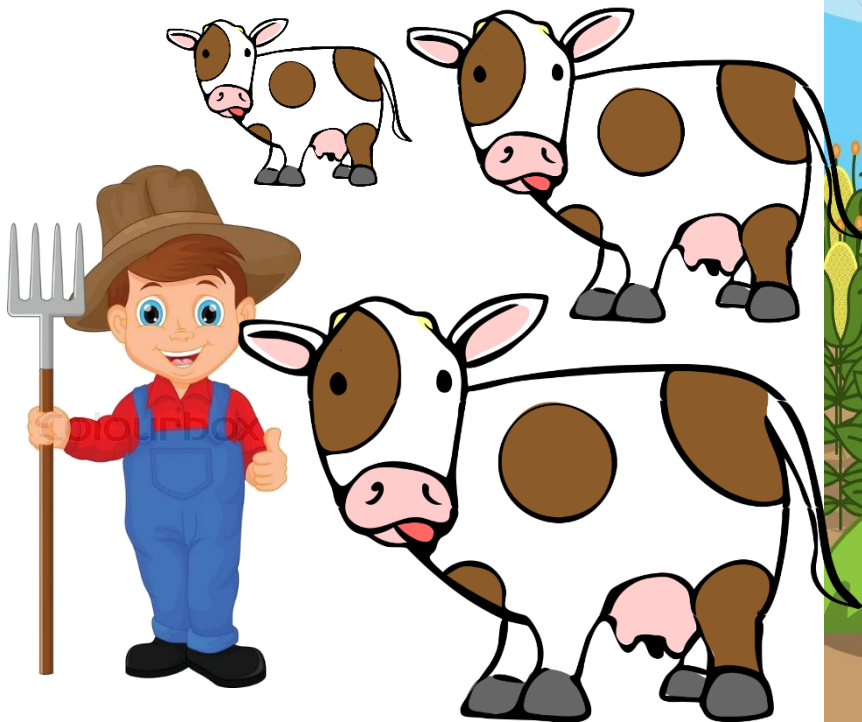
The general formula is: $\delta \geq \frac{\Pi^D - \Pi^C}{\Pi^D - \Pi^P}$

Cooperative Game Theory: Coase Theorem

- The type of cooperation considered thus far can emerge tacitly (no need for negotiations)
- But (cooperative) game theory has an alternative proposition to offer: If the countries understand there is a Pareto efficient solution, they will bargain and make a binding agreement.
- Question: Does it matter *for efficiency* what is the initial allocation before negotiations commence?
 - We've seen a negative answer to this when discussing the "second theorem of welfare economics".

Coase Theorem: Rancher vs. Farmer

Rancher

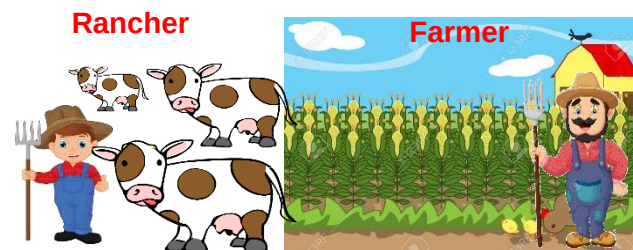


Farmer



Coase Theorem: Which rule to adopt?

- Should we protect the Farmer from the Rancher's cows? Who should bear the cost of building a fence?
- Two possible rules:
 - **“Rancher's rights”**: The farmer is responsible for keeping cows off his property. If they get in and eat his corn – it's his problem and he bears the damages.
 - **“Farmer's rights”**: The rancher is responsible for keeping cows on his property. If they get out, he must pay for the damages.
- Which rule should we adopt?



Coase Theorem: Numerical example

- Suppose that “invasion by cows” costs the farmer 100\$ in lost profits.
- Cost of installing a fence: 50\$ for farmer; 75\$ for rancher.
- Thus: we can avoid a damage of 100\$ at a cost of 50\$ or 75\$, depends who pays. Efficiency requires that the one with lower costs pays: i.e. the farmer.
- What would happen under each liability rule (“farmer’s rights” and “rancher’s rights”)?
 - **Recall: non-cooperative game theory:** Everyone just maximize their own profit.
 - Non-cooperation: farmer will refuse to pay under “farmer’s rights”.

Coase Theorem: Numerical example

- **COASE Theorem:** Parties negotiate and reach the efficient outcome.
- If the farmer is liable – he pays, because his benefit (100\$) exceeds the cost (50\$).
 - The efficient outcome.
- Suppose that the rancher is liable – what can he do?

RANCHER: “The law makes me responsible for any damage that my cattle do to your crops. There would be no damage if there was a fence. I can fence my ranch for \$75 per year, whereas you can fence your cornfields for \$50 per year. Let’s make a deal. I’ll pay you a bit more than \$50 per year to fence your cornfield.”

Farmer is “bribed” by the (liable) rancher to pay him between 50\$-75\$ and efficient outcome is still reached

Coase Theorem: Necessary Conditions

- What are the necessary conditions for this to work?
 - (1) **Property rights must be well-defined** – it must be clear who is liable under the law. Property rights must also be transferrable.
 - (2) **Transaction costs must be sufficiently low (preferably zero):**
 - Suppose that the farmer and rancher must hire a lawyer to draft up a contract between them, and the lawyer costs 10\$. The farmer can then offer to also pay for the lawyer (so that he pays 60\$ altogether, instead of 75\$). But if the lawyer costs too much, then an agreement is no longer feasible.
 - If transaction costs are zero, ('Coasean') bargaining will lead to a highest valued use of resources, regardless of the law (= the efficient outcome!)

Coase Theorem: Formally stated

Coase Theorem:

“When transaction costs are zero, an efficient use of resources results from private bargaining, regardless of the legal assignment of property rights.”

Transaction costs

- Back to international trade, there are different types of transaction costs:
 - **Search & Information costs** – costs incurred to find a trading partner.
 - **Bargaining costs** – costs incurred during negotiations (inc. figuring out the WTP of the other side) or for completing the transaction.
 - **Enforcement costs** – cost incurred to ensure compliance (e.g. monitoring costs)
- Law & Economics perspective:
 - The role of the law is to **lubricate** transactions by reducing transaction costs
 - ..and to **allocate** efficiently (replicate the efficient result, had transaction costs were zero) when lubrication is too costly.

Classifying goods

Public goods: (definition taken from *Ulen & Cooter, p. 40*)

- *Non-rivalrous consumption*: consumption of a public good by one person does not leave less for any other consumer.
- *Non-excludability*: the costs of excluding non-paying beneficiaries who consume the good are so high that no private profit-maximizing firm is willing to supply the good

	Non-rivalrous	Rivalrous
Non-excludable	Public goods (e.g. security)	<u>Common</u> goods (e.g. highways, nature)
Excludable	Club goods (e.g. netflix, gym)	Private goods (e.g. apartments)

Public goods: Free rider problem

- Adam, Bella, and Chris are economics students. They find out that their professor published a book from which he usually takes his exam questions.
- Each student values access to the book by (almost) 2\$. The price of the book is also 2\$. They agree to “buy the book together” where each must donate 1\$.
 - For simplicity, assume that the book store doesn’t give any change back.
 - If they reach 2\$ the book is bought. If not, each person gets her money back.
- Decisions are simultaneous and anonymous.
- If students behave strategically, what is the equilibrium of this game?

Public goods: Free rider problem

My decision / Others' decision	No one else donates	1 other donates	Both others donate
Don't donate	0 , 0, 0	0, 0, 0	2, 1, 1 (book is bought)
Donate	0 , 0, 0	1, 1, 2	1, 1, 1

If exactly one more person donates, I'm better off donating as well (I get a payoff of $1 > 0$)

But if two persons donate, I'm better off not donating (I still get the book but pay nothing).

If I believe the probability that others donate is sufficiently large, I don't donate.

And if everyone believes the same like me, no one donates.

Public goods: Free rider problem - generalization

- There is a group of n (> 2) individuals;
- Each individual i is endowed with income e , which can be either privately consumed or contributed to a group activity;
- All decisions are simultaneous and anonymous;
- If individual i contributes b_i ($0 \leq b_i \leq e$), his/her monetary payoff is:

$$\pi_i = e - b_i + \alpha \sum_{j=1}^n b_j$$

- The social welfare function is thus $SW = n * \pi_i$
- We want to check two things:
 - 1) When is it efficient that everyone donates?
 - 2) What are the incentives to donate?

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Public goods: Free rider problem - generalization

1) When is it efficient to donate?

- It's efficient to donate if donating increases social welfare, i.e. $\frac{\partial SW}{\partial b_i} > 0$.

$$SW = n\pi_i = n * \left(e - b_i + \alpha \sum_{j=1}^n b_j \right)$$

- As everyone is symmetrical, we can simplify to $SW = n(e - b_i + \alpha n b_i)$.

We get that $\frac{\partial SW}{\partial b_i} = n(-1 + \alpha n) > 0 \Leftrightarrow \alpha > \frac{1}{n}$

2) When is it profit-maximizing to donate?

- Slightly rewrite the profit function of the individual: $\pi_i = e - b_i + \alpha b_i + \alpha \sum_{j=-i}^n b_j$
- Donating is profit-maximizing only if $\frac{\partial \pi_i}{\partial b_i} > 0 \Leftrightarrow \alpha > 1$.

Note: there is no internal solution where the partial derivative is equal to zero

Public goods: Free rider problem - generalization

The thought exercise is as usual in game theory:
“Suppose everyone does X”, is it profitable for me to unilaterally deviate and do something else.

The free rider problem arises if $1 > \alpha > \frac{1}{n}$
due to $\alpha < 1$, i 's dominant strategy is $b_i = 0$;
due to $\alpha > 1/n$, efficiency requires $b_i = e$ for all i .

What should be privately owned?

- Because of the free-rider problem, **public goods should be publically owned.**
- But **private goods should be privately owned**
 - Trade ensures that the party who values the product the most eventually gets the product (Coasean bargaining).
 - Importance of the right to exclude: people can sell their property because they aren't afraid it will be stolen/destroyed by others.
 - This also gives an incentive to invest, in order to increase the value of the property.
- What happens if the government owns private goods?
 - Fear of misallocation - consumption by someone who doesn't value it the most (e.g. political economy – see next topic).
 - High transaction costs: allocation requires elaborate public decision making mechanisms.

Tragedy of the Commons

Overuse of common goods: "a fountain in the desert"



Which type of product is this?

Efficient use = as long as the total demand of all users is less than the total amount of water (**simplified*)

Tragedy of the Commons

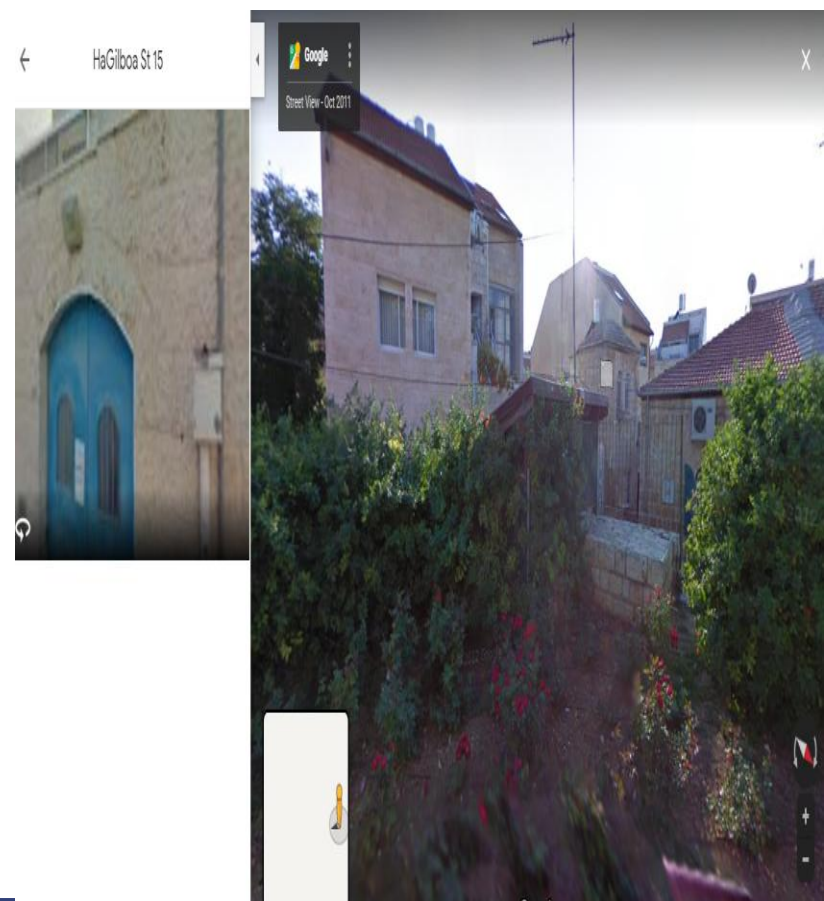


- The problem:
 - Individuals don't internalize the cost to others from over-consumption of the good.
 - Non-excludible: individual's can't exclude others from consuming.
- Possible Solutions:
 - Private right: give control to one individual, who will then have enough incentive to invest in preservation and exclude others.
 - → Will bargaining then be efficient? TC of one individual bargaining with many others.
 - Public right: central planner (or collective decision making mechanism) will be in charge of efficiently restricting access to the common good

Tragedy of the Anti-Commons

- The problem:
 - Fragmented / Overlapping property rights (“too much ownership”)
 - Individuals don’t internalize the cost to others from under-consumption of the good.
 - Full excludability – too many people have a veto right and can exclude.
- Possible Solutions:
 - Private right: give control to one individual, who isn’t bounded by other’s veto.
 - → Will bargaining then be efficient?
 - TC of one individual bargaining with many others.
 - But also – what if people veto because they have a preference against consumption? (e.g. nature reserve)

Real-life example



Public or private? summary

- Private ownership:
 - Assigns the resource to a person who owns it and can control access.
 - The owner bears the cost of exclusion and maintenance
 - Works well when externalities impact only a few people (so that TC are low)
- Public ownership: three options:
 - Open access – anyone can use, nothing spent on boundaries / exclusion. Works well when resource is uncongested (whereas congestion causes overuse = tragedy of the commons).
 - Political control – lawmakers set rules that limit access (e.g. via majority vote)
 - Unanimity – no access unless everyone agrees (veto rights). May cause underuse – either tragic (tragedy of anti-commons) or desirable (when individuals have strong preferences against consumption).

Part 1: Principles of Economics

1. Basics
2. Necessary
Conditions for
Functioning Markets
3. Market Failure &
Market intervention
4. International Trade
5. Political Economy



Rent-seeking

Political Economy

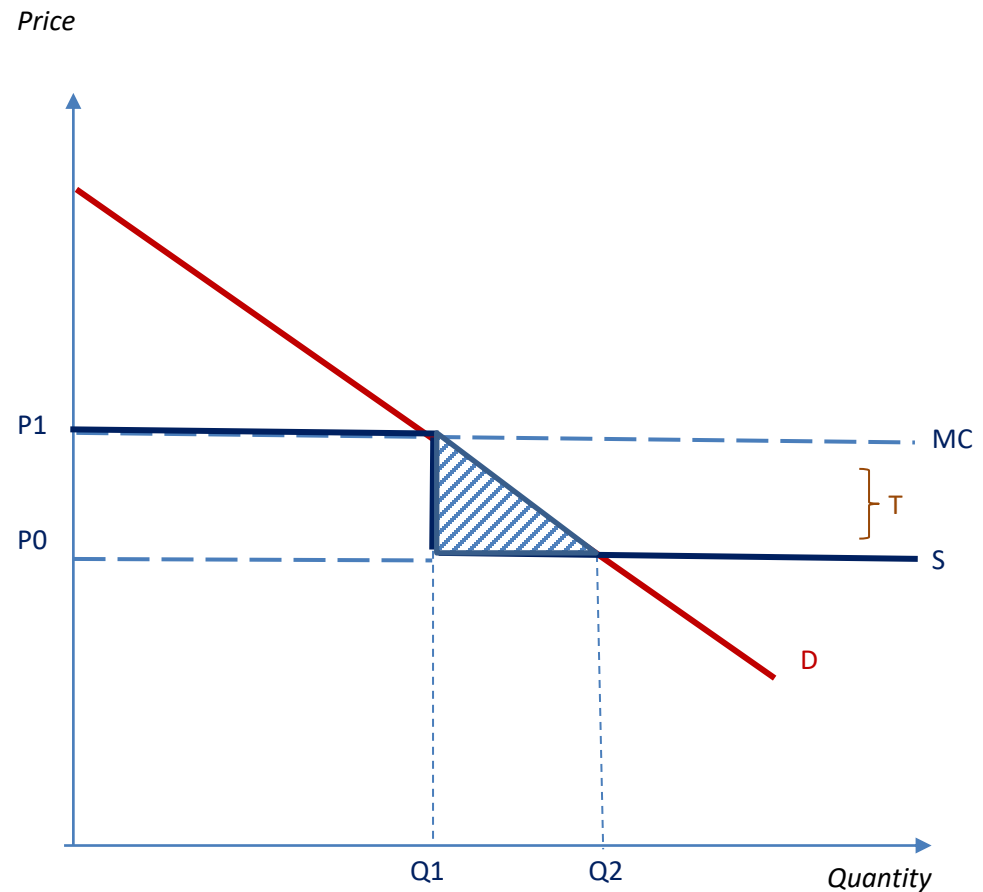
- We've seen that with Kaldor-Hicks efficiency there are winners and losers.
- Recall that KH efficiency is based on the idea that the winners *could in theory* compensate the losers.
- The question is whether *they actually do compensate* the losers?
- In other words: the problem is “rent-seeking”

Rent seeking

- The concept of “economic rents” dates back to Adam Smith: Entities earn money from three sources: Wages, Profits, and “rents”
- Economic rent = payment that exceeds the costs involved in the associated resource.
 - E.g. land owners charge a monopoly price on their own land, which is not proportional to the costs of the landlord.
- The identification of “rent-seeking” is often attributed to a 1967 paper by Gordon Tullock (see next slides), but gained popularity through later work of Anne Krueger (1974).

Rent-seeking (Tullock 1967)

- Local Supplier has constant MC
- Product can be imported at price p_0
- Government imposes a tariff $T = p_1 - p_0$
- Dashed triangle is the “usual” DWL.
 - But is it the whole social cost?

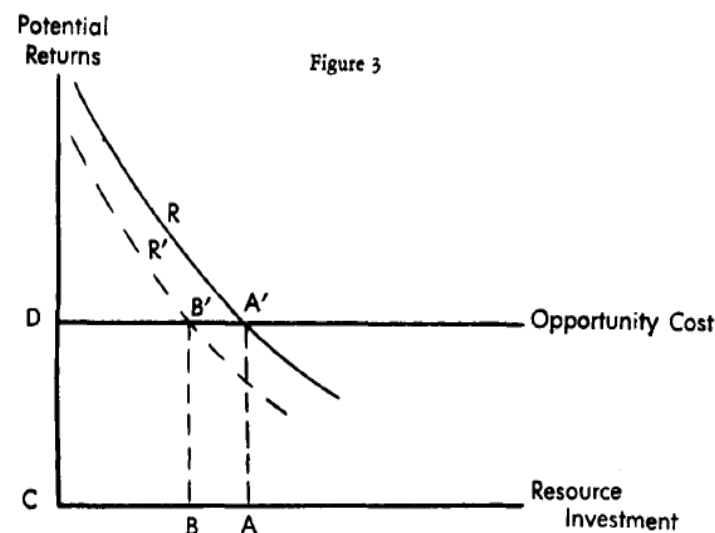


Rent-seeking (Tullock 1967)

- The diagram ignores...
 - The cost of collecting the tax (e.g. expenditure on tax collectors)
 - Salary of coast guards who prevent smuggling (e.g. fighting tax evasion)
 - Cost of middlemen hired by companies to manage their taxes
 - How the taxes are used post-collection (maybe politicians transfer them to their friends)
 - The fact that the government subsidizes inefficient production (we've pointed this out before)
 - The government usually imposes a tax only under pressure from lobbyists (whose salary is a cost).
- → These might be larger than the triangle!

Rent-seeking (Tullock 1967)

- Consider Theft: Why are thefts problematic? Isn't it just a transfer from V (the victim) to C (the criminal)?
 - X-axis: C can invest resources to commit the crime (e.g. buy burglary equipment)
 - Y-axis: returns to crime
 - D-line: opportunity costs
 - Equilibrium level of theft: When marginal return = marginal cost
 - Dashed line: what happens when the returns to crime decrease (e.g. because victims buy locks)
- Social cost = equilibrium resource investments.



*“The problem with income transfers is not that they directly inflict welfare losses, but that they lead people to employ resources in attempting to obtain or prevent such transfers.”
(Tullock 1967, p. 8)*

Rent-seeking (Tullock 1967)

- Tullock makes the argument that *monopolies* have the same problem: when a monopoly can extract a large surplus from consumer, there is:
 - A large incentive to invest to become a monopoly
 - A large incentive to fight monopolies
 - Associated costs would be a social waste
- Note: The costs might be spread out (e.g. across firms that compete to become a monopoly)

Rent seeking (Krueger 1974)

- Rent-seeking can occur also in a competitive market, where participants compete for economic rents.
- Example: suppose governments regulate import by requiring a license.
 - Importers then compete for the scarce licenses and waste resources along the way.
 - For example, if licenses are given in proportion to production capacity, firms might inefficiently expand capacity just to get a license.
 - ..or take illegal actions such as bribery

Some implications of Rent-seeking (Krueger 1974)

- Kruger suggests that with rent seeking:
 - It might be better to prohibiting import altogether rather than restricting it through licenses
 - Because licenses generate a rent-seeking competition.
 - It might be better to have a monopoly rather than licenses
 - Because less resources are wasted chasing a license.

Public Choice

Public Choice

- Rent-seeking plays a role also in political decision making.
- The Economic Theory of Politics – or Public Choice – began to emerge in the late 1950s
- Since its roots were in the U.S. and the U.K., public choice scholars were primarily interested in understanding the functioning of democracies
- Today, Public Choice has its own journals and thousands of papers on pertinent issues have been published
- Having some intuition regarding the incentives that drive politicians is important to understand their decision to implement – or not – specific kinds of regulation

Public Choice: Median Voter Theorem

- Assumptions:
 - Unidimensional space
 - Voting population is equally distributed along the space
 - Voter's decision to vote will depend on how far it is from the candidate
 - All voters vote
- Equilibrium: Where do politicians position themselves?



https://www.youtube.com/watch?v=cFt0k6n_HKc

Public Choice: the median voter theorem

- Example: Israel's election in 2021:

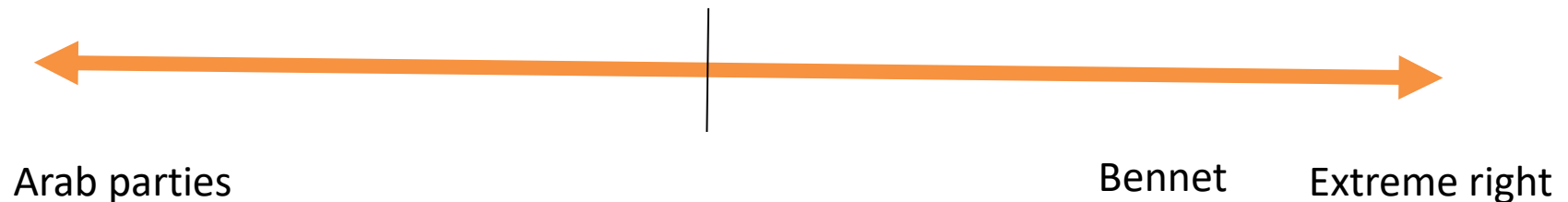
Israel is divided on many topics, but the customary division is between right and left.

In 2021, there was a fourth election (in a very short time frame).

Naftali Bennet had only 6 parliament members (out of 120) and yet managed to become the prime minister – can the median voter theorem explain this?

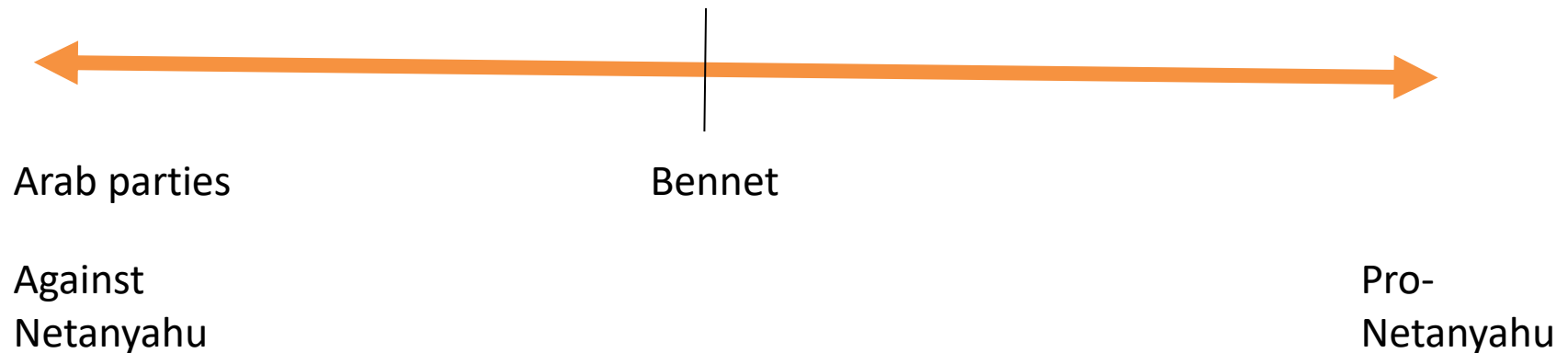
PUBLIC CHOICE: THE MEDIAN VOTER THEOREM

- Bennet was more or less located to the far right, but the elections had a different unimodal theme..



PUBLIC CHOICE: THE MEDIAN VOTER THEOREM

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PUBLIC CHOICE: THE MEDIAN VOTER THEOREM

- Is the median voter theorem about rent-seeking?
- The problem starts if someone adjusts their policy just to get elected – a pessimist view of politicians

Beyond the median voter theorem

- Median voter theorem places emphasis on the *demand side* for politics.
- But the supply side include bureaucrats that have more complex utility functions, e.g. they want..
 - To be re-elected
 - A larger office budget
 - Higher income
 - Reputation
 - To relax rather than work
 - Getting jobs for their friends & relatives (=corruption)
- There are many models of bureaucratic behaviour, mostly predicting that rational behaviour by the bureaucrats would lead to inefficiency (buros that are too large, insufficient control, low accountability etc.).

Public choice

- Central questions of the economic theory of bureaucracy:
 - *When is social welfare (including preferences of voters and politicians) neglected in favor of bureaucrats' preferences?*
 - How much influence do bureaucrats exert on the bundle of public goods provided? On the supplied quantities? On the amount of resources spent for their provision?
 - *What possibilities do politicians have to monitor the behavior of bureaucrats?*
 - Under what conditions is ex ante monitoring (~ constitutional level) more likely
 - Under what conditions is ex post monitoring (~ post-constitutional level) more likely?

Economic theory of bureaucracy: approaches

- 1) Bureaucrats only care about maximizing their budget, and dominate the legislature (the Niskanen model)
- 2) Regulators are “selling” regulation, which firms are “buying” (Stigler – Economic Theory of Regulation; Peltzman with extensions), but political power is not infinite (e.g. because of procedural safeguards like slow bureaucracy).
- 3) Legislature dominates the bureaucracy (Weingast) – it builds committees that implement the new law; can monitor the bureaucrats.
- 4) Endogenous administrative structure - Legislature only delegates for credit-claiming or blame-shifting.

Economic theory of bureaucracy: example

- What do judges maximize?
 - Posner (1973): The same thing everyone else does!
- Public choice says we need to treat judges as economic agents.
- What might be included in an (appellant) judge's utility function?
 - Popularity – they want to be liked by litigants and peers.
 - Prestige – they want to feel superior
 - Avoiding reversals / Reputation
 - Public interest – they assign some weight to the public interest
- Other factors:
 - Effort cost
 - Ideology
- If we want to get judges to behave in line with social welfare, what should we do?

Public choice: Example (Posner, 1973)

- Public choice theory says we need to approach the problem in the same way we approach e.g. the free-rider problem: We need to check (1) what is the social welfare, (2) how the judge's utility function differs from it, and (3) what can we change to align the judge's interest and social welfare.
- For example:
 - We could set up an accurate appeal system, which causes the judge to be more careful, as this reduces the chance of reversal.
 - We could measure effort, then reward hard work and punish shirking
 - But It's hard to find a good proxy. E.g. what happens if we reward fast-decision making?
 - We could pay the judge a monetary bonus when he gets it right – but is that a good idea?
- The same solution might work in some places, but backfire in other places (see next slide)

Public choice: Example (Feess & Sarel, 2018)

- To simplify, suppose the judge only cares about three things: (1) reputation, (2) getting it right, and (3) effort costs.
- If the effort costs are too high, the judge might shirk (even though he does care about getting it right).
- The government decides to construct an appeal system. Clearly, this can enhance reputation concerns – leading to more effort, if effort reduces the probability of being reversed.
- But how does it affect the concern of getting it right?

Public choice: Example (Feess & Sarel, 2018)

- But how does it affect the concern of getting it right?
 - Common law systems: The appeal court's accuracy depends on the effort at trial (appellate judges typically cannot review facts) → This will cause the trial judge to work harder, to “help” the appeal court.
 - Civil law systems: the appeal court is free to review the case from scratch (i.e. depends less on what the first court did) → a free-riding incentive for the trial judge: if he gets it wrong, his superior can fix it.
- This is a typical example of a public choice issue:
 - The public agent is assumed to be a selfish utility-maximizer and not a “benevolent regulator”
 - Sometimes, the agent has “bad” incentives, because he does not fully internalized the social welfare.
 - We can try to correct these incentives through legal intervention
 - But we need to make sure the intervention doesn't backfire.

Public choice: Practice

Can you think of examples of a public choice issue in the context of Covid-19 policies?

- (1) What is the social welfare?
- (2) What is the utility-function of the public agent?
- (3) How can we intervene to make sure the utility function overlaps with social welfare?
- (4) What can go wrong (or right) when we intervene?

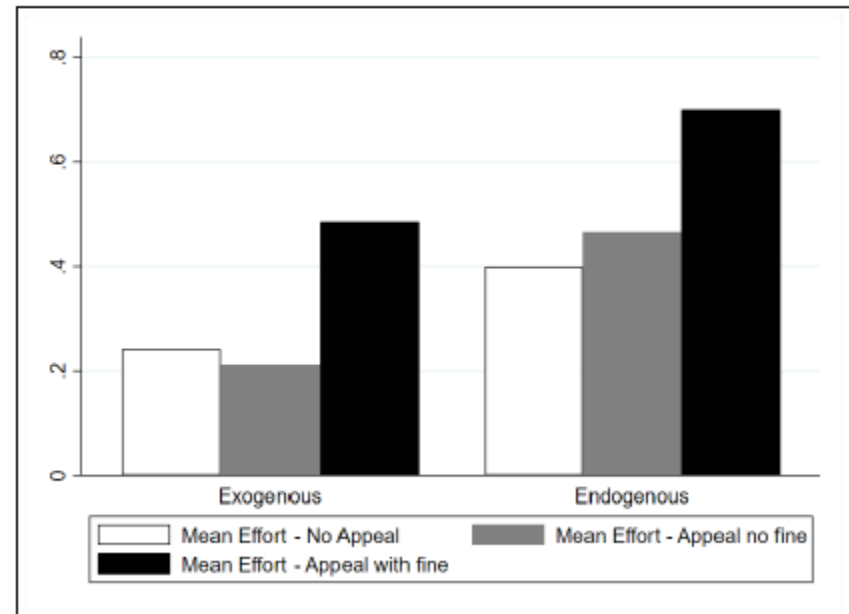
Public choice: How to test?

- In many cases, it can be challenging to check whether public officials really deviate from the social interest:
 - Difficulty of calculating what the social interest is
 - Difficulty of estimating the utility functions
 - Lack of control group: if we see a public official X doing Y, we never observe him doing anything else at the same time – what should we compare it to?
- Solutions:
 - 1) Run an experiment – with either actual public officials (difficult to get) or students
 - 2) Analyze data using rigorous econometrics, applying “identification strategies”

Public choice: How to test?

- Experiment: Example (Feess & Sarel 2018)
 - Students in the lab are randomly assigned to an “appeal system” – a computer that either reverses or affirms their decisions.
 - The appeal’s accuracy is either affected by the subjects (“endogenous”) or not (“exogenous”)
 - The decisions affect real people (other subjects who may or may not have taken money designated to a charity)
 - Reputation is proxied by payment – if they are reversed, they lose money.

Figure 1: Mean effort - full sample

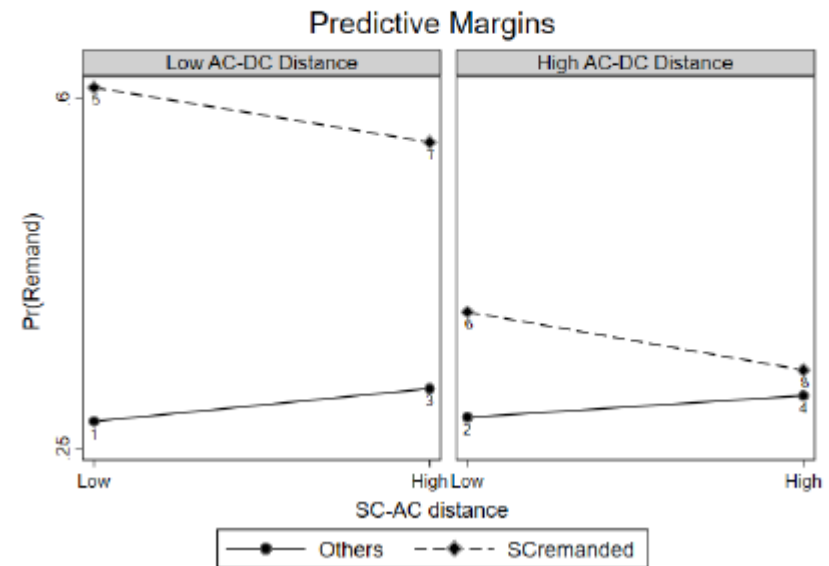


Public choice: How to test?

- Empirical (Sarel & Demirtas, 2021)
 - Suppose you want to test if ideology affects the probability that an appellate judge sends a case back to the lower court
 - Ideology is measured via “scores” based on who appointed the judge / which decisions the judge made.
 - Then, take the distance in scores to those of judges in lower or upper courts

Figure 6: Marginal effects

(a) Unweighted



POLITICAL BARGAINING

POLITICAL BARGAINING

Politicians tend to bargain with each other when creating law: “ I will vote for A if you vote for B”

Example (Cooter & Gilbert 2022*, p. 12): three legislators bargain on how to spend money on three projects (A,B,C).

A prefers (100, 0, 0)

B prefers (0, 100, 0)

C prefers (0, 0, 100)

A majority is needed for a deal to happen. Suppose the first proposal is an equal spend on all projects (33, 33, 33). Is this a Nash equilibrium?

*Cooter, Robert, and Michael Gilbert. *Public Law and Economics*. Oxford University Press, 2022, <https://library.oapen.org/handle/20.500.12657/59253>

POLITICAL BARGAINING

A is offered to get 33% for his project, But then he thinks to himself: “wait, maybe I should offer to split everything 50:50 with B (or C, doesn’t matter), we are both better off then!”

B likes that. But then B thinks: “Wait a minute... I could offer C, say, 25% for his project and then I keep even more!”

C likes that, but then.... He thinks: “WAIT!!!! I could offer A to give him 25%, and cut-out B”

And this goes on and on in circles..

The problem here is that this is a purely distributive game, with an inefficient institutional design (simple majority). Can we solve this by redesigning the bargaining mechanism?

POLITICAL BARGAINING

The Pirate game

Five pirates find a treasure of 100 coins. They bargain on who gets what with the following rules:

- 1) The most senior pirate makes an offer.
- 2) The offer is accepted if a majority votes in favor
- 3) If there is a tie, the most senior pirate gets an extra vote
- 4) If the offer is rejected, the most senior pirate is thrown overboard and the next in seniority makes an offer

What is the Nash Equilibrium of this game?

(this will be solved in-class on the board, in order not to spoil it)

POLITICAL BARGAINING

It is easy to see how one can readjust politicians' voting mechanism to reflect a pirate game:

- the politicians have some ranking (seniority)
- Each get one chance to make an offer
- If the offer fails, they are out of the negotiations (figuratively, thrown overboard)

So now we have a stable solution – but is it an efficient one? That depends – does the most senior pirate maximize efficiency?

POLITICAL BARGAINING

Example: The Dark Knight Rises Opening scene



POLITICAL BARGAINING

A different set of bargaining games concern “production” rather than “distribution”, e.g. consider the game of whether to drive on the left lane or the right lane in a highway.

	Left	Right
Left	(1, 1)	(0, 0)
Right	(0,0)	(1,1)

Here parties just want to be coordinated – produce a better outcome for everyone.

POLITICAL BARGAINING: MIXED BARGAINING

Most bargaining games are “mixed”, in the sense that there is both production and distribution: the parties can produce if they agree on distribution.

Example (Cooter & Gilbert 2022, p. 14):

- Adam is the warden of a state prison. He values having some empty cells at 3000
- Blair works in a federal prison which is overcrowded. She prefers sending the prisoners to a state prison and pay a “jail day” fee.
- There is scope for negotiations because Blair’s WTP > Adam’s WTA.
- They will agree on a fee between 3000-4000.

The production here is making sure there are prisoners inside the prisons (higher welfare), but it is only achieved if they agree on distribution.

POLITICAL BARGAINING IN VOTES

In politics, the currency is usually not money but rather votes.

Example (Cooter & Gilbert 2022, p. 17): C and D bargain over votes, where C wants to invest in schools whereas D wants to invest in the police.

C proposes raising taxes by 100K and spending them on schools. D proposes to raise taxes by 50K and hire more police. If both get what they want, taxes increase by 150K.

Even if they can't hire "half a policeman" (i.e. the choices are "lumpy"), they should agree to the unequal split (C gets more of the pie) if TC are low.

This split will entail assisting each other by giving a vote in favor of the other person's agenda.

The same point can be made on the state-level (**recall our example of tariffs & the Coase theorem**)

THE PUBLIC COASE THEOREM

“The Public Coase Theorem asserts that as the transaction costs of bargaining among lawmakers approach zero, they will cooperate with each other and allocate public entitlements to the lawmakers who value them the most”

(Cooter & Gilbert 2022, p. 22)

THE PUBLIC COASE THEOREM

The Public Coase Theorem is a positive prediction, but it has a normative dimension.

Slightly restated (Cooter & Gilbert 2022, p. 27): **as TC of political bargaining approach zero, laws will become socially efficient.**

Example: C and D are facing an opposition from G, they will “buy him out” if it is efficient to do so.

THE PUBLIC COASE THEOREM

Why does public bargaining fail sometimes?

Market failures (market power, asymmetric information, externalities, behavioral)

Is the Public Coase Theorem consistent with our previous discussion of rent-seeking? Discuss!

BEHAVIORAL ECONOMICS

WHAT IS BEHAVIORAL ECONOMICS?

- Traditional economics relies on somewhat strong assumptions, in particular – Homo(o)economicus.
- How do we know if these assumptions make sense?
- → Run experiments, or collect quasi-experimental evidence
- → look at the data!
- → check if it's consistent with the assumptions
- → if it is not – revise the assumptions? Or provide a good reason for why the assumptions are ok.

WHAT IS BEHAVIORAL ECONOMICS?

- Historically:
 - First, acceptance of 'bounded rationality' (e.g. lack resources to fully calculate payoffs)
 - Second, experiments challenging rationality and selfishness become mainstream
- We will consider several examples
 - Ultimatum game
 - Pen & Mug experiment
 - Biases & Heuristics
 - Prospect theory
- And then connect them back to L&E



Daniel Kahneman
Nobel prize winner 2002



Amos Tversky
(died before Nobel prize)



Esther Duflo, Abhijit Banerjee,
and Michael Kremer.
Nobel prize winners 2019



Richard Thaler
Nobel prize winner 2017

EXAMPLE: THE ULTIMATUM GAME EXPERIMENT

- Subjects are randomly divided into pairs and must decide how to divide a good (or sum of money) between them.
- One subject (“Proposer”) proposes a distribution to the other subject (“Responder”).
- If the responder says yes, the division is realized. But if the responder says no, they both receive zero.

EXAMPLE: THE ULTIMATUM GAME EXPERIMENT

Before we reveal the outcome – what would a Homo(o)economics do?

	Response s	Offers							
		1	2	3	4	5	6	7	8
Payoffs (Responder, Proposer)	Accept	(1,7)	(2,6)	(3,5)	(4,4)	(5,3)	(6,2)	(7,1)	(8,0)
	Reject	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0

- Equilibrium of the game is that the proposers offers the minimum, and the responder accepts (because 1 is better than 0).
- Is this what happens in “real life”?

EXAMPLE: THE ULTIMATUM GAME EXPERIMENT

- Evidence show that responders reject small offers
-and that proposers often offer a 50:50 split.
- Possible reasons for rejections?
 - Responders want to punish proposers
 - → Compare with “impunity game” – people still reject.
 - Responders want to signal their anger to the other side
 - → Compare with UG + Communication
- Possible reasons for higher offers?
 - Strategic: proposers anticipate rejection (compare to Dictator Game)
 - Social concerns (altruism): proposers care about others
 - Inequality aversions: proposers dislike unequal payoffs

EXAMPLE: THE ULTIMATUM GAME EXPERIMENT

- Suppose we apply the insights of the ultimatum game to the Pirate game (remember that game?)
- The senior pirate can no longer bribe the junior pirates with a small amount (as they will reject) nor do we expect him to offer them so little.
- This is an example of how behavioral insights should cause us to adjust our predictions.

EXAMPLE: PEN & MUG

- Subjects are each given either a mug or a pen.
 - No reason to assume these have a different value
- They must answer for how much they are willing to sell their own product, and for how much they are willing to buy the other product.
- What do you think happened?

EXAMPLE: PEN & MUG

- **“The endowment effect”**: people value products *more* if they own them, so there is a gap between WTP and WTA.
- Possible reasons (partial list..)
 - Mere-possession: you value it more because you own it.
 - Mere-touch: you value it more because you touched it.
 - Expectation: you value it more because you expect to keep it.
- Most common explanation is „loss aversion“: people hate losing 1 euro more than they enjoy earning 1 euro. And Selling is perceived as losing the product.
 - We will discuss this a bit more later

EXAMPLE 3: COMMON HEURISTICS (1/3)

- Which of these is more likely to kill you?



EXAMPLE 3: COMMON HEURISTICS (1/3)

- Which of these is more likely to kill you?



13 people per year



450 people per year



150 people per year

Were you correct? What was your guess based on?

EXAMPLE 3: COMMON HEURISTICS (2/3)

“Linda is thirty-one years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in antinuclear demonstrations.” (Khaneman, 2011)

What do you think is Linda doing today?

1. Linda is a teacher in elementary school.
2. Linda works in a bookstore and takes yoga classes.
3. Linda is active in the feminist movement.
4. Linda is a psychiatric social worker.
5. Linda is a member of the League of Women Voters.
6. Linda is a bank teller.
7. Linda is an insurance salesperson.
8. Linda is a bank teller and is active in the feminist movement

EXAMPLE 3: COMMON HEURISTICS (2/3)

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EXAMPLE 4: AFRICAN STATES

- I've chosen a number between 0-100. Please guess what it is and write it down on a piece of paper.

EXAMPLE 4: AFRICAN STATES

- Now, I would like you to guess something else: how many African countries are members of the UN Human Rights Council?
- Please write down your guess next to your guess for the first question.

EXAMPLE 5: COGNITIVE REFLECTION TASK

- You have 5 minutes to answer the following three questions:
 - (1) A bat and a ball cost together 1.10\$. The bat costs one dollar more than the ball. How much does the ball cost?
 - (2) Five machines need 5 minutes to produce 5 parts. How long do 100 machines need to produce 100 parts?
 - (3) There are waterlilies on a lake. The space they occupy doubles itself every day. If it takes 48 days for the lillies to cover the whole lake, how long does it take until they cover one half of the lake?

COGNITIVE REFLECTION TEST

CONTRASTING WITH OTHER JUDGES AND STUDENTS (MEAN)

- Task designed to measure a person's tendency to override an incorrect intuitive response and engage in further reflection to find a correct answer; correlates highly with various measures of mental heuristics

- MIT Students 2.18 (n=61)
- St. Gallen L&E Students 2.15 (n=32)
- German Administrative Law Judges 2.02 (n=86)
- German Non Law Students 1.87 (n=121)
- German Law Students 1.80 (n=110)
- ICCA Arbitrators 1.47 (n=239)
- US Trial Judges 1.23 (n=252)
- Florida Judges 1.23 (n=192)
- University of Michigan Students 1.18 (n=1267)

SYSTEMS 1 AND 2



SYSTEM 1

Intuition & instinct

95%

Unconscious
Fast
Associative
Automatic pilot

SYSTEM 2

Rational thinking

5%

Takes effort
Slow
Logical
Lazy
Indecisive



Source: Daniel Kahneman

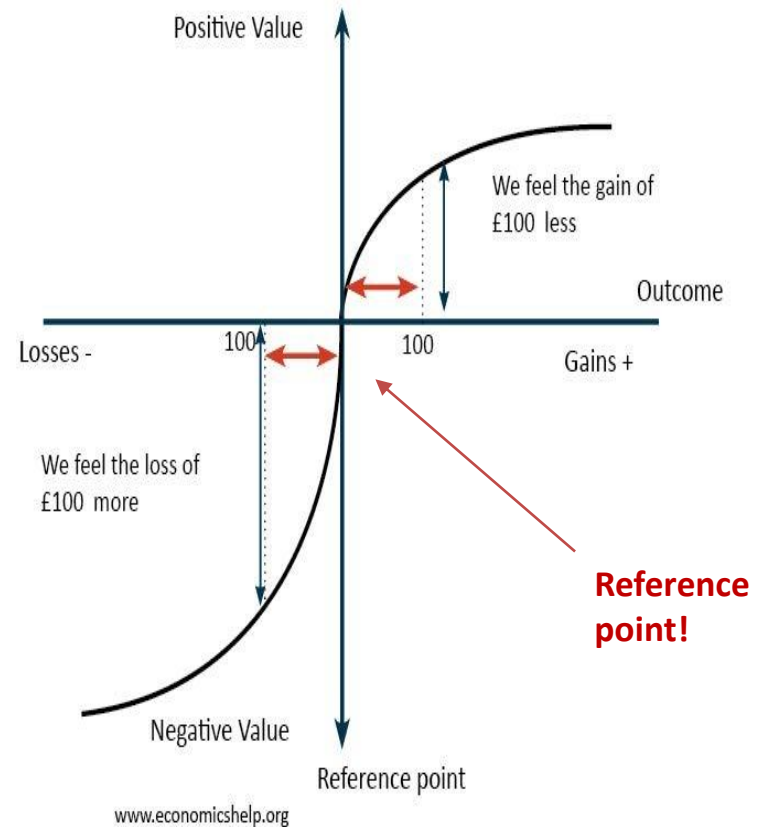
<https://www.youtube.com/watch?v=btTZA0XLd2k>

PROSPECT THEORY

- Situation 1:
 - Option 1: Earn 100,000 Euros for sure
 - Option 2: A lottery: earn 0 with 99% and 10,000,000 with 1%.
- Situation 2:
 - Option 1: Pay 100,000 Euros for sure.
 - Option 2: A lottery: pay 0 with 99% and 10,000,000 with 1%.
- Which option do you prefer in each situation?

PROSPECT THEORY

- Loss aversion: “*Losses loom larger than gains*”
- Risk aversion in gains, risk seeking in losses
- Under-weighting or overweighting of small probabilities
- “Cumulative prospect theory”: Classification as a ‘loss’ or ‘gain’ depends on the **reference point**



PROSPECT THEORY

Summarizing, Prospect Theory has three main features:

- (1) Reference-dependent utility
- (2) Loss aversion
- (3) Diminishing sensitivity to losses and to gains (→ risk aversion in gains, risk seeking in losses)

FRAMING

- Prospect theory is related to another major point identified in behavioral economics: Framing.
- Framing effect = individuals respond differently to identical decision-making situations if they are described differently.

KHANEMAN & TVERSKY 1981 – ASIAN DISEASE

“Imagine that the US is preparing for the outbreak of an Asian disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume the exact scientific estimates of the consequences of the programs are as follows:

- (1) Program A: 200 people will be saved.
- (2) Program B: $\frac{1}{3}$ probability that 600 are saved, and $\frac{2}{3}$ probability that no one is saved.

What do you chose?

FRAMING (KHANEMAN & TVERSKY 1982)

Now assume instead that you must choose between the following two options:

(1) Program C: 400 people will die.

(2) Program D: $\frac{1}{3}$ probability that nobody would die, $\frac{2}{3}$ probability that 600 will die.

What do you chose now?

FRAMING (VAN AAKEN & SAREL)

We recently tested an alternate version of the Asian disease problem in the context of proportionality.

We found the following ranking of being affected:
Laymen > Law students > Judges.

Why might expertise mitigate framing?

LAST EXAMPLE

How well do you think you will do in the exam for this course?

- (1) I will be in the top 25%
- (2) I will be in the 25%-50% quartile
- (3) I will be in the 50%-75% quartile
- (4) I will be in the bottom 25%

Please take a piece of paper and write on it a number (1-4) that you think best predicts your performance. Don't worry, it will be anonymous.

BEHAVIORAL ECONOMICS: SUMMARY

- We saw several deviations from perfect rationality
 - Social preferences (Ultimatum game)
 - Loss aversion (Prospect theory, endowment effect, framing)
 - Availability heuristic (coconuts)
 - Representativeness heuristic / Conjunction fallacy (Linda problem)
 - Anchoring (African States)
 - Framing (Asian Disease)
 - Over-optimism (quartiles)

BEHAVIORAL ECONOMICS: EFFICIENCY?

- What do these behavioral insights mean for efficiency?
 - Should we care about non-selfishness?
 - Do biases cancel out in the general population? (some are optimistic, some are pessimistic)
 - Do market forces overcome the biases?
 - Do laws already account for behavioral insights? If not, can we change them to account for these insights?
- These are the types of questions we are interested in
Behavioral law and economics!

BEHAVIORAL L&E

WHAT IS BEHAVIORAL L&E?

“The key difference between neoclassical and behavioral law and economics is the **set of assumptions** the analyst makes about the utility function of a representative member of society. While first-generation law and economics models often have assumed that individuals exclusively maximize personal monetary profit, **behavioral models allow for richer definitions of utility, and in particular for social preferences.**”

(Engel, WP 2018)

BEHAVIORAL L&E: AVAILABILITY HEURISTIC

- Are judges too harsh or too lenient on sentencing? (*Diamond & Stalans 1989*)
 - Experiment: asked laymen and judges to decide on sentence in different cases.
 - No significant differences were found. Then why do people feel judges are too lenient?
 - Availability heuristic: people think of newspaper articles & repeat offenders, but don't think about other cases.
- Other examples:
 - Are judges biased when estimating probability of guilt / liability?
 - Do criminals misperceive the probability of conviction? ("I know someone who downloaded a song but wasn't caught = must be a low probability of punishment")
 - Do parties to a contract under/over estimate risk?

BEHAVIORAL L&E: REPRESENTATIVENESS

Example from *Guthrie, Inside the judicial mind (2001)*: ■

The plaintiff was passing by a warehouse owned by the defendant when he was struck by a barrel, resulting in severe injuries. At the time, the barrel was in the final stages of being hoisted from the ground and loaded into the warehouse. The defendant's employees are not sure how the barrel broke loose and fell, but they agree that either the barrel was negligently secured or the rope was faulty. Government safety inspectors conducted an investigation of the warehouse and determined that in this warehouse: (1) when barrels are negligently secured, there is a 90% chance that they will break loose; (2) when barrels are safely secured, they break loose only 1% of the time; (3) workers negligently secure barrels only 1 in 1,000 times.

What is the probability that the defendant was negligent?

(1) 0-25% (2) 26%-50% (3) 51%-75% (4) 76%-100%

BEHAVIORAL L&E: REPRESENTATIVENESS

- Most participants said it's high (90%),
- But the actual probability of guilt is only 8.26%!
- Confusion is caused when you overlook the information in (3), i.e. that “only 0.1% of workers are negligent”.

TABLE 3: CONDITIONAL PROBABILITIES FOR RES IPSA LOQUITUR PROBLEM

		Event		Total
		Injury	No Injury	
Actual Condition	Negligent	0.090%	0.010%	0.10%
	Not Negligent	0.999%	98.901%	99.90%
Total		1.089%	98.911%	100%

$$[\text{Bayes' rule } \Pr(t = g | \text{Harm} = 1) = \frac{\Pr(\text{Harm} = 1 | t = g) \cdot \Pr(t = g)}{\Pr(\text{Harm} = 1 | t = g) \cdot \Pr(t = g) + \Pr(\text{Harm} = 1 | t = l) \cdot \Pr(t = l)} = 0.09\% / 1.089\% = 8.26\%]$$

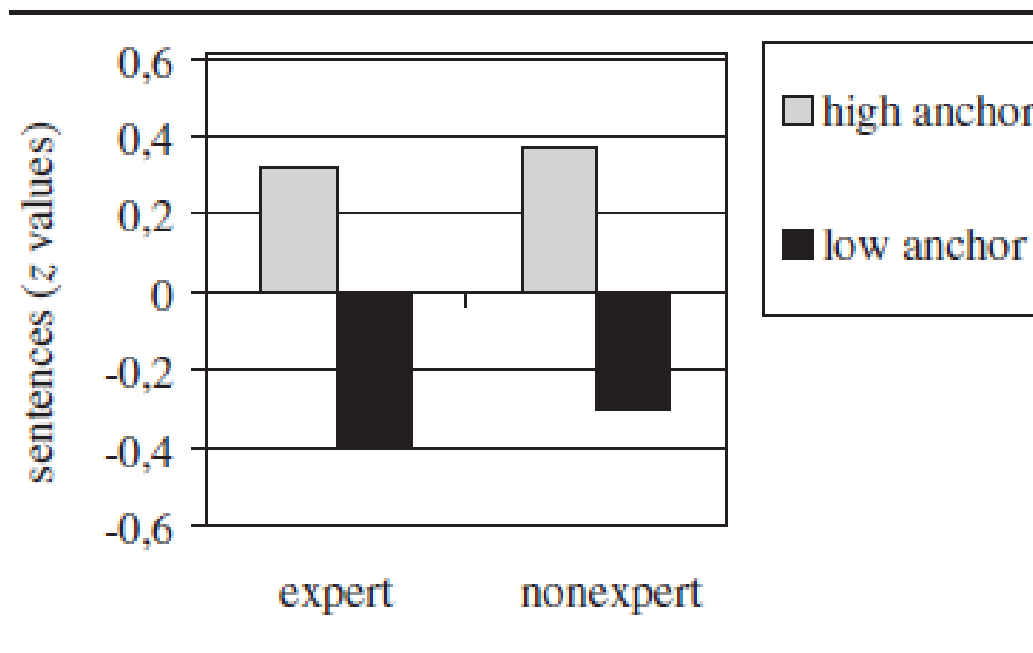
You don't need to calculate using bayes' rule for the exam!

BEHAVIORAL L&E: ANCHORING *(ENGLISCH ET AL 2006)*

- Study 1: Journalism
 - experts and non-experts receive facts about a legal case (rape). They are told that a journalist calls them and asks “do you think the penalty X is appropriate?”
 - Participants then decide on the sentence they would give.
 - → When X was high, participants gave higher sentences, and vice versa.
- Study 2: Prosecutor’s demand / Random chance
 - Same idea (different case – shoplifting), but now prosecutors demand X (either high or low)
 - → BUT: they were told that the demand of the prosecutor was **random**

BEHAVIORAL L&E: ANCHORING *(ENGLISCH ET AL 2006)*

- Experts and non-experts are both susceptible!



BEHAVIORAL L&E: ANCHORING *(ENGLISCH ET AL 2006)*

- Study 3: Dice!
 - Same idea, but participants throw a dice by themselves
 - → Anchoring still led to the same result
- Study 4: Incriminating vs. exculpatory evidence
 - First, a dice anchor (as before).
 - Then, subjects read a set of statements, some exculpatory and some inculpatory.
 - They were asked to classify which is which, and the researcher measured the time it took to classify.
 - How do you think the anchor affected the task?

BEHAVIORAL L&E: ANCHORING *(ENGLISCH ET AL 2006)*

- Those with a high anchor classified incriminating evidence faster
- → Thinking of a high sentence reminded them of the guilt and made it easier to classify these
- No difference in classifying exculpatory evidence
 - Authors say it's because negative information is processed faster

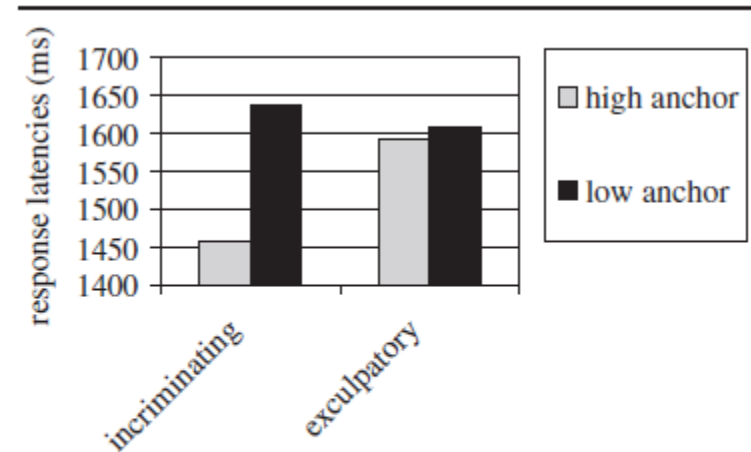


Figure 2 Response latencies (milliseconds) for categorizations of incriminating versus exculpatory arguments by judges who had been exposed to a high versus low sentencing demand (Study 4).

BEHAVIORAL L&E: ENDOWMENT EFFECT *(WOOD 2001)*

- Consumers who are dissatisfied with a product can usually return it, at least during a “cool-off period”.
 - Even if the product isn’t faulty
- How does the endowment effect relate to product returns?
- → People get attached to the product after they buy, and therefore return less than anticipated by rational choice.
- Explains why companies offer lenient return policies: they know that (some) people won’t return anyway

BEHAVIORAL L&E: ENDOWMENT EFFECT *(ARLEN & TONTRUP 2015, BECHER, FEESS, AND SAREL 2019)*

- Does the endowment effect justify legal intervention?
- Experimental evidence: endowment effect is driven by anticipation of regret from selling. Delegating to an agent prevents that, and most sales in the world are done through agents.
- Theory: endowment effect may be a justification for non-intervention, as it prevents people from being exploited by firms (e.g. airlines) that charge high cancellation fees.

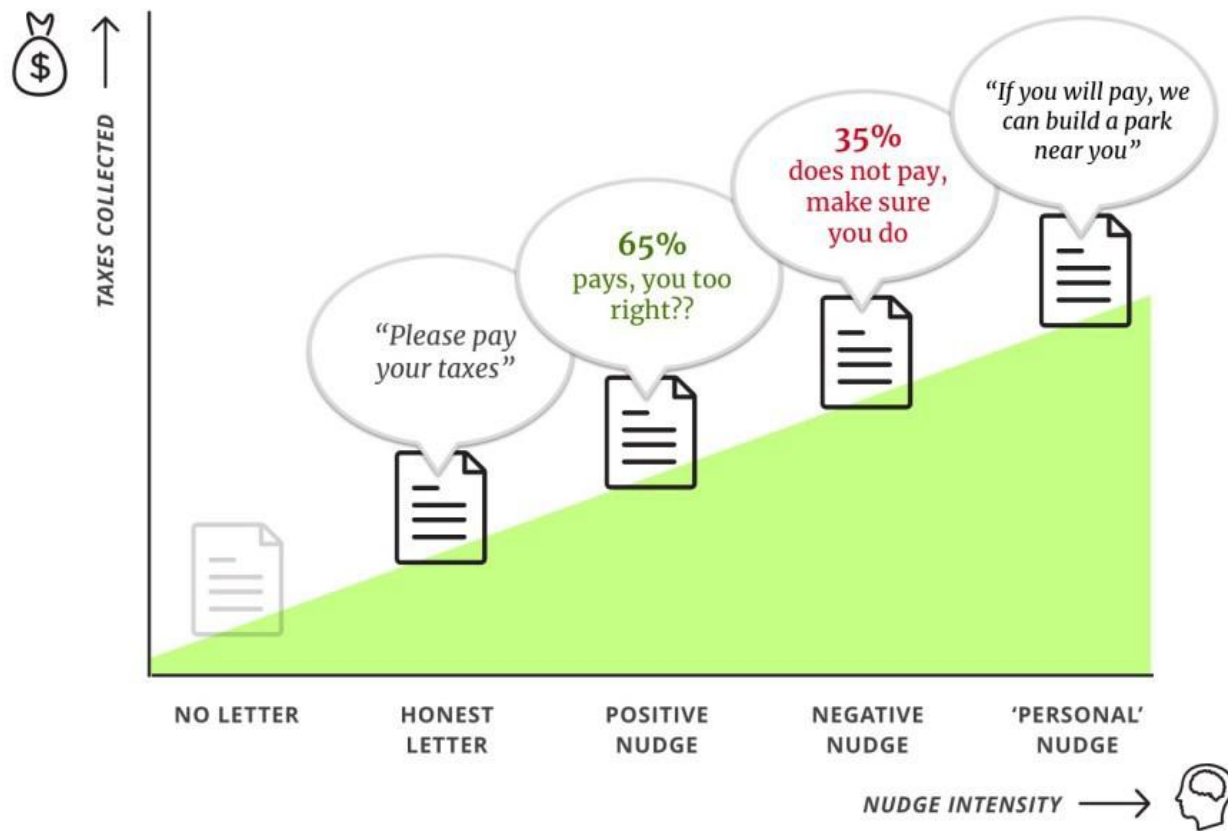
BEHAVIORAL L&E: WHAT TO DO WITH THE INSIGHTS?

- These insights provide *positive* descriptions, but can we use them for *normative* purposes to impact people's behavior?
- **Nudging:** “*any aspect of the choice architecture that alters people's behavior in a predictable way without forbidding any options or significantly changing their economic incentives*”
(*Thaler & Sunstein, pp 6*).
 - E.g. changing the default, when cost of altering is zero.
- Improving theoretical models: adding a behavioral component

NUDGING: EXAMPLE 1



NUDGING: EXAMPLE 2



<https://medium.com/@frankvandeven/nudging-making-tax-payers-actually-pay-af36fda7e434>

NUDGING: EXAMPLE 3 - STICKY DEFAULTS

- Laws can be mandatory or they can set a *default*
- Defaults may be “sticky” for various reasons:
 - Rational: default is perceived as the social norm (declaratory role of the law), and deviation includes cost (e.g. reputation)
 - Behavioral: people tend to stick to the default, for non-rational reasons
 - E.g. because of reference points (we will discuss this later in behavioral law & economics)
 - Or because of lack of salience of alternatives

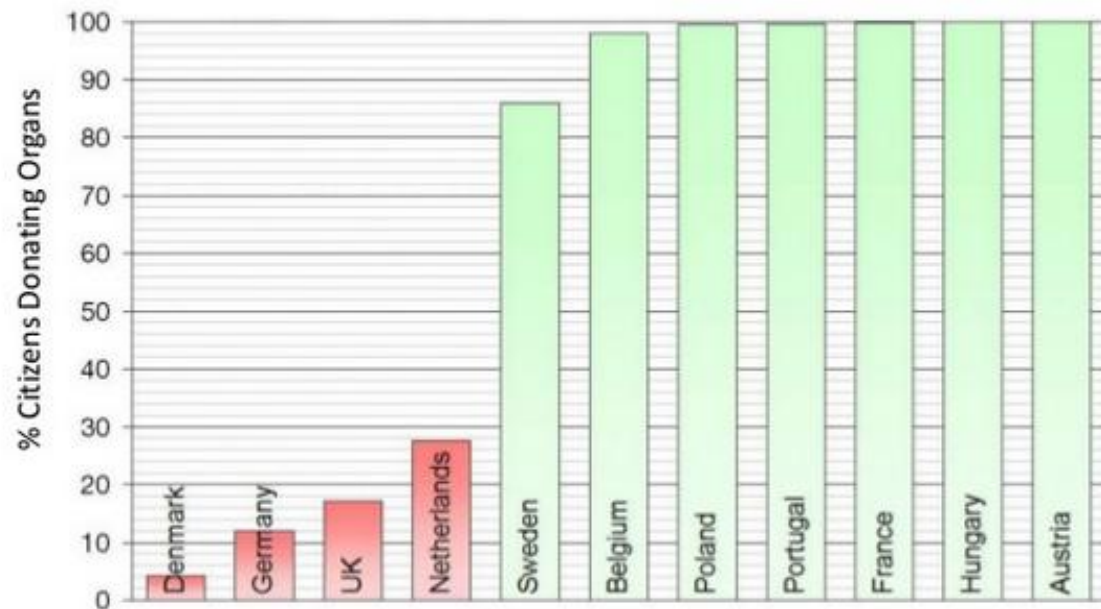
NUDGING: EXAMPLE 3 - STICKY DEFAULTS

<http://www.slideshare.net/theHCCA/regulations-and-ethics-on-using-ehr-data-for-research-2014-hcca-research-compliance-conference>

People seem to stick with the first choice that is presented.

Example: organ donation

Example: Opt-In Vs. Opt Out Organ Donation



Source: Johnson, Eric J. and Daniel Goldstein. 2003. "Do Defaults Save Lives?" *Science* 302: 1338-1339

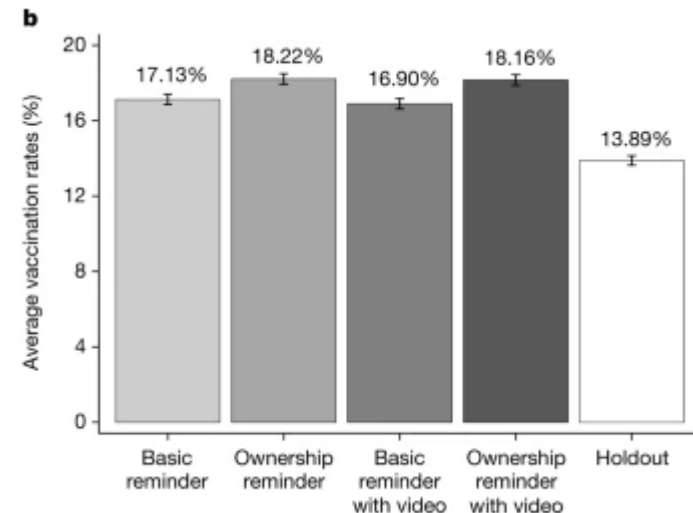
Fundamental cultural difference...or just laziness to not check boxes?

WHAT DETERMINES WHETHER NUDGES WORK?

- The literature is still trying to figure that out... Some distinctions:
 - 1) Personal v. non-personal nudges
 - 2) Substitutes vs. complements of other laws
 - 3) Cognitive Dissonance
 - 4) Susceptibility
 - 5) Transparency

1) Personal v. non-personal nudges

- Dai et. Al (2021): experiment on getting people to take a COVID vaccine
 - Basic reminder: e.g. “You have an appointment”
 - Ownership reminder: “A vaccine has been saved for you”
 - Use of Video in the reminder
- Result: **personal nudges work best**



Proportion of participants who scheduled
An appointment for the 1st Covid shot within 4 weeks
Source: Dai et al. (2021)

2) Substitutes vs. complements

- Elm & Sarel (AJLM, 2022): nudges would perform better if they *complement* rather than substitute other rules.
- E.g. a sign “please don’t meet others at home” won’t work well if the regular policy is to allow meeting at the office



3) Cognitive dissonance (Cialdini, Robert B., et al. "Managing social norms for persuasive impact." *Social influence* 1.1 (2006): 3-15)

- Refresher: The petrified forest in Arizona suffered thefts
- A field experiment: add signs
 - 1) Injunctive – “please don’t take pieces” (=strong) or “please keep pieces in the park” (=weak)
 - 2) Descriptive – “many people have stolen before..” + images, v. no description.
- Remember what happened?

Petrified Forest National Park

4,7 ★ ⓘ (13K) · National park in Arizona · Closed ⋮



3) Cognitive dissonance

- Descriptive (mentioning many people stole) led to *more* theft
 - “Focus theory” (people focused on that)
 - “Cognitive Dissonance” – it signals that the prohibited action “must be profitable”, as otherwise why bother prohibiting it
- Strong injunction (“please don’t”) was more effective

4) Susceptibility (Patel, Mitesh S. "Text-message nudges encourage COVID vaccination." (2021): 336-337.)

- Are all people equally susceptible to nudges?
- Patel (2021) highlights the conditional for nudges of vaccinations to work
 - Prompt action
 - Increase motivation
 - Make action easier

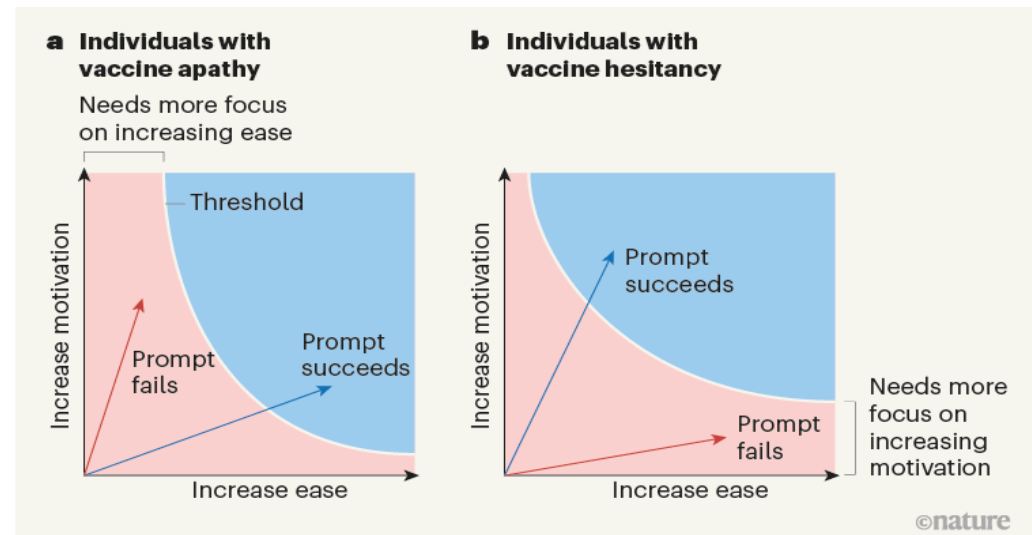


Figure 1 | Nudges to encourage vaccination uptake will depend on individuals' attitudes. Nudges are ways in which information is presented or choices offered that drive behavioural change. Dai *et al.*¹ report the effects of sending individuals text-message reminders to encourage them to book an appointment to receive their vaccine against COVID-19. According to the Fogg Behavior Model, nudges should combine three aspects for success: they should prompt an action, increase motivation and make action easier. But nudges need to be tailored to individuals with different attitudes. **a**, If individuals are apathetic about getting a vaccine, prompting nudges will need to focus more on making it easier to arrange an appointment. Dai *et al.* achieved this by including a link to the booking website in the text messages. **b**, If individuals are more hesitant about getting vaccinated, such nudges will need to focus on increasing motivation.

4) Susceptibility (van Aaken & Sarel (2025). "Framing effects in proportionality analysis: experimental evidence." Journal of Law & Empirical Analysis, 2(2) 174-205, (2025))

- Expertise may matter for susceptibility!
- Example: using Framing to influence judges

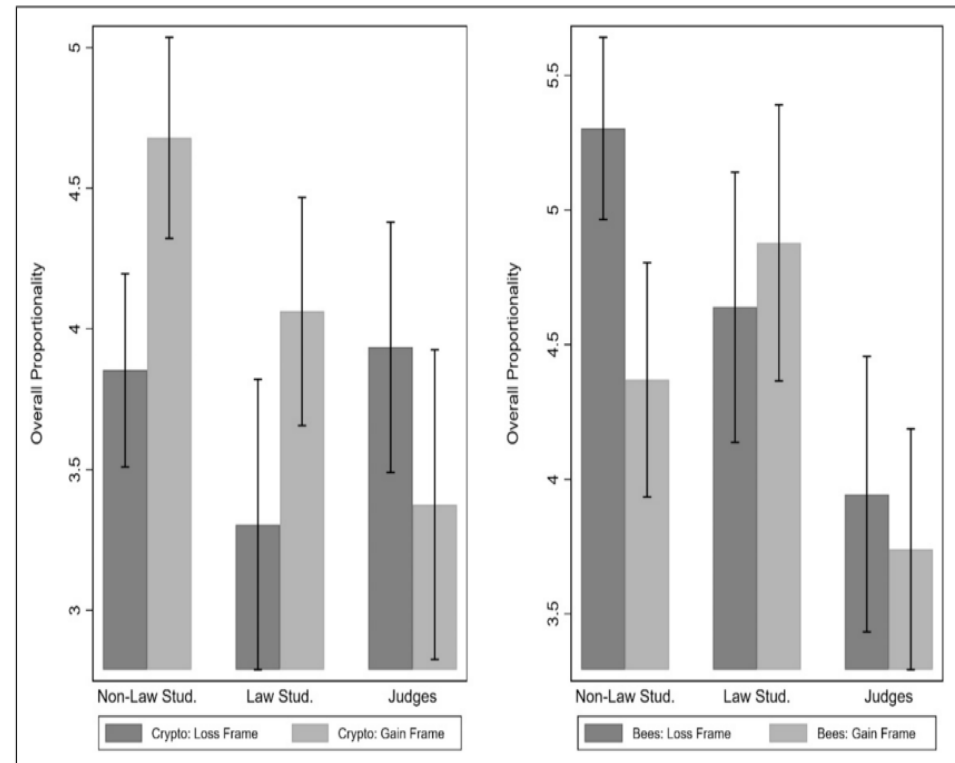


Figure 1. Comparison of Mean Agreement Rates with Overall Proportionality.

Note. This figure compares the mean rate of agreement with the statement that the act is overall proportional. The dark gray bars correspond to the loss frames, whereas the light gray bars correspond to the gain frames. The lines on top of the bars represent 95% confidence intervals

4) Susceptibility (van Aaken & Sarel (2025). "Framing effects in proportionality analysis: experimental evidence." Journal of Law & Empirical Analysis, 2(2) 174-205, (2025))

- But:
susceptibility
persists outside
the scope of
expertise!

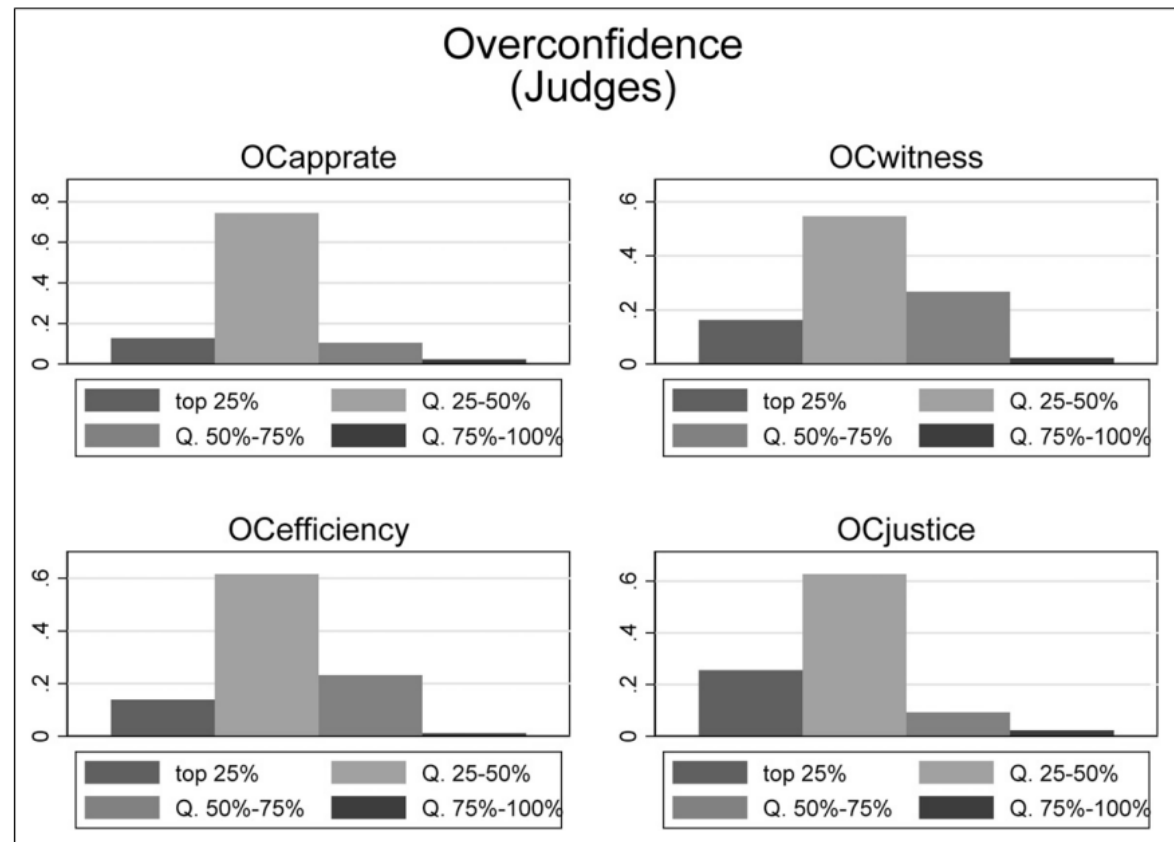


Figure 2. Overconfidence Among Judges

5) Transparency (Bruns, Hendrik, et al. "Can nudges be transparent and yet effective?." *Journal of Economic Psychology* 65 (2018): 41-59.)

- Does it matter if I tell you “you are being nudged”?
- Bruns et al ran an experiment where they set a default on carbon emissions, but were transparent about it.
- Result: it didn’t matter – nudge worked even when it was transparent

NUDGES V. SLUDGES

Nudges are usually ■
about making
certain choice
easier to make

In the past few ■
years: discussion
on making choices
harder to make



NUDGES V. SLUDGES

- Importantly: nudges are not “good” and sludges are not “bad”.
 - Thaler (Science, 2018) sort of disagrees – he says sludges are often used to achieve undesirable behaviors:

“sludge can take two forms. It can discourage behavior that is in a person's best interest such as claiming a rebate or tax credit, and it can encourage self-defeating behavior such as investing in a deal that is too good to be true.”
- Nudges and sludges don't necessarily operate using the same mechanism (one is psychology, the other is effort cost + psychology).

BEHAVIORAL L&E VS. STANDARD L&E

Behavioral L&E

- Wider assumptions (non-rationality, biases, social preferences)
- Longer models, complex to derive
- Supported by experimental evidence
- Sensitive to choice of parameters, and specific assumptions

Traditional L&E

- Assumptions of traditional economics (homo(o)economicus)
- Parsimonious models
- Mostly consistent with empirical evidence (aggregate behavior “as if” maximizes utility).
- Usually provides clear(er) predictions

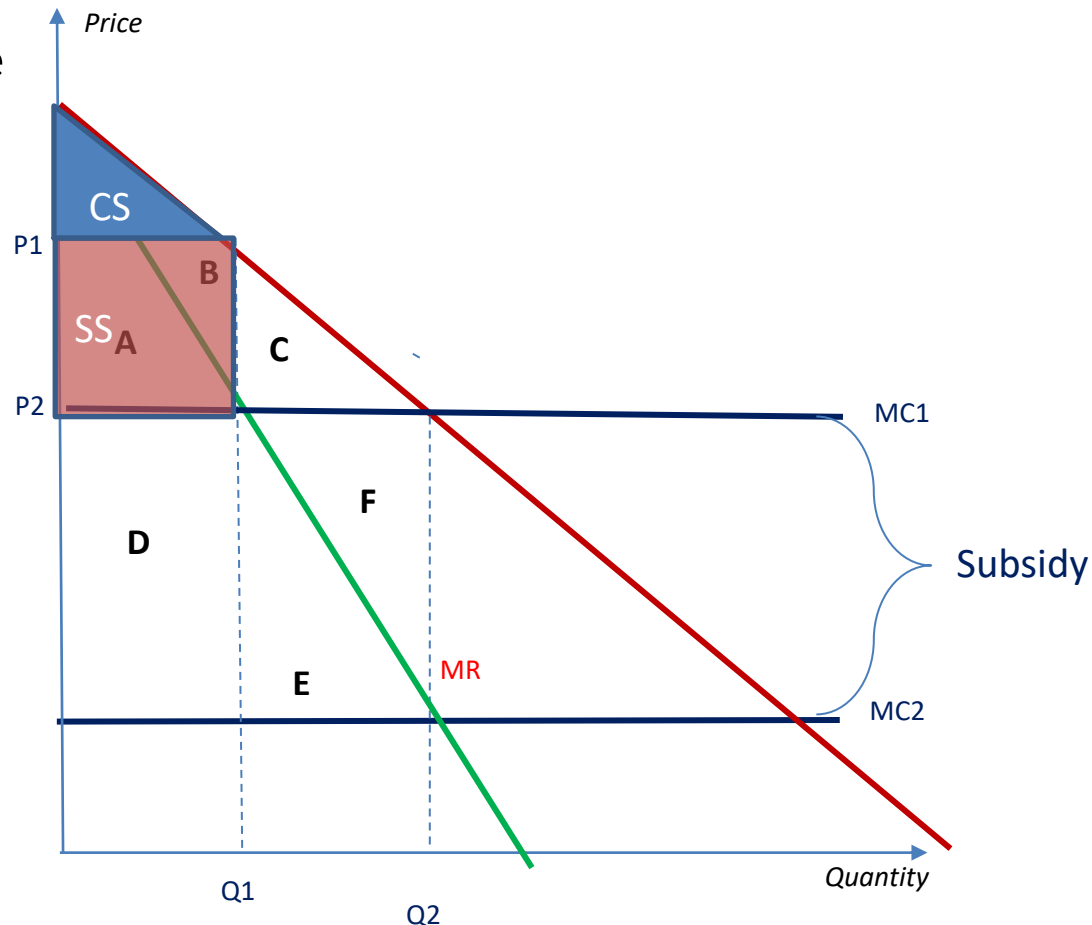
The End



Appendix

Subsidy – Monopolies- clarification

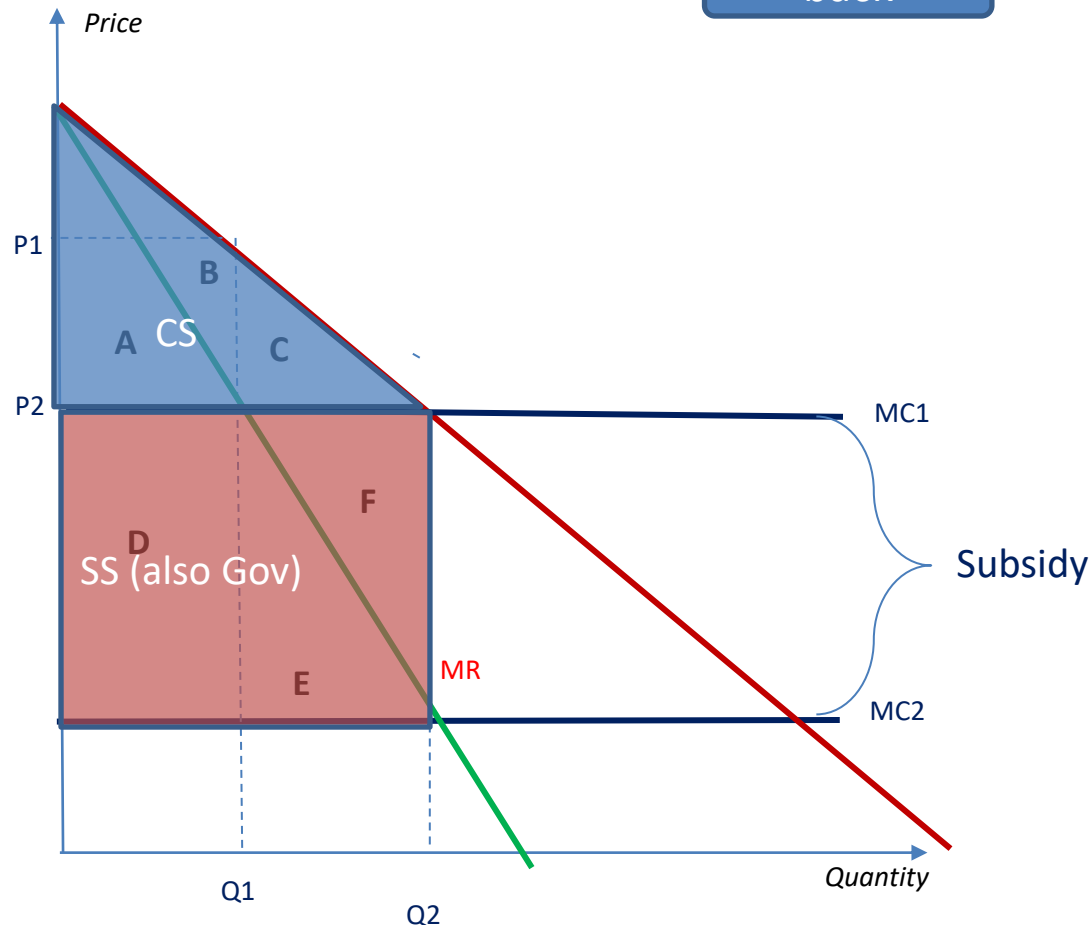
- Here are the surpluses before the subsidy.
- The blue triangle is the CS before the subsidy
- The pink square is the SS before the subsidy
- Without a subsidy, the government spending (GOV) is, of course, zero.



Subsidy – Monopolies- clarification

back

- Here are the surpluses after the subsidy.
- The blue triangle is the CS. As can be seen, it now includes, in addition, also areas A, B and C.
- The pink square is the SS after the subsidy. It includes D, E, and F – but no longer includes A and B.
- The pink square is also the government spending: size of subsidy times number of units sold.
- The comparison yields the changes in the main slides.



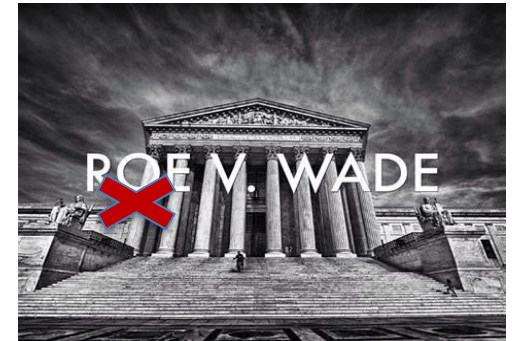
What is N-o-t Roee's name



~~Roo~~



~~Row~~



back